

# Dimensional hyperreduction of nonlinear finite element models via empirical cubature with manifold-adaptive weights

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## Abstract

Nonlinear-manifold reduced-order models for parametrized finite element problems can achieve substantial compression both in the number of generalized (latent) coordinates and, through sampling-and-weighting hyperreduction, in the number of sampled elements/integration points. Yet current sampling-and-weighting approaches employ weights that remain fixed over the entire solution manifold. We contend that this restriction leaves part of the hyperreduction potential untapped: allowing the weights to vary continuously and nonlinearly with the latent coordinates can further decrease the number of spatial entities that must be sampled. To exploit this possibility, we propose the Manifold-Adaptive-Weight Empirical Cubature Method (MAW-ECM). Starting from a feasible fixed-weight ECM rule, a greedy pruning strategy progressively removes sampled entities through a sequence of convex quadratic weight-redistribution problems that enforce the local conditions and positivity. The methodology is assessed on two nonlinear benchmarks: a hyperelastic metamaterial and a history-dependent continuum-damage problem. In both cases, the nonlinear manifold is constructed from an initial linear compression followed by an input-informed identification of the latent coordinates as general linear combinations of the retained modal amplitudes, incorporating graph information when relevant to seek the intrinsic dimensionality of the solution manifold. We show that combining the nonlinear-manifold representation with MAW-ECM reduces the number of sampled integration points by more than two orders of magnitude relative to the corresponding standard linear reduced model. Relative to the fixed-weight manifold models alone, the adaptive weights eliminate approximately 80% of the remaining points in the hyperelastic benchmark and more than 97% in the damage benchmark, while essentially preserving their accuracy.

*Keywords:* Nonlinear manifold reduced-order models; Hyperreduction; Empirical Cubature Method; Adaptive cubature weights; Computational mechanics

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## 1. Introduction

### 1.1. Manifold HROMs using standard sampling-and-weighting hyperreduction

Recent years have witnessed rapid development of projection-based hyperreduced-order models built on nonlinear solution manifolds, hereafter referred to as manifold HROMs. In these models, the high-dimensional state is represented as a nonlinear function of a small set of latent coordinates. Galerkin formulations enforce the governing equations on the tangent space of the manifold [1, 2], whereas least-squares Petrov-Galerkin formulations determine the reduced state through residual minimization [1, 3, 4, 5, 6]. These approaches constitute a natural evolution of classical projection-based HROMs [7, 8], in which the state is confined to a fixed linear space, typically obtained from solution snapshots through proper orthogonal decomposition (POD), computed in turn via the singular value decomposition (SVD). Among the various

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strategies encompassed by the term “hyperreduction”, we focus here on *sampling-and-weighting* procedures, in which the full-order evaluation of the nonlinear terms entering the reduced equations is replaced by a positive linear combination of a subset of local contributions. This class includes the Energy-Conserving Sampling and Weighting method (ECSW) [7, 9], the Empirical Cubature Method (ECM) [8, 10], and the Empirical Quadrature Procedure (EQP) [11, 12].

Of the two levels of reduction involved in manifold HROMs—the nonlinear parametrization of the state and the subsequent hyperreduction by sampling-and-weighting—research efforts have so far been concentrated almost entirely on the former. Representative examples (the list is not exhaustive) include methods that learn the nonlinear manifold directly in the high-dimensional state space via autoencoders [1, 13], as well as approaches that first compress the state linearly, typically by POD, and subsequently construct a nonlinear parametrization in the resulting reduced coordinate space [14, 3, 4, 6, 15]. Despite their different architectures, all these approaches share the same objective: to identify the smallest set possible of latent variables and reconstruct with maximum accuracy the high-dimensional training states from them.

By contrast, the second compression stage has so far been carried out essentially with the same sampling-and-weighting machinery developed for standard HROMs; in particular, the manifold HROM formulations in Refs. [2, 4, 5, 6] all rely on ECSW. Neither specific algorithms nor refinements of existing ones have been devised to exploit the new nonlinear-manifold framework. ECSW (and, in essence, the same applies to ECM and EQP) may be viewed as seeking a sparse vector of nonnegative weights whose local contributions reproduce, within a prescribed accuracy, those of the high-dimensional model over the training set. These contributions are assembled into a matrix that collects, for every candidate spatial entity, the projected contributions generated over the sampled states. Passing from a linear HROM to a manifold HROM reduces the number of such conditions generated at each state (and hence the size of the matrix), because projection is now performed onto a much smaller state-dependent tangent space; this reduction in the number of conditions per state helps explain the smaller sampled meshes reported for manifold HROMs using ECSW, in comparison with standard HROMs, in Refs. [2, 4, 6, 5]. Yet the underlying approach remains essentially unchanged: a single weight vector is required to satisfy the conditions from all training states simultaneously. Since the constraints are linear, the resulting rule should be able to reproduce, not only the conditions obtained from the training data, but the entire linear span of such conditions, including combinations that need not correspond to any state on the solution manifold.

### 1.2. Motivation for using adaptive weights

It is precisely this global linear character of the standard sampling-and-weighting construction that we challenge in the present paper. Our central claim is that this standard approach does not fully exploit the nonlinear-manifold structure underlying the reduced model. If the state is parameterized on a curved manifold, and quantities entering the reduced equations, such as the test functions in a Galerkin projection, change as the solution moves over that manifold, then the positive weights used for hyperreduction should change as well, rather than remain fixed over all latent states. The expected consequence of this adaptiveness is a further reduction in the number of spatial entities, such as elements, control volumes, or integration points, that must be retained to reproduce the full-order contributions entering the reduced equations.

The mechanism by which adaptive weights can reduce the number of required spatial entities can be readily illustrated by a simple quadrature problem: sampling-and-weighting hyperreduction can be interpreted, in essence, as a quadrature/cubature problem, as made explicit by the ECM and EQP [8, 11], as well as by the pioneering work of Ref. [16]. Consider the one-dimensional family  $f_q(x) = x^q$ , with  $x \in [0, 1]$  and  $q \in \{0, 1, \dots, P\}$ , whose exact integral is  $I(q) = \int_0^1 f_q(x) dx = 1/(q+1)$ . Here, the monomial exponent  $q$  may be regarded as a discrete latent coordinate, while any  $x \in [0, 1]$  is a potential sampling point. It can be readily seen that, if a single fixed-weight rule is required to integrate every latent state exactly,  $I(q) = \sum_{g=1}^m f_q(x_g) \omega_g$ , then it must also integrate their complete linear span, namely, the polynomial space of degree at most  $P$ . Even the optimal Gauss–Legendre rule therefore requires  $m \geq (P+1)/2$  sampling points: the required number of integration points grows linearly with the number of latent states. By contrast, if one retains a single sampling point, say  $x_1 = 1$ , and allows its weight to depend on the latent coordinate, then  $I(q) = \omega_1(q) f_q(x_1)$ , with  $\omega_1(q) = 1/(q+1)$ ; thus, one point suffices for every  $q$ , irrespective of  $P$ . The gain in efficiency (from  $m \geq (P+1)/2$  points to just  $m = 1$  point) comes thus from allowing the weight to adapt to the latent state, as asserted.

In a more practical context, further evidence that adaptive weights can substantially reduce the number of sampled spatial entities is provided by our previous work [17] on local HROMs. In local HROMs, the solution manifold is represented piecewise by a collection of local linear subspaces, each associated with a different region of the solution space. Sampling-and-weighting hyperreduction in this setting was first explicitly addressed in Ref. [18]. Building on this setting, and using the Empirical Cubature Method, ECM, as the basic point-selection procedure, our proposal in Ref. [17] was to endow each subspace with its own nonnegative weights while retaining a common set of cubature (i.e., integration) points; we termed the resulting approach the Subspace-Adaptive Weights ECM (SAW-ECM). Numerical results in Ref. [17] showed that, as the local representation is progressively refined, the required common support decreases markedly, approaching in the limiting regime a size governed by the dimension of the parameter space rather than by the dimension of the global span of all subspaces.

### 1.3. Manifold-adaptive weights ECM

The main novel contribution of the present work is the extension of the SAW-ECM described above from a discrete collection of subspaces to a continuously parameterized nonlinear manifold. In direct analogy with the terminology of Ref. [17], we term the resulting method the *Manifold-Adaptive Weights ECM* (MAW-ECM).

In passing from a discrete collection of subspaces in the SAW-ECM to a continuous manifold in the proposed MAW-ECM, the subspace index associated to each local subspace is replaced by the latent coordinates, so that each spatial entity is endowed with a positive weight field over the latent space. This transition introduces, however, an essential additional requirement: the corresponding weight fields should be as smooth as possible over the latent space. Such smoothness is important because the discrete weight samples must ultimately be represented by regression over the latent domain; strong local oscillations would compromise both the accuracy of this regression and the robustness of the nonlinear solution procedure (Newton-Raphson in the examples considered here). To enforce this smoothness requirement, we propose to modify the sparsification-type optimization problem of Ref. [17] by adding a graph-based regularization term into the objective function, thereby penalizing irregular variations of the weight fields.

As in Ref. [17], and also in the standard ECSW, ECM, and EQP, the resulting sparsification problem is NP-hard [19] because of the presence of the  $\ell_0$  pseudo-norm, which measures the number of nonzero entries of the weight vectors. Practical approaches to  $\ell_0$ -minimization typically rely either on suboptimal greedy or active-set procedures, as in ECM and ECSW, or on suitable convex relaxations, most commonly obtained by replacing the  $\ell_0$  objective with an  $\ell_1$ -based surrogate promoting sparsity, as in EQP [12, 11]. The latter strategy was explored by the authors in Ref. [17] for the SAW-ECM. The main conclusion drawn there, which carries over to the problem at hand, is that this  $\ell_1$ -based convexification route is not viable in practice, since it requires stacking all adaptive weight vectors into a single global optimization variable from the outset and, in doing so, fails to exploit the fundamentally local structure of the constraints associated with each sampled latent state.

Accordingly, the present work does not pursue this  $\ell_1$ -based route, but instead adopts a greedy iterative strategy that avoids treating the entire global optimization variable simultaneously at all iterations. The underlying idea is inspired by the *pruning* strategies employed in the derivation of generalized quadrature rules [20, 21, 22], where the goal is likewise to identify the smallest possible set of integration points, including both their positions and weights. The analogy with the present MAW-ECM setting is the following. In both cases, sparsification starts from an already feasible cubature rule and proceeds through successive pruning steps in which integration points are removed while the remaining weights are redistributed so as to preserve the constraints. The essential difference lies in the nature of the design variables. In generalized quadrature, such as in the continuous version of the ECM proposed by the authors in Ref. [22], both the positions and the weights of the cubature points evolve continuously during pruning. In the present MAW-ECM setting, by contrast, the candidate entities are fixed in space (i.e., Gauss points inherited from the FE discretization), and only the manifold-dependent weights are modified. At each tentative elimination, redistributing the remaining weights leads to a convex quadratic program with linear equality and positivity constraints. A key ingredient of the proposed procedure is an algorithm that exploits the local structure of the state-wise constraints and activates the graph regularization, the most computationally demanding component of the method, only when required to preserve positivity.

#### 1.4. Input-informed construction of the nonlinear manifold

The proposed methodology is assessed on two nonlinear finite element (FE) solid-mechanics benchmarks: a large-strain hyperelastic homogenization problem (Section 6) and a history-dependent damage problem (Section 7). A prerequisite for applying the MAW-ECM is the identification of suitable latent coordinates and the corresponding nonlinear function between high-dimensional state and such latent variables (we call it here the *decoder*). As in Refs. [4, 6, 15], we first apply POD to obtain a linear compression of the state and then construct the nonlinear decoder in terms of the resulting modal coefficients. Our approach differs in the definition of the latent coordinates. Whereas Refs. [4, 6] identify them with the leading POD coefficients, and Ref. [15] selects an informative subset not necessarily following the singular-value ordering, here the latent coordinates are allowed to be arbitrary linear combinations of the retained modal coefficients. Our proposal is that the identification of these linear combinations should also exploit the associated problem inputs, as well as, when relevant, their relative arrangement, encoded through a graph over the training states. More specifically, in the hyperelastic problem of Section 6, the input parameters are used directly in a least-squares identification, while in the damage problem of Section 7, the nonlinear component of the latent vector is identified from the damage-evolving states through a graph-based eigenvalue problem informed by the loading history. In both cases, we manage to identify the intrinsic dimensionality of the solution manifold, i.e., the minimum number of independent latent coordinates required for its representation.

To the best of the authors' knowledge, this input-informed approach to constructing the POD-based decoder constitutes an original contribution of the present work, particularly for history-dependent deformations, where the loading history, its graph structure, and constitutive information guide the identification of the latent coordinates. A related recent work [23] has considered manifold HROMs for history-dependent problems, with POD-based decoder and fixed-weight ECSW hyperreduction; however, they do not seek an intrinsically minimal latent representation (latent variables are chosen as leading POD coefficients), nor do they construct the decoder using problem-specific graph-based, input information (the nonlinear component is quadratic).

#### 1.5. Scope

We choose parametrized, quasi-static nonlinear solid finite element problems subject to affine kinematic constraints as the vehicle for presenting and assessing the proposed methodology. More specifically, the formulation is cast in terms of interior Gauss-point sampling (i.e., as a true cubature problem), with the projected internal forces and, when applicable, volumetric quantities of interest as the terms to be hyperreduced. These choices are not intrinsic to MAW-ECM, which can in principle be extended to other scenarios, including dynamic solid problems, other governing equations, and discretizations other than finite elements, provided that the corresponding local contributions and invariant constraints to be preserved by the adaptive rule are defined appropriately. Likewise, within the finite element setting, Gauss-point selection is only one possible realization: MAW-ECM may also be posed as a mesh-sampling procedure, as already done with fixed-weight ECM for standard HROMs in Refs. [24, 25].

As for the input-informed POD-based decoders, the particular constructions used in the two benchmarks are tailored to the problems considered, and no claim is made here regarding their generality as manifold-learning procedures. Finally, given the exploratory character of this study, the numerical assessment is deliberately restricted to low-dimensional parameter and latent spaces, for which the adaptive weight fields can be directly visualized and interpreted. Scalability to substantially higher-dimensional spaces, particularly regarding the construction of the latent graph used for regularization, lies outside the scope of the present work.

## 2. Full-order finite element model

Let us consider a general, parametrized, quasi-static nonlinear solid FE problem subject to affine kinematic constraints, in which the solution depends on an input parameter vector  $\boldsymbol{\mu} \in \mathcal{P} \subset \mathbb{R}^{n_\mu}$ , which may represent prescribed displacements and/or applied forces. In history-dependent problems, this dependence is understood incrementally, with the current response also depending on the previously converged internal variables. The FE nodal displacement vector  $\mathbf{d}^{\text{FE}} \in \mathbb{R}^{N_{\text{def}}}$  is written as

$$\mathbf{d}^{\text{FE}} = \mathbf{T}\mathbf{d} + \hat{\mathbf{d}}, \quad (1)$$

where  $\mathbf{d} \in \mathbb{R}^N$  ( $N < N_{\text{dof}}$ ) collects the independent degrees of freedom,  $\mathbf{T}$  is the lifting operator associated with the homogeneous constrained space, and  $\hat{\mathbf{d}} = \hat{\mathbf{d}}(\boldsymbol{\mu})$  accounts for the affine prescribed kinematic data. We define the full finite element residual  $\mathbf{R}^{\text{FE}} \in \mathbb{R}^{N_{\text{dof}}}$  as

$$\mathbf{R}^{\text{FE}}(\mathbf{d}; \boldsymbol{\mu}) = \mathbf{F}^{\text{int}}(\mathbf{d}; \boldsymbol{\mu}) - \mathbf{F}^{\text{ext}}(\boldsymbol{\mu}), \quad (2)$$

where  $\mathbf{F}^{\text{int}}$  and  $\mathbf{F}^{\text{ext}}$  are the nodal internal and external force vectors, respectively (the latter is assumed to be independent of  $\mathbf{d}$ ). The constrained parametrization (1) induces the admissible virtual displacements

$$\delta \mathbf{d}^{\text{FE}} = \mathbf{T} \delta \mathbf{d}. \quad (3)$$

Substitution of the preceding expression into the discrete virtual-work statement

$$\delta \mathbf{d}^{\text{FE}T} \mathbf{R}^{\text{FE}} = 0, \quad (4)$$

and exploiting the arbitrariness of  $\delta \mathbf{d}$ , leads to the FE equilibrium conditions

$$\mathbf{T}^T (\mathbf{F}^{\text{int}} - \mathbf{F}^{\text{ext}}) = \mathbf{0}. \quad (5)$$

### 2.1. Internal forces

By hypothesis, the only term in the above equilibrium equations that depends on the unknown displacement vector  $\mathbf{d}$  is the nodal internal force  $\mathbf{F}^{\text{int}}$ . This is therefore the term to be approximated by hyperreduction via the proposed adaptive empirical cubature rule. Since such a rule will be constructed from Gauss-point contributions, it is necessary to derive the expression for  $\mathbf{F}^{\text{int}}$  so that these contributions appear explicitly. We start from its total Lagrangian expression

$$\mathbf{F}^{\text{int}} = \int_{\Omega_0} \mathbf{B}^T \mathbf{P} d\Omega, \quad (6)$$

where  $\Omega_0$  denotes the reference domain,  $\mathbf{P} \in \mathbb{R}^{s_\sigma}$  is the first Piola–Kirchhoff stress vector ( $s_\sigma = 9$  for 3D and  $s_\sigma = 4$  for 2D), and  $\mathbf{B} \in \mathbb{R}^{s_\sigma \times N_{\text{dof}}}$  is the global operator relating nodal displacement variations to the corresponding variations of the deformation gradient. In the damage problem considered in Section 7, the formulation is small-strain; consequently, in this case,  $\mathbf{P}$  is to be replaced by the Cauchy stress vector, and  $\mathbf{B}$  becomes the standard finite element strain–displacement matrix. We next expand the integral in Eq. 6 as the sum of the contributions of each finite element, and then approximate such element contributions by Gauss quadrature; this leads to:

$$\mathbf{F}^{\text{int}} = \sum_{e=1}^{N_{\text{el}}} \mathbf{L}_e^T \left( \sum_{i=1}^{m_{\text{gp}}} \mathbf{B}_{e,i}^T \mathbf{P}_{e,i} w_{e,i} \right). \quad (7)$$

Here  $\mathbf{L}_e$  denotes the Boolean assembly operator of element  $e$ ,  $\mathbf{B}_{e,i}$  is the element-level counterpart of  $\mathbf{B}$  evaluated at Gauss point  $i$ ,  $\mathbf{P}_{e,i}$  is the corresponding stress value, and  $w_{e,i}$  is the physical quadrature weight, including both the parent-domain Gauss weight and the Jacobian determinant of the isoparametric mapping.

Finally, we reindex the element/Gauss-point pairs  $(e, i)$  by a single global index  $g = 1, \dots, M_{\text{gp}}$  (here  $M_{\text{gp}}$  denotes the total number of Gauss points). This numbering defines a bijection between  $g$  and the corresponding pair  $(e(g), i(g))$ , where  $e(g)$  identifies the element containing the point and  $i(g)$  its local Gauss-point index. The constrained internal force appearing in Eq. 5 can then be written as

$$\mathbf{T}^T \mathbf{F}^{\text{int}} = \sum_{g=1}^{M_{\text{gp}}} \mathbf{r}_g^{\text{FE}} w_g^{\text{FE}}, \quad (8)$$

where

$$\mathbf{r}_g^{\text{FE}}(\mathbf{d}; \boldsymbol{\mu}) := \mathbf{T}^T \mathbf{L}_{e(g)}^T \mathbf{B}_{e(g), i(g)}^T \mathbf{P}_{e(g), i(g)}(\mathbf{d}; \boldsymbol{\mu}) \quad (9)$$

stands for the internal-force density associated with global Gauss point  $g$ , whereas  $w_g^{\text{FE}} = w_{e(g), i(g)} > 0$  denotes its integration weight.

## 2.2. External actions

The vector of external actions  $\mathbf{F}^{\text{ext}} \in \mathbb{R}^{N_{\text{dof}}}$  in Eq. 2 contains the contributions induced by prescribed tractions on the Neumann boundary and by body forces. As already pointed out, these external actions are assumed here to be independent of the unknown displacement vector  $\mathbf{d}$ , and therefore will not be subjected to hyperreduction via the proposed cubature scheme. Since in the problem under consideration the input parameters control the amplitude of the external forces, we can write the projection of this term into the constrained space as a linear function of  $\boldsymbol{\mu}$ , that is:

$$\mathbf{T}^T \mathbf{F}^{\text{ext}}(\boldsymbol{\mu}) = \hat{\mathbf{f}}^{\text{ext}} \boldsymbol{\mu}, \quad (10)$$

the columns of  $\hat{\mathbf{f}}^{\text{ext}} \in \mathbb{R}^{N \times n_\mu}$  being the nodal patterns associated with unit values of the components of  $\boldsymbol{\mu}$ .

Finally, substituting Eqs. 8 and 10 into Eq. 5 gives the full-order equilibrium equations in a form that makes explicit the loop over all Gauss points of the mesh:

$$\overbrace{\sum_{g=1}^{M_{\text{gp}}} \mathbf{r}_g^{\text{FE}}(\mathbf{d}^{\text{FE}}; \boldsymbol{\mu}) w_g^{\text{FE}}}^{\text{Internal forces}} - \overbrace{\hat{\mathbf{f}}^{\text{ext}} \boldsymbol{\mu}}^{\text{External forces}} = \mathbf{0}. \quad (11)$$

This is a system of  $N$  nonlinear algebraic equations for the independent unknowns  $\mathbf{d} \in \mathbb{R}^N$ , to be solved for prescribed values of  $\boldsymbol{\mu}$ . In history-dependent problems, the solution also depends on the internal variables from the previous pseudo-time step; for simplicity, this dependence is omitted in the formulation that follows.

## 2.3. Volumetric outputs of interest

Some outputs of interest, depending nonlinearly on the unknown  $\mathbf{d}$ , may also involve volumetric integration and may therefore benefit from the same cubature viewpoint. We denote by  $\mathbf{y}_y \in \mathbb{R}^{n_y}$  a generic output of interest of this type and write its full finite element quadrature form as

$$\mathbf{y}_y = \int_{\Omega_0} \mathbf{r}^y d\Omega = \sum_{g=1}^{M_{\text{gp}}} \mathbf{r}_g^y w_g^{\text{FE}}, \quad (12)$$

where  $\mathbf{r}^y \in \mathbb{R}^{n_y}$  is the volumetric integrand defining the output of interest, and  $\mathbf{r}_g^y(\mathbf{d}; \boldsymbol{\mu})$  denotes its evaluation at Gauss point  $g$ . When such an output is present, the cubature rule used for hyperreduction is also applied to (12), so that both the internal-force term in Eq. 11 and the output functional are approximated over the same reduced set of integration points. This is the case of the metamaterial benchmark of Section 6, where the output of interest is the volumetric average of the first Piola–Kirchhoff stress  $\mathbf{P}$ .

## 3. Manifold hyperreduced-order model

The goal in this section is to construct the hyperreduced-order counterpart of the equilibrium equation 11. This involves two approximations. The first is a kinematic compression: we approximate the unknown displacement  $\mathbf{d} \in \mathbb{R}^N$  as a nonlinear function of a vector of latent coordinates<sup>1</sup>  $\mathbf{q}_M \in \mathbb{R}^n$ :

$$\mathbf{d} \approx \mathcal{D}(\mathbf{q}_M). \quad (13)$$

Ideally, the number of latent coordinates  $n$  should be as close as possible to the number of input parameters  $n_\mu$ . The second approximation is the hyperreduction per se: the loop over the  $M_{\text{gp}}$  Gauss points in Eq. 11 is to be replaced by a loop over a much smaller set of selected integration points, with tailored, positive weights. The adaptive-weight formulation introduced later in Section 4 will modify only this second reduction step, by allowing the cubature weights to depend on the latent coordinate  $\mathbf{q}_M$ .

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<sup>1</sup>The subscript  $M$  in  $\mathbf{q}_M$  stands for “master”, anticipating the master–slave partition of modal basis coordinates used later to construct the benchmark-specific decoders. This terminology is borrowed from nonlinear structural dynamics [26, 27]. Other terms found in the related literature refer to the master coordinates as active [28, 29] or primary [6], and to the remaining coordinates as passive [28] or secondary [6].

### 3.1. Decoder and encoder

We shall refer to  $\mathcal{D} : \mathbb{R}^n \rightarrow \mathbb{R}^N$  in Eq. 13 as the *decoder*, and to the associated approximate inverse map

$$\mathbf{q}_M \approx \mathcal{E}(\mathbf{d}), \quad (14)$$

as the *encoder*. These maps may be represented through general-purpose autoencoder architectures, as in Refs. [1, 13]; here, however, the terminology is used in a broader descriptive sense and does not prescribe any particular architecture. The only condition imposed on the decoder is that it be twice differentiable with respect to  $\mathbf{q}_M$ . This regularity is required because the first derivative, represented by the Jacobian matrix:

$$\Phi_{\mathcal{D}} := \frac{\partial \mathcal{D}}{\partial \mathbf{q}_M} \in \mathbb{R}^{N \times n}, \quad (15)$$

defines the tangent space used in the projected equilibrium equations, while the second derivative enters the consistent Newton–Raphson linearization required by the solution procedure.

### 3.2. Factorized decoder and identification of latent coordinates

The decoder is constructed offline from displacement snapshots associated with a representative training set

$$\mathcal{P}^{\text{tr}} = \{\boldsymbol{\mu}_i\}_{i=1}^P \subset \mathcal{P}. \quad (16)$$

We collect these snapshots in the matrix

$$\mathbf{D} = [\mathbf{d}(\boldsymbol{\mu}_1) \quad \mathbf{d}(\boldsymbol{\mu}_2) \quad \cdots \quad \mathbf{d}(\boldsymbol{\mu}_P)] \in \mathbb{R}^{N \times P}. \quad (17)$$

As already pointed out in Section 1.4, we adopt a two-level decoder representation, combining a preliminary linear compression with a nonlinear coefficient map. With this representation, the decoder is expressed in the factorized form

$$\mathcal{D}(\mathbf{q}_M) = \Phi \boldsymbol{\tau}(\mathbf{q}_M), \quad (18)$$

where  $\Phi \in \mathbb{R}^{N \times n_\Phi}$  is an orthonormal basis matrix for a low-rank approximation of the column space of  $\mathbf{D}$ , and  $\boldsymbol{\tau} : \mathbb{R}^n \rightarrow \mathbb{R}^{n_\Phi}$  maps the latent coordinates to the coefficients in this linear basis. As customary, the number of modes  $n_\Phi$  in  $\Phi$  is controlled by a relative tolerance  $\varepsilon^D \in [0, 1)$  such that

$$\frac{\|\mathbf{D} - \Phi \Phi^T \mathbf{D}\|_F}{\|\mathbf{D}\|_F} \leq \varepsilon^D, \quad \Phi^T \Phi = \mathbf{I}, \quad (19)$$

where  $\|\cdot\|_F$  stands for the Frobenius norm ( $\Phi$  may be obtained as the left singular vectors of a truncated SVD of  $\mathbf{D}$ , for instance). If the latent coordinates  $\mathbf{q}_M$  have already been identified, the map  $\boldsymbol{\tau}$  can be fitted by standard regression procedures. By contrast, if the latent coordinates are not known a priori, they may instead be learned jointly with the coefficient map through an autoencoder-based construction, as in Ref. [14].

In the examples presented later, we follow the former approach. More specifically, let

$$\mathbf{q} := \Phi^T \mathbf{d} \quad (20)$$

denote the modal coefficients of a displacement state in the linear basis  $\Phi$ . Following [3, 4, 6, 15], we seek latent coordinates obtained as linear combinations of  $\mathbf{q}$ :

$$\mathbf{q}_M = \mathbf{H}_M \mathbf{q}. \quad (21)$$

The problem of selecting the latent coordinates is thus recast as that of determining  $\mathbf{H}_M \in \mathbb{R}^{n \times n_\Phi}$ . We assume that  $\mathbf{H}_M$  has full row rank  $n$ , so that the latent coordinates are linearly independent. References [3, 4, 6] use the SVD to determine  $\Phi$ , and adopt  $\mathbf{H}_M = [\mathbf{I}_n \mathbf{0}]$ , where  $\mathbf{I}_n$  is the  $n \times n$  identity matrix; the latent coordinates are therefore the leading entries of  $\mathbf{q}$ , associated with the largest singular values. Reference [15], on the other hand, allows  $\mathbf{H}_M$  to select an arbitrary sparse subset of the modal coefficients.

The identification adopted in the examples discussed later departs from this Boolean selection strategy, in the sense that the matrix  $\mathbf{H}_M$  is not constrained to select individual entries of  $\mathbf{q}$ , but may be dense, allowing each latent coordinate to combine all the modal coefficients. Its identification is instead input-informed, in

the sense discussed in Section 1.4. Further details on the identification of  $\mathbf{H}_M$  will be given in Sections 6.4.1 and 7.3.1.

Regardless of the particular expression for  $\mathbf{H}_M$ , on the approximation manifold, the relation in Eq. 21 together with Eq. 20 defines the following master–slave representation:

$$\mathbf{d} = \Phi_M \mathbf{Z}_M \mathbf{q}_M + \Phi_S \mathcal{N}_S(\mathbf{q}_M). \quad (22)$$

Here,  $\Phi_M \in \mathbb{R}^{N \times n}$  denotes the matrix of master modes, whereas  $\Phi_S \in \mathbb{R}^{N \times (n_\Phi - n)}$  contains the slave modes. They are chosen so that

$$\Phi_M^T \Phi_M = \mathbf{I}, \quad \Phi_S^T \Phi_S = \mathbf{I}, \quad \Phi_M^T \Phi_S = \mathbf{0} \quad (23)$$

( $\Phi_S$  is therefore an orthonormal basis for the orthogonal complement of  $\text{col}(\Phi_M)$  within  $\text{col}(\Phi)$ , and hence  $[\Phi_M \ \Phi_S]$  spans  $\text{col}(\Phi)$ ). On the other hand,  $\mathbf{Z}_M \in \mathbb{R}^{n \times n}$  is a scaling matrix. The matrices  $\Phi_M$  and  $\mathbf{Z}_M$  are obtained from the compact SVD of  $\Phi \mathbf{H}_M^T$ ; in particular  $\Phi_M$  contains the corresponding left singular vectors, i.e.:

$$\Phi \mathbf{H}_M^T = \Phi_M \Sigma_M \mathbf{V}_M^T. \quad (24)$$

whereas

$$\mathbf{Z}_M = \Sigma_M^{-1} \mathbf{V}_M^T. \quad (25)$$

The latter expression follows by premultiplying Eq. 22 by  $\Phi_M^T$ , using the orthogonality conditions in Eq. 23, and combining the resulting relation with Eqs. 20, 21 and the SVD above. The slave amplitudes associated with the training snapshots are then computed as

$$\mathbf{q}_{Sj} = \Phi_S^T \mathbf{d}(\mu_j), \quad j = 1, 2, \dots, P$$

and the nonlinear closure relation

$$\mathbf{q}_S \approx \mathcal{N}_S(\mathbf{q}_M) \quad (26)$$

is learned from the training pairs  $(\mathbf{q}_{Mj}, \mathbf{q}_{Sj})$  ( $j = 1, 2, \dots, P$ ).

With a slight abuse of notation, we henceforth redefine the reduced basis as  $\Phi \leftarrow [\Phi_M, \Phi_S]$ . This change of basis leaves  $\text{col}(\Phi)$  unchanged and yields the following particular expression for the coefficient map in Eq. 18:

$$\boldsymbol{\tau}(\mathbf{q}_M) = \begin{bmatrix} \mathbf{Z}_M \mathbf{q}_M \\ \mathcal{N}_S(\mathbf{q}_M) \end{bmatrix}, \quad (27)$$

and for its Jacobian:

$$\mathbf{J}_\tau := \frac{\partial \boldsymbol{\tau}}{\partial \mathbf{q}_M} = \begin{bmatrix} \mathbf{Z}_M \\ \frac{\partial \mathcal{N}_S(\mathbf{q}_M)}{\partial \mathbf{q}_M} \end{bmatrix} \quad (28)$$

Lastly, we particularize the encoder introduced in Eq. 14 for the master–slave decoder in Eq. 22. We adopt the explicit projection-based map

$$\mathcal{E}(\mathbf{d}) := \mathbf{Z}_M^{-1} \Phi_M^T \mathbf{d}. \quad (29)$$

By virtue of the orthogonality conditions in Eq. 23, this map is an exact left inverse of the decoder on the approximation manifold, i.e.:  $\mathcal{E}(\mathcal{D}(\mathbf{q}_M)) = \mathbf{q}_M$ . For states outside the approximation manifold, however, the coordinates returned by Eq. 29 need not coincide with those obtained from the nonlinear least-squares encoder used, for instance, in Ref. [6]. We have nevertheless verified numerically that this discrepancy has no appreciable effect on the results reported here, and therefore adopt the simplified encoder in Eq. 29 henceforth.

### 3.3. Tangent-space projection of the equilibrium equations

The nonlinear approximation in Eq. 13 restricts the admissible variations of the independent displacement vector to the tangent space of the manifold:

$$\delta \mathbf{d} = \Phi_{\mathcal{D}}(\mathbf{q}_M) \delta \mathbf{q}_M, \quad (30)$$



where  $\Phi_{\mathcal{D}}$  is the Jacobian of the decoder, introduced in Eq. 15. Substituting Eq. 30 into Eq. 3, and the resulting expression into the virtual work statement 4, yields, by virtue of the arbitrariness of  $\delta \mathbf{q}_M$ , the reduced-order equilibrium equations:

$$\underbrace{\mathbf{F}_{\mathcal{D}}^{\text{int}} : \text{Internal forces}}_{M_{\text{gp}}} - \underbrace{\mathbf{F}_{\mathcal{D}}^{\text{ext}} : \text{External forces}}_{\Phi_{\mathcal{D}}(\mathbf{q}_M)^T \hat{\mathbf{f}}^{\text{ext}} \boldsymbol{\mu}} = \mathbf{0}, \quad (31)$$

where

$$\mathbf{r}_g^{\mathcal{D}}(\mathbf{q}_M; \boldsymbol{\mu}) := \Phi_{\mathcal{D}}(\mathbf{q}_M)^T \mathbf{r}_g^{\text{FE}}(\mathbf{q}_M; \boldsymbol{\mu}) \in \mathbb{R}^n \quad (32)$$

is the projected internal force density associated with Gauss point  $g$ .

In the form given above, both terms in Eq. 31 appear, in principle, to require online operations whose cost depends on the size of the underlying finite element model. This dependence arises, on the one hand, from the products involving the tangent basis  $\Phi_{\mathcal{D}} \in \mathbb{R}^{N \times n}$ , in the expressions of both  $\mathbf{F}_{\mathcal{D}}^{\text{int}}$  and  $\mathbf{F}_{\mathcal{D}}^{\text{ext}}$ , and, on the other hand, from the summation over all Gauss points of the mesh in  $\mathbf{F}_{\mathcal{D}}^{\text{int}}$ .

The former dependence can be eliminated in both terms by exploiting the factorized decoder in Eq. 18. Indeed, in terms of this factorization, the Jacobian matrix defined in Eq. 15 adopts the form

$$\Phi_{\mathcal{D}}(\mathbf{q}_M) = \Phi \mathbf{J}_{\tau}(\mathbf{q}_M), \quad (33)$$

where  $\mathbf{J}_{\tau}$  is given in Eq. 28. Combining Eqs. 32, 9, and 33 (and reverting temporarily to the local Gauss-point indices  $(e, i)$  for clarity), we obtain

$$\mathbf{r}_{e,i}^{\mathcal{D}}(\mathbf{q}_M; \boldsymbol{\mu}) = \mathbf{J}_{\tau}(\mathbf{q}_M)^T \mathbf{B}_{e,i}^{\Phi} \mathbf{P}_{e,i}(\mathbf{q}_M; \boldsymbol{\mu}), \quad (34)$$

where

$$\mathbf{B}_{e,i}^{\Phi} := \mathbf{B}_{e,i} \mathbf{L}_e \mathbf{T} \Phi \in \mathbb{R}^{s_{\sigma} \times n_{\Phi}} \quad (35)$$

maps variations of the auxiliary coordinates (amplitude of the basis vectors  $\Phi$ ) to the displacement gradient components at Gauss point  $i$  of element  $e$ . This matrix is independent of both the latent coordinates and the input parameters and can therefore be precomputed offline for every Gauss point. Thus, once  $\mathbf{J}_{\tau}(\mathbf{q}_M)$  is available, evaluation of  $\mathbf{r}_{e,i}^{\mathcal{D}}$  requires only the constitutive update at the corresponding Gauss point—whose kinematic input is likewise obtained through  $\mathbf{B}_{e,i}^{\Phi}$ —together with matrix–vector products in spaces of dimensions  $n_{\Phi}$  and  $n$ . No operation involving the ambient dimension  $N$  is required.

The same conclusion holds for the external-force term  $\mathbf{F}_{\mathcal{D}}^{\text{ext}}$ . Substitution of Eq. 33 into the expression for  $\mathbf{F}_{\mathcal{D}}^{\text{ext}}$  in Eq. 31 gives

$$\mathbf{F}_{\mathcal{D}}^{\text{ext}}(\mathbf{q}_M; \boldsymbol{\mu}) = \mathbf{J}_{\tau}(\mathbf{q}_M)^T \hat{\mathbf{f}}_{\text{ext}}^{\tau} \boldsymbol{\mu}, \quad (36)$$

where

$$\hat{\mathbf{f}}_{\text{ext}}^{\tau} := \Phi^T \hat{\mathbf{f}}^{\text{ext}} \quad (37)$$

is independent of both the latent coordinates and the parameter values and can therefore be computed offline.

The only remaining source of mesh-dependent online cost is thus the summation over the complete set of Gauss points required to evaluate  $\mathbf{F}_{\mathcal{D}}^{\text{int}}$  in Eq. 31 and, when present, the volumetric output of interest in Eq. 12. As previously mentioned, this bottleneck is addressed by replacing the full finite element quadrature sums with sums over a much smaller set of Gauss points, equipped with tailored positive weights. We first consider a fixed-weight rule constructed by the Empirical Cubature Method (ECM) [8, 10]. Although its derivation is similar to that of a standard HROM, it is presented here because it provides both the conceptual basis and the initialization for the manifold-adaptive rule introduced in Section 4.

#### 3.4. Fixed-weight cubature approximation

We seek the smallest set of Gauss points  $\mathcal{Z}_{\mathcal{D}} \subset \{1, \dots, M_{\text{gp}}\}$  and positive weights  $w_g^{\mathcal{D}}$ ,  $g \in \mathcal{Z}_{\mathcal{D}}$ , such that the resulting cubature rule reproduces, to prescribed accuracy, the projected internal forces at the training states  $\{(\mathbf{q}_{Mj}, \boldsymbol{\mu}_j)\}_{j=1}^P$ :

$$\mathbf{F}_{\mathcal{D}}^{\text{int}}(\mathbf{q}_{Mj}; \boldsymbol{\mu}_j) = \sum_{g=1}^{M_{\text{gp}}} \mathbf{r}_g^{\mathcal{D}}(\mathbf{q}_{Mj}; \boldsymbol{\mu}_j) w_g^{\text{FE}} \approx \sum_{g \in \mathcal{Z}_{\mathcal{D}}} \mathbf{r}_g^{\mathcal{D}}(\mathbf{q}_{Mj}; \boldsymbol{\mu}_j) w_g^{\mathcal{D}}, \quad j = 1, \dots, P. \quad (38)$$

When the output of interest in Eq. 12 is present, the same rule is also required to satisfy

$$\mathbf{y}_y(\mathbf{q}_{Mj}; \boldsymbol{\mu}_j) = \sum_{g=1}^{M_{\text{gp}}} \mathbf{r}_g^y(\mathbf{q}_{Mj}; \boldsymbol{\mu}_j) w_g^{\text{FE}} \approx \sum_{g \in \mathcal{Z}_{\mathcal{D}}} \mathbf{r}_g^y(\mathbf{q}_{Mj}; \boldsymbol{\mu}_j) w_g^{\mathcal{D}}, \quad j = 1, \dots, P. \quad (39)$$

For the sake of generality, we formulate the above requirements in terms of a generic vector-valued integrand  $\mathbf{r}_g(\mathbf{q}_{Mj}; \boldsymbol{\mu}_j) \in \mathbb{R}^{n_c}$ :

$$\sum_{g=1}^{M_{\text{gp}}} \mathbf{r}_g(\mathbf{q}_{Mj}; \boldsymbol{\mu}_j) w_g^{\text{FE}} \approx \sum_{g \in \mathcal{Z}_{\mathcal{D}}} \mathbf{r}_g(\mathbf{q}_{Mj}; \boldsymbol{\mu}_j) w_g^{\mathcal{D}}, \quad j = 1, \dots, P. \quad (40)$$

If the rule is employed only to evaluate the projected equilibrium equations, then  $\mathbf{r}_g = \mathbf{r}_g^{\mathcal{D}}$  and  $n_c = n$ . If equilibrium and output evaluation are both hyperreduced using the same rule,  $\mathbf{r}_g$  is formed by stacking  $\mathbf{r}_g^{\mathcal{D}}$  and  $\mathbf{r}_g^y$ , so that  $n_c = n + n_y$ . A third case arises when the cubature rule is required only for the evaluation of the output of interest, i.e.,  $\mathbf{r}_g = \mathbf{r}_g^y$  and  $n_c = n_y$ . This occurs when the latent state is obtained directly from the input parameters, without the need for solving the projected equilibrium equations.

To apply the ECM, Eq. 40 must first be cast in matrix form. To this end, we define

$$\mathbf{A}_j := \begin{bmatrix} \mathbf{r}_1(\mathbf{q}_{Mj}; \boldsymbol{\mu}_j)^T \\ \mathbf{r}_2(\mathbf{q}_{Mj}; \boldsymbol{\mu}_j)^T \\ \vdots \\ \mathbf{r}_{M_{\text{gp}}}(\mathbf{q}_{Mj}; \boldsymbol{\mu}_j)^T \end{bmatrix} \in \mathbb{R}^{M_{\text{gp}} \times n_c}, \quad j = 1, 2 \dots P. \quad (41)$$

The columns of the preceding matrix contain the values of the  $n_c$  scalar-valued integrands at the  $M_{\text{gp}}$  Gauss points of the finite element mesh, for the  $j$ -th training parameter. Concatenating these matrices over the complete training set gives

$$\mathbf{A} := [\mathbf{A}_1 \quad \mathbf{A}_2 \quad \dots \quad \mathbf{A}_P] \in \mathbb{R}^{M_{\text{gp}} \times (n_c P)}. \quad (42)$$

Next we introduce the vectors of finite element weights and reduced integration weights

$$\mathbf{w}^{\text{FE}} = \begin{bmatrix} w_1^{\text{FE}} \\ \vdots \\ w_{M_{\text{gp}}}^{\text{FE}} \end{bmatrix}, \quad \mathbf{w}^{\mathcal{D}} = \begin{bmatrix} w_1^{\mathcal{D}} \\ \vdots \\ w_{M_{\text{gp}}}^{\mathcal{D}} \end{bmatrix} \quad (43)$$

respectively. With the preceding definitions, the integration conditions in Eq. 40 take the compact matrix form

$$\mathbf{A}^T \mathbf{w}^{\text{FE}} \approx \mathbf{A}^T \mathbf{w}^{\mathcal{D}}. \quad (44)$$

The fixed-weight cubature problem therefore amounts to determining the sparsest nonnegative vector  $\mathbf{w}^{\mathcal{D}}$  that satisfies the above conditions to the prescribed accuracy.

### 3.5. Construction of the fixed-weight ECM rule

The formulation presented above is common to other point-selection and weighting procedures, notably the pioneering cubature method of An et al. [16] and the ECSW [7]. Unlike these approaches, however, the ECM does not address the selection and weighting directly in terms of the sampled training matrix  $\mathbf{A}$ . Instead, it first constructs a low-dimensional orthonormal basis  $\mathbf{V} \in \mathbb{R}^{M_{\text{gp}} \times n_U}$  of the column space of  $\mathbf{A}$ , controlled by a truncation tolerance  $\varepsilon^A \in [0, 1)$ , i.e.:

$$\frac{\|\mathbf{A} - \mathbf{V} \mathbf{V}^T \mathbf{A}\|_F}{\|\mathbf{A}\|_F} \leq \varepsilon^A, \quad \mathbf{V}^T \mathbf{V} = \mathbf{I}, \quad (45)$$

and then requires<sup>2</sup> the reduced cubature rule to integrate exactly the basis functions represented by  $\mathbf{V}$ . Together with the requirement that the selected set  $\mathcal{Z}_{\mathcal{D}}$  be as small as possible, this condition leads formally

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<sup>2</sup>For simplicity of notation, orthogonality is stated here with respect to the Euclidean inner product. Other inner products may be used; in particular, the natural choice for the approximation of volume integrals is the inner product induced by  $\text{diag}(\mathbf{w}^{\text{FE}})$ , together with its associated norm. This is the convention employed in the original ECM formulation in Refs. [8, 10, 22].

to the optimization problem

$$\begin{aligned} \min_{\mathbf{w}^{\mathcal{D}}} \quad & \|\mathbf{w}^{\mathcal{D}}\|_0, \\ \text{s.t.} \quad & \mathbf{V}^T \mathbf{w}^{\mathcal{D}} = \mathbf{b}^{\text{FE}}, \\ & \mathbf{w}^{\mathcal{D}} \geq \mathbf{0}, \end{aligned} \tag{46}$$

where the  $\ell_0$  pseudo-norm  $\|\bullet\|_0$  counts the number of nonzero entries of its argument, and

$$\mathbf{b}^{\text{FE}} := \mathbf{V}^T \mathbf{w}^{\text{FE}} \tag{47}$$

contains the reference integrals of the retained basis functions, evaluated using the full finite element integration rule. The selected set of Gauss points is then given by the support of  $\mathbf{w}^{\mathcal{D}}$ , i.e.:

$$\mathcal{Z}_{\mathcal{D}} = \text{supp}(\mathbf{w}^{\mathcal{D}}) := \{g \in \{1, \dots, M_{\text{gp}}\} : w_g^{\mathcal{D}} > 0\}. \tag{48}$$

A few remarks concerning the preceding optimization problem are in order here. First, the entries of  $\mathbf{A}$  are constructed from the FE training data as follows. For each FE displacement snapshot  $\mathbf{d}(\mu_j)$ , the associated latent coordinates are obtained through the encoder in Eq. 14, i.e.,  $\mathbf{q}_M^{\text{train}}(\mu_j) = \mathcal{E}(\mathbf{d}(\mu_j))$ , and are used to evaluate  $\mathbf{J}_{\tau}$  in Eq. 34. The stresses entering the same equation, together with any other quantities required by the output integrand  $\mathbf{r}_g^y$ , are then reevaluated at all Gauss points from the reconstructed<sup>3</sup> displacement vector  $\mathbf{d}^{\text{train}}(\mu_j) = \mathcal{D}(\mathbf{q}_M^{\text{train}}(\mu_j))$ .

Second,  $\mathbf{V}$  may be obtained from a standard, truncated SVD of  $\mathbf{A}$ . For large problems, however, storing and factorizing the complete matrix may be computationally prohibitive. In the examples shown later, we use the Sequential Randomized SVD (SRSVD) proposed by the authors in Ref. [22], which processes the column blocks  $\mathbf{A}_j$  sequentially and uses randomization to reduce the cost of the basis construction. Parallel and distributed randomized SVD algorithms, such as those in Ref. [30], provide an alternative when distributed resources are available.

Third, to avoid potential degeneracies for the case  $\mathbf{b}^{\text{FE}} \approx \mathbf{0}$ , the ECM further imposes the exact integration of the volume of the domain  $|\Omega_0|$ :

$$\mathbf{1}^T \mathbf{w}^{\mathcal{D}} = |\Omega_0|, \tag{49}$$

where  $\mathbf{1} \in \mathbb{R}^{M_{\text{gp}}}$  is the all-ones vector. In order to preserve the orthogonality of  $\mathbf{V}$ , the constraint is incorporated by augmenting  $\mathbf{V}$  with the component of  $\mathbf{1}$  orthogonal to the column space of  $\mathbf{V}$ ; see Ref. [22] for more details.

Lastly, the  $\ell_0$ -minimization problem in Eq. 46 is introduced only as a conceptual statement of the sparsification objective. Because its exact solution is NP-hard and therefore computationally intractable, the ECM does not attempt to solve the  $\ell_0$  problem directly. Instead, it constructs a sparse nonnegative solution of the underdetermined system  $\mathbf{V}^T \mathbf{w}^{\mathcal{D}} = \mathbf{b}^{\text{FE}}$ . The authors show in Ref. [17] that the ECM invariably achieves a solution with a number of positive weights, denoted hereafter by  $m_0$ , equal to the number of columns of  $\mathbf{V}$ :

$$m_0 = |\mathcal{Z}_{\mathcal{D}}| = \text{ncol}(\mathbf{V}) \tag{50}$$

(this follows from Carathéodory's theorem for convex cones [31, § 17], and the fact that the FE weights themselves are positive, and thus  $\mathbf{b}^{\text{FE}} = \mathbf{V}^T \mathbf{w}^{\text{FE}}$  lies in the conical hull of the rows of  $\mathbf{V}$ ). Ref. [17] also shows that the solution is generally nonunique: distinct rules can be elicited by varying the initial candidate set, thereby allowing selected regions of the domain to be favored. We do not exploit this possibility in the examples presented later, but instead adopt the default choice of considering all Gauss points as candidates.

## 4. Cubature with manifold-adaptive weights

### 4.1. Problem statement and requirements

We introduce next the main novel contribution of the present work: the cubature scheme with manifold-adaptive weights. The idea is to select a subset of the Gauss points provided by the preceding fixed-weight rule

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<sup>3</sup>For path-dependent constitutive models, the snapshots cannot be processed independently. Each snapshot must retain its position within the training trajectory, because the constitutive update requires the internal variables of the preceding state. Stress reconstruction must therefore proceed in trajectory order.

and assign them positive, state-dependent weights, in such a way that the resulting cubature rule reproduces, with the same accuracy as the fixed-weight rule, the integration conditions in Eq. 40, that is:

$$\sum_{g \in \mathcal{Z}_{\mathcal{D}}} \mathbf{r}_g(\mathbf{q}_M^j; \boldsymbol{\mu}^j) w_g^{\mathcal{D}} = \sum_{g \in \mathcal{Z}_{\omega}} \mathbf{r}_g(\mathbf{q}_M^j; \boldsymbol{\mu}^j) \omega_g(\mathbf{q}_M^j), \quad j = 1, \dots, P_{\mathcal{M}}, \quad (51)$$

where

$$\mathcal{Z}_{\omega} \subseteq \mathcal{Z}_{\mathcal{D}}, \quad (52)$$

and

$$\omega_g : \mathbb{R}^n \rightarrow \mathbb{R}_+, \quad g \in \mathcal{Z}_{\omega}. \quad (53)$$

Thus, the novelty lies in that each retained Gauss point  $g \in \mathcal{Z}_{\omega}$  is endowed with a positive scalar weight field over the latent space. This is the sense in which the method is termed manifold-adaptive weights (MAW) cubature: the locations of the integration points remain fixed in the reference configuration  $\Omega_0$ , whereas their positive weights adapt to the current state on the nonlinear solution manifold.

The reduced equilibrium equations in this new setting are obtained by replacing the full finite element rule for internal forces in Eq. 31 with the manifold-adaptive rule:

$$\underbrace{\mathbf{F}_{\mathcal{D}}^{\text{int}} : \text{Internal forces}}_{\sum_{g \in \mathcal{Z}_{\omega}} \mathbf{r}_g^{\mathcal{D}}(\mathbf{q}_M; \boldsymbol{\mu}) \omega_g(\mathbf{q}_M)} - \underbrace{\mathbf{F}_{\mathcal{D}}^{\text{ext}} : \text{External forces}}_{\boldsymbol{\Phi}_{\mathcal{D}}(\mathbf{q}_M)^T \hat{\mathbf{f}}^{\text{ext}} \boldsymbol{\mu}} = \mathbf{0}. \quad (54)$$

Because the cubature weights now depend on  $\mathbf{q}_M$ , their variation must also be included in the consistent Newton–Raphson linearization. The resulting tangent operator is derived in Appendix A.

To construct the adaptive weight fields, we first extract from the input parameter set  $\mathcal{P}^{\text{tr}}$ , see Eq. 16, a subset

$$\mathcal{P}^{\mathcal{M}} = \{\boldsymbol{\mu}^j\}_{j=1}^{P_{\mathcal{M}}} \subseteq \mathcal{P}^{\text{tr}}, \quad (55)$$

where, for simplicity and with a slight abuse of notation, the retained parameters have been reindexed from 1 to  $P_{\mathcal{M}}$ . The corresponding latent states are taken as the nodes of a graph  $\mathcal{G}_{\mathcal{M}}$ , with edges encoding neighborhood relations on the latent manifold. The Laplacian of  $\mathcal{G}_{\mathcal{M}}$  will later be used to measure and penalize distortions of the weight fields between neighboring states.

The subsampling represented by expression 55 may serve the usual purpose of avoiding an unnecessarily fine representation of the weight fields and the associated risk of overfitting. In some problems, however, restricting the state-dependent conditions to  $\mathcal{P}^{\mathcal{M}}$  does not necessarily discard the integration content associated with the eliminated states. Integrand components that must be reproduced independently of the current latent state are instead collected in an invariant matrix and enforced for every  $\boldsymbol{\mu}^j \in \mathcal{P}^{\mathcal{M}}$ , as described in the ensuing subsection. In the damage benchmark of Section 7, for instance,  $\mathcal{P}^{\mathcal{M}}$  contains only states from the damaged regime, while the integrand space associated with the undamaged elastic response is incorporated into the invariant matrix.

With this setting, the selected integration points  $\mathcal{Z}_{\omega}$  and the associated state-dependent weights must satisfy four requirements:

1. The selected set should be as small as possible, i.e.,  $m = |\mathcal{Z}_{\omega}|$  is to be minimized.
2. The adaptive rule must satisfy exactly the state-wise integration conditions in Eq. 51 for  $\boldsymbol{\mu}^j \in \mathcal{P}^{\mathcal{M}}$ .
3. The weights must remain nonnegative over the sampled latent domain, and also preserve the total volume; that is, for a given  $\mathbf{q}_M$

$$\omega_g(\mathbf{q}_M) \geq 0, \quad \sum_{g \in \mathcal{Z}_{\omega}} \omega_g(\mathbf{q}_M) = |\Omega_0|.$$

The second condition is the local counterpart of Eq. 49.

4. Each scalar field  $\omega_g = \omega_g(\mathbf{q}_M)$ ,  $g \in \mathcal{Z}_{\omega}$ , should vary as smoothly as possible over the manifold.

The preceding requirements are cast as a constrained optimization problem over the next two subsections. The first subsection develops the initialization and local constraints associated with requirements 2 and 3. Section 4.3 then introduces the objective function incorporating requirements 1 and 4.

#### 4.2. Initialization and local integration constraints

As established in expression 52, the reference integration rule is not the original FE Gauss-point rule, but the fixed-weight ECM rule discussed in Section 3.5, characterized by the sparse weight vector  $\mathbf{w}^{\mathcal{D}} \in \mathbb{R}^{M_{\text{gp}}}$  and its support  $\mathcal{Z}_{\mathcal{D}}$ , of size  $m_0$ . Accordingly, we set as design variables of the optimization problem a collection of weight vectors (one for each sampled state) of size  $m_0$

$$\{\boldsymbol{\omega}_1, \boldsymbol{\omega}_2, \dots, \boldsymbol{\omega}_{P_{\mathcal{M}}}\},$$

where

$$\boldsymbol{\omega}_j \in \mathbb{R}^{m_0}, \quad j = 1, \dots, P_{\mathcal{M}},$$

contains the cubature weights associated with the  $j$ -th latent state, expressed on the initial candidate support  $\mathcal{Z}_{\mathcal{D}}$ . We anticipate that the proposed solution algorithm initializes all such state-dependent weight vectors with the initial weight vector

$$\boldsymbol{\omega}_0 := \mathbf{w}_{\mathcal{Z}_{\mathcal{D}}}^{\mathcal{D}} \in \mathbb{R}^{m_0}, \quad (56)$$

where  $\mathbf{w}_{\mathcal{Z}_{\mathcal{D}}}^{\mathcal{D}}$  denotes the restriction of the sparse fixed-weight vector  $\mathbf{w}^{\mathcal{D}}$  to its support  $\mathcal{Z}_{\mathcal{D}}$ .

For each  $\boldsymbol{\mu}^j \in \mathcal{P}^{\mathcal{M}}$ , let  $\mathbf{A}_j$  denote the corresponding local integrand block introduced in Eq. 41. The fixed ECM rule is constructed to integrate exactly those functions whose discrete representations lie in the subspace spanned by  $\mathbf{V}$ . Since  $\mathbf{V}$  is obtained from the truncated SVD of the global matrix  $\mathbf{A}$  (or a similar low-rank decomposition), the integrand information retained at the  $j$ -th state is represented by the projected block

$$\tilde{\mathbf{A}}_j := \mathbf{V}\mathbf{V}^T \mathbf{A}_j \in \mathbb{R}^{M_{\text{gp}} \times n_c}, \quad j = 1, \dots, P_{\mathcal{M}}. \quad (57)$$

Only in the absence of truncation, i.e., when  $\varepsilon^A = 0$  in expression 45, does the basis  $\mathbf{V}$  span the full column space of  $\mathbf{A}$ , in which case  $\tilde{\mathbf{A}}_j = \mathbf{A}_j$  for every  $j$ .

As previously pointed out, the local projected blocks are complemented with state-independent integration conditions. These are collected in the invariant matrix

$$\mathbf{U}_{\text{inv}} \in \mathbb{R}^{M_{\text{gp}} \times n_{\text{inv}}}. \quad (58)$$

At minimum,  $\mathbf{U}_{\text{inv}}$  contains the all-ones vector  $\mathbf{1} \in \mathbb{R}^{M_{\text{gp}}}$ , so that the volume constraint in Eq. 49 is enforced at every state. To remove linearly dependent conditions, the invariant and projected integrand blocks are combined and factorized by an SVD:

$$\left[ \mathbf{U}_{\text{inv}}, \tilde{\mathbf{A}}_j \right] = \overline{\mathbf{U}}_j \boldsymbol{\Sigma}_j \mathbf{V}_j^T, \quad j = 1, \dots, P_{\mathcal{M}}, \quad (59)$$

where only the singular vectors associated with nonzero singular values are retained. The columns of  $\overline{\mathbf{U}}_j \in \mathbb{R}^{M_{\text{gp}} \times \bar{n}_c^j}$  therefore form an orthonormal basis for the combined invariant and state-dependent constraint space. Note that

$$\bar{n}_c^j \leq n_c + n_{\text{inv}} \quad (60)$$

for all  $j = 1, 2, \dots, P_{\mathcal{M}}$ .

Since the fixed-weight ECM rule is supported on  $\mathcal{Z}_{\mathcal{D}}$ , only the rows of  $\overline{\mathbf{U}}_j$  associated with the initial candidate points contribute to the reference integrals. We accordingly define

$$\mathbf{U}_j := \overline{\mathbf{U}}_j|_{\mathcal{Z}_{\mathcal{D}}} \in \mathbb{R}^{m_0 \times \bar{n}_c^j}, \quad (61)$$

and evaluate the corresponding reference integrals using the initial weight vector defined in Eq. 56:

$$\mathbf{b}_j := \mathbf{U}_j^T \boldsymbol{\omega}_0 \in \mathbb{R}^{\bar{n}_c^j}. \quad (62)$$

The desired local equality constraints then follow directly: at every sampled latent state, the adaptive weights must satisfy

$$\mathbf{U}_j^T \boldsymbol{\omega}_j = \mathbf{b}_j, \quad j = 1, \dots, P_{\mathcal{M}}. \quad (63)$$

#### 4.3. Objective function

To accommodate requirements 1 (minimization of the number of integration points) and 4 (smoothness of the sampled weight fields), we propose the following objective function:

$$\mathcal{J} := \left\| \sum_{j=1}^{P_{\mathcal{M}}} \omega_j \right\|_0 + \mathcal{R}(\mathcal{W}, \mathcal{G}_{\mathcal{M}}), \quad (64)$$

where

$$\mathcal{W} := [\omega_1 \quad \omega_2 \quad \cdots \quad \omega_{P_{\mathcal{M}}}] \in \mathbb{R}^{m_0 \times P_{\mathcal{M}}}. \quad (65)$$

The first term on the right-hand side of Eq. (64) is precisely the number of integration points  $m$ —this follows from the fact that nonnegativity prevents cancellations in the sum of the local weight vectors. The second term,  $\mathcal{R} = \mathcal{R}(\mathcal{W}, \mathcal{G}_{\mathcal{M}})$ , regularizes the sampled weight fields, represented by the rows of  $\mathcal{W}$ . It will be defined as a quadratic functional involving, among other contributions, the graph Laplacian associated with  $\mathcal{G}_{\mathcal{M}}$ , thereby penalizing irregular variations of the weights over the sampled manifold. At this stage,  $\mathcal{R}(\mathcal{W}, \mathcal{G}_{\mathcal{M}})$  is introduced only in abstract form; its specific definition, together with the associated solution strategy, is presented later.

#### 4.4. Optimization problem and solution strategy

With all the ingredients introduced in the preceding subsections, the manifold-adaptive weights ECM problem can now be formulated as the following constrained optimization problem: find the weight vectors  $\{\omega_j\}_{j=1}^{P_{\mathcal{M}}}$ , collected in  $\mathcal{W}$ , as the solution of

$$\begin{aligned} \min_{\omega_1, \dots, \omega_{P_{\mathcal{M}}}} \quad & \left\| \sum_{j=1}^{P_{\mathcal{M}}} \omega_j \right\|_0 + \mathcal{R}(\mathcal{W}, \mathcal{G}_{\mathcal{M}}) \\ \text{subject to} \quad & \mathbf{U}_j^T \omega_j = \mathbf{b}_j, \quad j = 1, \dots, P_{\mathcal{M}}, \\ & \omega_j \geq 0, \quad j = 1, \dots, P_{\mathcal{M}}. \end{aligned} \quad (66)$$

The solution determines simultaneously the state-dependent weights and the final common support  $\mathcal{Z}_{\omega}$ , identified by the nonzero rows of  $\mathcal{W}$ . As discussed in Section 1.3, we adopt a greedy pruning strategy. Starting from the feasible fixed-weight ECM rule, integration points are removed successively while the remaining manifold-dependent weights are redistributed so as to preserve the local equality and positivity constraints and control the regularity of the resulting weight fields. The resulting pruning strategy is developed in the next section.

### 5. Greedy pruning strategy

To describe the proposed greedy pruning procedure, we proceed hierarchically from its most elementary operation to the final global algorithm. We begin by isolating the fundamental building block of the construction, namely a single tentative pruning step. Assume that one integration point is selected for removal from the current active support. The central question then becomes whether the remaining adaptive weights can be redistributed so as to preserve the local exactness constraints over the sampled manifold. This leads to the *local redistribution problem* introduced in Section 5.1, and whose solution procedure is summarized in Algorithm 1. Subsequently, Section 5.2 introduces Algorithm 2, whose purpose is to partially mitigate the path dependence intrinsic to the greedy pruning strategy by exploring several feasible candidate eliminations before committing to one of them. Finally, the complete greedy pruning loop is described in Section 5.3 through Algorithm 3, which repeatedly applies the candidate-sweep procedure. For convenience, the complete offline workflow is summarized in Section 5.5, Box 5.1. Readers primarily interested in the practical implementation may jump directly to this summary and navigate from there to the corresponding algorithms.

### 5.1. Redistribution problem

Assume that, at a given pruning stage, the current active support is  $\mathcal{I}_{\text{act}} \subset \{1, \dots, m_0\}$ . Here, active refers to the integration points that have not yet been removed from the initial candidate support. For each sampled latent state  $\mathbf{q}_M^{(j)}$ , the current adaptive weights are denoted by  $\boldsymbol{\omega}_{\text{old}}^{(j)} \in \mathbb{R}^{|\mathcal{I}_{\text{act}}|}$ . By construction, they satisfy the local linear equality constraints in Eq. 63:

$$\mathbf{U}_{j\mathcal{I}_{\text{act}}}^T \boldsymbol{\omega}_{\text{old}}^{(j)} = \mathbf{b}_j, \quad j = 1, \dots, P_{\mathcal{M}}, \quad (67)$$

where  $\mathbf{U}_{j\mathcal{I}_{\text{act}}}$  denotes the restriction of  $\mathbf{U}_j$  to the active support. A pruning trial begins by selecting one candidate integration point (the selection strategy will be discussed later),  $p \in \mathcal{I}_{\text{act}}$ , and attempts to remove it. The remaining active support is  $\mathcal{I}_{\text{rem}} = \mathcal{I}_{\text{act}} \setminus \{p\}$ . Equivalently, in the local numbering induced by the current active support  $\mathcal{I}_{\text{act}}$ ,  $\tilde{\mathcal{I}}_{\text{rem}} = \{1, \dots, |\mathcal{I}_{\text{act}}| \setminus \{\text{loc}(p, \mathcal{I}_{\text{act}})\}\}$ . Here,  $\text{loc}(p, \mathcal{I}_{\text{act}})$  denotes the position occupied by the global index  $p$  within the ordered set  $\mathcal{I}_{\text{act}}$ . The question is whether, for every sampled latent state, one can construct redistributed weights  $\boldsymbol{\omega}_{\text{new}}^{(j)} \in \mathbb{R}^{|\mathcal{I}_{\text{rem}}|}$  such that

$$\mathbf{U}_{j\mathcal{I}_{\text{rem}}}^T \boldsymbol{\omega}_{\text{new}}^{(j)} = \mathbf{b}_j, \quad j = 1, \dots, P_{\mathcal{M}}. \quad (68)$$

If such weights exist and satisfy the required admissibility conditions, the candidate removal is accepted.

#### 5.1.1. Regularized optimization problem

The weights we seek are the values of  $|\mathcal{I}_{\text{rem}}|$  scalar fields defined at  $P_{\mathcal{M}}$  samples of the latent space:

$$\mathcal{W}_{\text{new}} = \begin{bmatrix} \left| \boldsymbol{\omega}_{\text{new}}^{(1)} \right| & \left| \boldsymbol{\omega}_{\text{new}}^{(2)} \right| & \dots & \left| \boldsymbol{\omega}_{\text{new}}^{(P_{\mathcal{M}})} \right| \end{bmatrix} \in \mathbb{R}^{|\mathcal{I}_{\text{rem}}| \times P_{\mathcal{M}}}, \quad (69)$$

(the column  $\boldsymbol{\omega}_{\text{new}}^{(j)} \in \mathbb{R}^{|\mathcal{I}_{\text{rem}}|}$  contains the weights associated with the sampled latent state  $\mathbf{q}_M^{(j)}$ ). Similarly, the weights from the previous pruning stage, restricted to the same remaining support, are denoted by

$$\overline{\mathcal{W}}_{\text{old}} = \begin{bmatrix} \left| \bar{\boldsymbol{\omega}}_{\text{old}}^{(1)} \right| & \left| \bar{\boldsymbol{\omega}}_{\text{old}}^{(2)} \right| & \dots & \left| \bar{\boldsymbol{\omega}}_{\text{old}}^{(P_{\mathcal{M}})} \right| \end{bmatrix} \in \mathbb{R}^{|\mathcal{I}_{\text{rem}}| \times P_{\mathcal{M}}},$$

where  $\bar{\boldsymbol{\omega}}_{\text{old}}^{(j)} = \boldsymbol{\omega}_{\text{old}\tilde{\mathcal{I}}_{\text{rem}}}^{(j)}$ . We propose to determine the redistributed weights  $\mathcal{W}_{\text{new}}$  as the solution of the following constrained quadratic optimization problem:

$$\begin{aligned} \min_{\mathcal{W}_{\text{new}}} \quad & \frac{1}{2} \|\mathcal{W}_{\text{new}} - \overline{\mathcal{W}}_{\text{old}}\|_F^2 + \frac{\alpha_{\mathcal{G}}}{2} \text{tr} \left[ (\mathcal{W}_{\text{new}} - \overline{\mathcal{W}}_{\text{old}}) \boldsymbol{\kappa}_{\mathcal{G}} (\mathcal{W}_{\text{new}} - \overline{\mathcal{W}}_{\text{old}})^T \right] \\ \text{subject to} \quad & (\mathbf{U}_{j\mathcal{I}_{\text{rem}}})^T \boldsymbol{\omega}_{\text{new}}^{(j)} = \mathbf{b}_j, \quad j = 1, \dots, P_{\mathcal{M}}, \\ & \boldsymbol{\omega}_{\text{new}}^{(j)} \geq 0, \quad j = 1, \dots, P_{\mathcal{M}}, \end{aligned} \quad (70)$$

where  $\text{tr}(\cdot)$  denotes the matrix trace operator.

The first contribution in the objective function is a least-change term written in Frobenius norm form; notice it does not introduce any coupling between different sampled latent states, since

$$\|\mathcal{W}_{\text{new}} - \overline{\mathcal{W}}_{\text{old}}\|_F^2 = \sum_{j=1}^{P_{\mathcal{M}}} \|\boldsymbol{\omega}_{\text{new}}^{(j)} - \bar{\boldsymbol{\omega}}_{\text{old}}^{(j)}\|_2^2.$$

Nevertheless, despite this decoupling, the local least-change redistribution still possesses an important implicit smoothing effect. Indeed, the redistribution is computed as the smallest admissible departure from the previously accepted weight fields. Since the initial adaptive weights are constant over the latent manifold, successive least-change redistributions tend naturally to preserve this smooth character and introduce only gradual local modifications. The smooth evolution described above may eventually be altered by the appearance of negative weights. In practice, this introduces localized irregularities, since solution algorithms

clamp these negative weights to zero. The second contribution in the objective function of Problem (70) is a graph-based, quadratic regularization term that naturally couples neighboring latent states and penalizes these localized irregularities, with  $\alpha_{\mathcal{G}} \geq 0$  controlling the strength of the regularization.

The notion of neighborhood entering this regularization is introduced through the already mentioned graph structure defined over the sampled latent manifold. More precisely, let

$$\mathcal{G}_{\mathcal{M}} = (\{\mathbf{q}_{\mathcal{M}}^{(j)}\}_{j=1}^{P_{\mathcal{M}}}, \mathcal{E}_{\mathcal{M}})$$

denote a graph whose nodes are the sampled latent states and whose edge set  $\mathcal{E}_{\mathcal{M}}$  encodes neighborhood relations over the sampled manifold. The matrix  $\mathbf{K}_{\mathcal{G}} \in \mathbb{R}^{P_{\mathcal{M}} \times P_{\mathcal{M}}}$  denotes the associated scalar graph operator defined on this graph, typically chosen as a graph Laplacian associated with the connectivity of the sampled latent states; see, e.g., Belkin and Niyogi [32].

The edge set  $\mathcal{E}_{\mathcal{M}}$  may be defined, for instance, from nearest-neighbor relations in latent space, as commonly done in manifold-learning methods based on graph operators [32]. In the benchmarks considered later, the sampled latent states admit particularly simple graph constructions. In the metamaterial-cell problem of Section 6, they lie on a structured two-dimensional Cartesian grid. The resulting connectivity defines an auxiliary quadrilateral mesh in latent space, and  $\mathbf{K}_{\mathcal{G}}$  is obtained from the standard finite element discretization of an isotropic Laplacian using bilinear shape functions. In the damage problem of Section 7, the adaptive weights depend on a single normalized nonlinear coordinate. The ordered samples therefore define a one-dimensional mesh, and  $\mathbf{K}_{\mathcal{G}}$  is constructed from the corresponding finite element discretization of the one-dimensional Laplacian using linear shape functions.

### 5.1.2. Solution strategy

Problem (70) is a standard convex quadratic program with linear equality and positivity constraints, and could in principle be solved using general-purpose quadratic optimization algorithms; see, e.g., Refs. [33, 34]. However, off-the-shelf quadratic optimization procedures do not exploit the particular structure of the present pruning process and therefore eventually become inefficient. In particular, enforcing the graph-regularization term (the one involving the graph matrix  $\mathbf{K}_{\mathcal{G}}$ ) at every pruning step is both unnecessary and computationally expensive. As discussed earlier, the graph-regularization term becomes truly necessary only when positivity is violated.

The proposed solution strategy, summarized in Algorithm 1, does exploit this feature of the pruning process, and distinguish two distinct redistribution regimes (controlled by the Boolean flag  $\chi_{\text{enf}}$  in Line 4). The uncoupled regime consists in the solution of individual least-norm problems:

$$\begin{aligned} \min_{\Delta \boldsymbol{\omega}^{(j)}} \quad & \frac{1}{2} \|\Delta \boldsymbol{\omega}^{(j)}\|_2^2 \\ \text{subject to} \quad & (\mathbf{U}_{j_{\mathcal{I}_{\text{rem}}}})^T \Delta \boldsymbol{\omega}^{(j)} = \mathbf{b}_j - (\mathbf{U}_{j_{\mathcal{I}_{\text{rem}}}})^T \bar{\boldsymbol{\omega}}_{\text{old}}^{(j)}, \quad j = 1, 2, \dots, P_{\mathcal{M}}, \end{aligned} \quad (71)$$

(this readily follows from setting  $\alpha_{\mathcal{G}} = 0$  in Problem (70)). The positivity-enforcement regime, on the other hand, is based on a null-space active-set strategy in which the local equality constraints are eliminated exactly through local null-space parametrizations, and positivity is enforced through latent-state-dependent active sets.

The first steps of the algorithm are common to both redistribution regimes and consist of a loop over the  $P_{\mathcal{M}}$  sampled latent states (Line 7). The key operation performed at each state  $j$  in this loop is the following SVD: (Line 9)

$$(\tilde{\mathbf{U}}_{\mathcal{F}^{(j)}}^j)^T = \boldsymbol{\Psi}^j \boldsymbol{\Sigma}^j (\tilde{\mathbf{V}}^j)^T, \quad \boldsymbol{\Sigma}^j = \text{diag}(\sigma_1^j, \dots, \sigma_{r_j}^j, 0, \dots, 0) \in \mathbb{R}^{\bar{n}_c^j \times |\mathcal{F}^{(j)}|}. \quad (72)$$

where  $\tilde{\mathbf{U}}_{\mathcal{F}^{(j)}}^j \in \mathbb{R}^{|\mathcal{F}^{(j)}| \times \bar{n}_c^j}$  denotes the local equality system restricted to the current free rows  $\mathcal{F}^{(j)} \subseteq \{1, \dots, |\mathcal{I}_{\text{act}}|\}$ , defined in Line 8 as the complementary set of the active rows  $\mathcal{D}^{(j)}$ . Initially,  $\mathcal{D}^{(j)}$  contains only the local index  $\bar{p}$  associated with the tentatively removed integration point (Line 3), although additional rows may subsequently be added during the positivity-enforcement iterations (Line 41).

The SVD outputs (left and right singular vectors,  $\boldsymbol{\Psi}^j$  and  $\tilde{\mathbf{V}}^j$ , respectively, and singular values  $\boldsymbol{\Sigma}^j$ ) play a threefold role within the algorithm: they are used to determine the numerical rank of the restricted local system (as the number of nonzero singular values in  $\boldsymbol{\Sigma}^j$ , Line 10), construct a particular feasible redistribution



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**Algorithm 1:** PRUNESTEP: Active-set strategy for solving the quadratic graph-regularized redistribution problem 70, associated with the tentative pruning of one integration point. This local redistribution procedure is repeatedly invoked within Algorithm 2 to evaluate different feasible candidate eliminations.

---

**Data:** Current active support  $\mathcal{I}_{\text{act}}$ ; candidate point  $p \in \mathcal{I}_{\text{act}}$ ; current weights  $\mathbf{W}_{\text{old}}$ ; local equality systems restricted to the current active support,  $\{(\tilde{\mathbf{U}}^j, \mathbf{b}_j)\}_{j=1}^{P_{\mathcal{M}}}$ , graph operator  $\mathcal{K}_{\mathcal{G}}$ ; regularization parameter  $\alpha_{\mathcal{G}} \geq 0$ .

**Result:** Acceptance flag  $\chi_{\text{feas}}$ , remaining support  $\mathcal{I}_{\text{rem}}$ , redistributed weights  $\mathbf{W}_{\text{new}}$ , and redistribution energy  $S_{\text{Dir}}$ .

```

1  $\bar{p} \leftarrow \text{loc}(p, \mathcal{I}_{\text{act}})$  // Local index of candidate point  $p$  in  $\mathcal{I}_{\text{act}}$ 
2  $\bar{\mathbf{W}}_{\text{old}} \leftarrow \mathbf{W}_{\text{old}}$ ;  $(\bar{\mathbf{W}}_{\text{old}})_{\bar{p},:} \leftarrow 0$  // Remove candidate point  $p$  by zeroing its weight field
3  $\mathcal{D}^{(j)} \leftarrow \{\bar{p}\}$ ,  $j = 1, \dots, P_{\mathcal{M}}$  // Initialize local active sets by tentatively clamping the candidate point
4  $\chi_{\text{feas}} \leftarrow \text{true}$ ;  $\chi_{\text{enf}} \leftarrow \text{false}$ ;  $S_{\text{Dir}} \leftarrow 0$  // Flags for feasibility/activation positivity. Energy initializ.
5  $k_{\text{act}} \leftarrow 0$  // Active-set iteration counter
6 while active sets change do
7   for  $j = 1, \dots, P_{\mathcal{M}}$  do
8      $\mathcal{F}^{(j)} \leftarrow \{1, \dots, |\mathcal{I}_{\text{act}}|\} \setminus \mathcal{D}^{(j)}$  // Free local rows at latent state  $j$ 
9      $(\Psi^j, \Sigma^j, \tilde{\mathbf{V}}^j) \leftarrow \text{SVD}((\tilde{\mathbf{U}}_{\mathcal{F}^{(j)}}^j)^T)$  // Full SVD, see Eq. 72
10     $r_j \leftarrow \text{rank}(\Sigma^j)$  // Numerical rank associated with nonzero singular values
11    if  $r_j < \bar{n}_c^j$  then
12       $\chi_{\text{feas}} \leftarrow \text{false}$  // Full-row-rank acceptance condition fails
13      Reject candidate and set  $\mathcal{I}_{\text{rem}} \leftarrow \mathcal{I}_{\text{act}}$ ,  $\mathbf{W}_{\text{new}} \leftarrow \mathbf{W}_{\text{old}}$ ; stop
14    end
15    Compute particular solution  $\omega_{\text{p,full}}^{(j)}$  // (see Eq. 74)
16    Compute null-space basis  $\mathbf{N}^{(j)}$  // (last  $|\mathcal{F}^{(j)}| - r_j$  columns of  $\tilde{\mathbf{V}}^j$ )
17    if  $\omega_{\text{p,full}}^{(j)}$  contains negative entries then
18       $\chi_{\text{enf}} \leftarrow \text{true}$  // Positivity enforcement required
19    end
20  end
21  if  $\chi_{\text{enf}} = \text{false}$  then
22     $\chi_{\text{feas}} \leftarrow \text{true}$  // Successful pruning step, no need for regularization
23     $\mathcal{I}_{\text{rem}} \leftarrow \mathcal{I}_{\text{act}} \setminus \{p\}$ , assemble  $\mathbf{W}_{\text{new}}$  from the local vectors  $\omega_{\text{p,full}}^{(j)}$ ; restrict  $\mathbf{W}_{\text{new}}$  to rows  $\{1, \dots, |\mathcal{I}_{\text{act}}|\} \setminus \{\bar{p}\}$ ; stop
24  else
25    end
26    if  $k_{\text{act}} = 0$  then
27       $\mathbf{z}_{\text{old}} \leftarrow \text{vec}(\bar{\mathbf{W}}_{\text{old}})$  // Vectorize by column stacking
28       $\mathbf{H} = \mathbf{I}_{|\mathcal{I}_{\text{act}}|P_{\mathcal{M}}} + \alpha_{\mathcal{G}}(\mathcal{K}_{\mathcal{G}} \otimes \mathbf{I}_{|\mathcal{I}_{\text{act}}|})$  // Graph-regularized operator (Eq. 77)
29       $\mathbf{g} \leftarrow \mathbf{H}\mathbf{z}_{\text{old}}$  // Reference state for the graph-regularized redistribution
30    end
31    Assemble global variables  $\mathbf{z}_{\text{p}}$  and  $\mathbf{N}$  // From  $\{\omega_{\text{p,full}}^{(j)}\}_{j=1}^{P_{\mathcal{M}}}$  and  $\{\mathbf{N}^{(j)}\}_{j=1}^{P_{\mathcal{M}}}$ 
32     $(\mathbf{N}^T \mathbf{H} \mathbf{N}) \mathbf{y} = \mathbf{N}^T (\mathbf{g} - \mathbf{H} \mathbf{z}_{\text{p}})$  // Solve for  $\mathbf{y}$  (see Eq. 78)
33     $\mathbf{z}_{\text{new}} \leftarrow \mathbf{z}_{\text{p}} + \mathbf{N} \mathbf{y}$  // Reconstruct redistributed weights
34    Reshape  $\mathbf{z}_{\text{new}}$  into  $\mathbf{W}_{\text{new}}$  and detect negative free entries of  $\mathbf{W}_{\text{new}}$ 
35    if no negative free entries are detected then
36       $\chi_{\text{feas}} \leftarrow \text{true}$  // Successful positivity-enforced redistribution
37       $S_{\text{Dir}} \leftarrow \text{tr}((\mathbf{W}_{\text{new}} - \bar{\mathbf{W}}_{\text{old}}) \mathcal{K}_{\mathcal{G}} (\mathbf{W}_{\text{new}} - \bar{\mathbf{W}}_{\text{old}})^T)$  // Graph Dirichlet energy
38       $\mathcal{I}_{\text{rem}} \leftarrow \mathcal{I}_{\text{act}} \setminus \{p\}$ ; restrict  $\mathbf{W}_{\text{new}}$  to rows  $\{1, \dots, |\mathcal{I}_{\text{act}}|\} \setminus \{\bar{p}\}$ ; stop
39    else
40      Update local active sets  $\mathcal{D}^{(j)}$  // Clamp negative weights to zero
41       $k_{\text{act}} \leftarrow k_{\text{act}} + 1$  // Update active-set iteration counter
42    end
43  end
44 end

```

---

(Line 15), and identify a basis for the associated null space (Line 16). The full-row-rank condition is what ultimately determines whether a tentative candidate point can be accepted by the algorithm. A single violation of this condition at any sampled latent state (Line 12) immediately causes rejection of the candidate and termination of the pruning step without modifying the current integration rule. It should be noted that such a violation may occur for two different reasons: either the restricted local system is numerically rank deficient, or the active-set updates have clamped so many weights that the number of remaining free design

variables  $|\mathcal{F}^{(j)}|$  becomes smaller than the number of local equality constraints  $\bar{n}_c^j$  ( $j = 1, 2 \dots P_{\mathcal{M}}$ ). Numerical experience indicates that the latter is the most frequent cause of rejection. This observation also identifies the natural stopping limit of the pruning process: the common support cannot contain fewer points than the maximum number of independent local constraints,

$$m_{\min} := \max_{j=1, \dots, P_{\mathcal{M}}} \bar{n}_c^j \leq n_c + n_{\text{inv}}. \quad (73)$$

The particular feasible redistribution computed in Line 15 is obtained in an incremental sense. Denoting by  $\Delta \tilde{\mathbf{w}}_F^j \in \mathbb{R}^{|\mathcal{F}^{(j)}|}$  the redistribution correction restricted to the current free rows, the local equality constraints become

$$(\tilde{\mathbf{U}}_{\mathcal{F}^{(j)}}^j)^T \Delta \tilde{\mathbf{w}}_F^j = \tilde{\mathbf{b}}^j,$$

where

$$\tilde{\mathbf{b}}^j = \mathbf{b}_j - (\tilde{\mathbf{U}}_{\mathcal{F}^{(j)}}^j)^T (\overline{\mathbf{W}}_{\text{old}, \mathcal{F}^{(j)}, j}),$$

The minimum-norm correction satisfying this system is then obtained through the Moore–Penrose pseudoinverse constructed from the SVD outputs associated with the nonzero singular values,

$$\Delta \tilde{\mathbf{w}}_F^j = \tilde{\mathbf{V}}_{:, 1:r_j}^j \text{diag}\left(\frac{1}{\sigma_1^j}, \dots, \frac{1}{\sigma_{r_j}^j}\right) (\Psi_{:, 1:r_j}^j)^T \tilde{\mathbf{b}}^j.$$

The updated free weights are subsequently reconstructed as

$$\mathbf{w}_F^j = (\overline{\mathbf{W}}_{\text{old}})_{\mathcal{F}^{(j)}, j} + \Delta \tilde{\mathbf{w}}_F^j. \quad (74)$$

At the first active-set iteration ( $k_{\text{act}} = 0$ ), the correction  $\Delta \tilde{\mathbf{w}}_F^j$  coincides with the solution of the incremental least-change problem 71. Consequently, if the vectors  $\mathbf{w}_F^j$  remain nonnegative for all  $j = 1, \dots, P_{\mathcal{M}}$ , the tentative pruning step already provides an admissible redistribution and the algorithm terminates without activating the graph-regularized stage (Line 22). Otherwise, the same local construction is still used as the affine origin for the constrained redistribution. For this purpose, each  $\mathbf{w}_F^j$  is lifted back to the full local ambient dimension by inserting zeros on the active rows, leading to the vector  $\omega_{\text{p, full}}^{(j)} \in \mathbb{R}^{|\mathcal{I}_{\text{act}}|}$  in Line 15. On the other hand, the global vector  $\mathbf{z}_p$  in Line 31 is obtained by stacking these lifted particular solutions over all sampled latent states.

The null-space basis computed in Line 16 is directly obtained from the matrix of right singular vectors (specifically, the last  $|\mathcal{F}^{(j)}| - r_j$  columns of  $\tilde{\mathbf{V}}^j$ ). Then it is subsequently lifted back to the full local ambient dimension, producing the basis matrix  $\mathbf{N}^{(j)} \in \mathbb{R}^{|\mathcal{I}_{\text{act}}| \times (|\mathcal{F}^{(j)}| - r_j)}$ , whose columns span all admissible local homogeneous redistributions compatible with the current clamping pattern. These local lifted bases are then assembled into the sparse block-diagonal matrix  $\mathbf{N}$ .

Using the global particular redistribution  $\mathbf{z}_p$  together with the global null-space basis  $\mathbf{N}$ , every admissible redistribution satisfying both the equality constraints and the current active-set pattern can be parametrized as

$$\mathbf{z}_{\text{new}} = \mathbf{z}_p + \mathbf{N} \mathbf{y}, \quad (75)$$

where  $\mathbf{y}$  collects the reduced null-space coordinates associated with all sampled latent states, whereas  $\mathbf{z}_{\text{new}} = \text{vec}(\mathbf{W}_{\text{new}})$ .

Using standard vectorization identities for Frobenius norms and trace operators, the matrix-valued objective function in Problem (70) can then be expressed as an equivalent quadratic functional in terms of  $\mathbf{z}_{\text{new}}$ . After expanding the resulting expression and discarding terms independent of the optimization variables, we obtain the reduced quadratic form

$$\frac{1}{2} \mathbf{z}_{\text{new}}^T \mathbf{H} \mathbf{z}_{\text{new}} - \mathbf{g}^T \mathbf{z}_{\text{new}}. \quad (76)$$

with

$$\mathbf{H} = \mathbf{I}_{|\mathcal{I}_{\text{act}}| P_{\mathcal{M}}} + \alpha_{\mathcal{G}} (\mathcal{K}_{\mathcal{G}} \otimes \mathbf{I}_{|\mathcal{I}_{\text{act}}|}), \quad \mathbf{g} = \mathbf{H} \mathbf{z}_{\text{old}}, \quad (77)$$

( $\mathbf{H} \in \mathbb{R}^{(|\mathcal{I}_{\text{act}}| P_{\mathcal{M}}) \times (|\mathcal{I}_{\text{act}}| P_{\mathcal{M}})}$  and  $\mathbf{g} \in \mathbb{R}^{|\mathcal{I}_{\text{act}}| P_{\mathcal{M}}}$  are introduced in Line 28). The Kronecker-product structure arises naturally from the vectorization of the trace term, since the same graph operator  $\mathcal{K}_{\mathcal{G}}$  acts independently on each one of the  $|\mathcal{I}_{\text{act}}|$  adaptive weight fields.

Substituting the null-space parametrization 75 into the quadratic expression 76 yields a reduced quadratic problem in the unknown reduced coordinates  $\mathbf{y}$ . Imposing stationarity with respect to  $\mathbf{y}$  in this problem leads to the linear system

$$(\mathbf{N}^T \mathbf{H} \mathbf{N}) \mathbf{y} = \mathbf{N}^T (\mathbf{g} - \mathbf{H} \mathbf{z}_p), \quad (78)$$

which corresponds precisely to the equation solved in Line 32.

After solving this system,  $\mathbf{z}_{\text{new}}$  is reconstructed from the reduced coordinates using Eq. 75 and reshaped into matrix  $\mathbf{W}_{\text{new}}$ . The algorithm then inspects  $\mathbf{W}_{\text{new}}$ . If no negative free entry is found the pruning step is accepted, and the final support is set to  $\mathcal{I}_{\text{rem}} = \mathcal{I}_{\text{act}} \setminus \{p\}$ . The matrix  $\mathbf{W}_{\text{new}}$  is then restricted to the locally retained rows  $\{1, \dots, |\mathcal{I}_{\text{act}}| \setminus \{\bar{p}\}\}$ , thereby returning the redistributed rule on the pruned support (Line 36). Besides the redistributed weights themselves, the algorithm also returns the graph Dirichlet energy of the admissible redistribution (Line 37). This quantity is subsequently used in the outer graph-based pruning stage to rank the feasible candidate eliminations.

If, on the contrary, negative free entries are detected, the corresponding rows are added to the appropriate local active sets  $\mathcal{D}^{(j)}$  and clamped to zero (Line 41). The active-set counter is then updated (Line 42), and the local SVD parametrizations are recomputed with the enlarged clamped sets. This process is repeated until either no new negative entries are detected or the candidate becomes infeasible because the local rank condition fails.

**Remark 1.** *Numerical experiments shown in Sections 6.5.1 and 7.4.1 indicate that most of weight fields are removed without activating the graph-coupled positivity-enforcement stage, which typically becomes necessary only during the final pruning steps, close to the final sparse solution. This favorable property may be explained by the presence of the volume-preservation constraint within the local equality systems (see Eq. 58). Indeed, after removing one integration point and its associated weight field, the remaining weights fields tend naturally to increase, on average, in order to compensate for the missing contribution, thereby favoring positivity. This compensation mechanism may be better grasped through the analogy of a statically redundant cable-supported structure of total self-weight  $T$ . The weights play the role of cable tensions, the latent states represent different loading configurations, and the equality constraints enforce equilibrium for each configuration. Thus, removing one integration point and its associated weight field is analogous to cutting one cable: the remaining cable tensions must then increase on average over all configurations in order to equilibrate the total self-weight  $T$ . As redundancy decreases, however, some cables may no longer be able to participate in the equilibrium rearrangement for certain loading configurations and simply go slack. In the present setting, this corresponds to weights attempting to become negative and therefore being clamped to zero (in Line 41 of Algorithm 1).*

## 5.2. Candidate sweep and branch selection

Algorithm 1 is inherently path-dependent: once a candidate elimination is accepted, the subsequent pruning sequence is conditioned by that choice, while the alternative supports associated with other feasible eliminations are no longer pursued. To mitigate this dependence on a single greedy decision, we refine the search through Algorithm 2, which repeatedly invokes Algorithm 1 (Line 9) for different candidate points, generating up to a user-prescribed number  $n_{\text{try}}$  of feasible pruning branches before committing to one elimination.

The sweep itself is driven by two successive indicators. First, the candidate points are prioritized according to the average value of their weights over the sampled latent manifold (Line 3). Candidates carrying smaller average weight are tested first, since they are expected to play a less significant role in satisfying the local exactness constraints.

The second indicator driving the sweep is the graph Dirichlet energy computed in Line 37 of Algorithm 1. Among all feasible pruning trials stored in Line 11, the sweep selects the candidate with smallest energy (Line 22). It should be noted that this secondary ranking becomes relevant only when the positivity-enforcement stage of Algorithm 1 has actually modified the redistribution, since otherwise the Dirichlet energy contribution does not participate in the objective function of Problem (70). In such cases, Algorithm 1 simply returns zero Dirichlet energy, and consequently Algorithm 2 explores no further branches (Line 13).

## 5.3. Global pruning loop

Finally, Algorithm 3 describes how the initial fixed ECM rule introduced in Section 4.2 (see, in particular, Eq. (56)) is progressively transformed into a more compact adaptive cubature rule through repeated

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**Algorithm 2:** PRUNESWEEP: exploration and ranking of feasible pruning candidates through repeated calls to Algorithm 1. This candidate-sweep procedure is in turn repeatedly invoked within Algorithm 3 during the global pruning process.

---

**Data:** Current active support  $\mathcal{I}_{\text{act}}$ ; current weights  $\mathbf{W}_{\text{old}}$ ; local equality systems restricted to the current active support,  $\{(\tilde{\mathbf{U}}^j, \mathbf{b}_j)\}_{j=1}^{P_{\mathcal{M}}}$  (where  $\tilde{\mathbf{U}}^j = (\mathbf{U}_j)_{\mathcal{I}_{\text{act}}} \in \mathbb{R}^{|\mathcal{I}_{\text{act}}| \times \bar{n}_c^j}$  and  $\mathbf{b}_j \in \mathbb{R}^{\bar{n}_c^j}$ ); graph operator  $\mathcal{K}_{\mathcal{G}}$ ; regularization parameter  $\alpha_{\mathcal{G}} \geq 0$ ; maximum number of feasible pruning trials  $n_{\text{try}}$ .

**Result:** Acceptance flag  $\chi_{\text{feas}}$ , selected candidate  $p^*$ , remaining support  $\mathcal{I}_{\text{rem}}$  and redistributed weights  $\mathbf{W}_{\text{new}}$

```

1  $\chi_{\text{feas}} \leftarrow \text{false}$ 
2 for  $\bar{p} = 1, \dots, |\mathcal{I}_{\text{act}}|$  do
3    $\bar{\omega}_{\bar{p}} \leftarrow \frac{1}{P_{\mathcal{M}}} \sum_{j=1}^{P_{\mathcal{M}}} (\mathbf{W}_{\text{old}})_{\bar{p},j}$  // Average weight carried by local candidate  $\bar{p}$  over the latent manifold
4 end
5 Sort  $\mathcal{I}_{\text{act}}$  in increasing order of  $\bar{\omega}_p$ , and denote the resulting ordered list by  $\mathcal{I}_{\text{ord}}$ 
6 Initialize empty lists  $\mathcal{J}_p, \mathcal{J}_I, \mathcal{Q}_W, \mathcal{R}_{\text{Dir}}$ ;  $k \leftarrow 1$ 
7 while  $k \leq |\mathcal{I}_{\text{ord}}|$  AND  $|\mathcal{J}_p| < n_{\text{try}}$  do
8    $p \leftarrow (\mathcal{I}_{\text{ord}})_k$ 
9    $(\chi_{\text{feas}}, \mathcal{I}_{\text{rem}}, \mathbf{W}_{\text{new}}, S_{\text{Dir}}^p) \leftarrow \text{PRUNESTEP}(\mathcal{I}_{\text{act}}, p, \mathbf{W}_{\text{old}}, \{(\tilde{\mathbf{U}}^j, \mathbf{b}_j)\}_{j=1}^{P_{\mathcal{M}}}, \mathcal{K}_{\mathcal{G}}, \alpha_{\mathcal{G}})$  // Algorithm 1
10  if  $\chi_{\text{feas}} = \text{true}$  then
11    Append  $p, \mathcal{I}_{\text{rem}}, \mathbf{W}_{\text{new}}$ , and  $S_{\text{Dir}}^p$  to  $\mathcal{J}_p, \mathcal{J}_I, \mathcal{Q}_W, \mathcal{R}_{\text{Dir}}$ , respectively
12    if  $S_{\text{Dir}}^p = 0$  then
13      break // No graph redistribution was needed; no smoother feasible trial is possible
14    end
15  end
16   $k \leftarrow k + 1$ 
17 end
18 if  $\mathcal{J}_p = \emptyset$  then
19    $\chi_{\text{feas}} \leftarrow \text{false}$  // No feasible pruning candidate was found
20    $\mathcal{I}_{\text{rem}} \leftarrow \mathcal{I}_{\text{act}}, \mathbf{W}_{\text{new}} \leftarrow \mathbf{W}_{\text{old}}$  // Preserve current support and weights
21 else
22    $a^* \leftarrow \arg \min_a (\mathcal{R}_{\text{Dir}})_a$  // Choose feasible pruning trial with smallest graph Dirichlet energy
23    $p^* \leftarrow (\mathcal{J}_p)_{a^*}; \mathcal{I}_{\text{rem}} \leftarrow (\mathcal{J}_I)_{a^*}; \mathbf{W}_{\text{new}} \leftarrow (\mathcal{Q}_W)_{a^*}; \chi_{\text{feas}} \leftarrow \text{true}$ 
24 end

```

---



---

**Algorithm 3:** MAWPRUNE: global feasible pruning of the initial fixed ECM rule through repeated calls to Algorithm 2. The algorithm progressively reduces the common support of the adaptive cubature rule while preserving the local exactness constraints over the sampled latent manifold.

---

**Data:** Initial fixed ECM support  $\mathcal{Z}_{\mathcal{D}} = \{z_1, \dots, z_{m_0}\} \subset \{1, \dots, M_{\text{gp}}\}$ ; initial fixed ECM weights  $\boldsymbol{\omega}_0 \in \mathbb{R}^{m_0}$ , defined on  $\mathcal{Z}_{\mathcal{D}}$ ; local equality systems  $\{(\mathbf{U}_j, \mathbf{b}_j)\}_{j=1}^{P_{\mathcal{M}}}$ , with  $\mathbf{U}_j \in \mathbb{R}^{m_0 \times \bar{n}_c^j}$  defined on  $\mathcal{Z}_{\mathcal{D}}$  and  $\mathbf{b}_j \in \mathbb{R}^{\bar{n}_c^j}$ ; graph operator  $\mathcal{K}_{\mathcal{G}} \in \mathbb{R}^{P_{\mathcal{M}} \times P_{\mathcal{M}}}$ ; regularization parameter  $\alpha_{\mathcal{G}} \geq 0$ ; maximum number of feasible pruning trials  $n_{\text{try}}$ .

**Result:** Subset of retained Gauss points  $\mathcal{Z}_{\omega} \subseteq \mathcal{Z}_{\mathcal{D}}$  and associated weights  $\mathbf{W} \in \mathbb{R}^{|\mathcal{Z}_{\omega}| \times P_{\mathcal{M}}}$  (the  $i$ -th row of  $\mathbf{W}$  contains samples over the latent space of the adaptive weight field associated with the  $i$ -th retained Gauss point).

```

1  $\mathcal{I}_{\text{act}} \leftarrow \{1, \dots, m_0\}$  // Active local indices with respect to the fixed support  $\mathcal{Z}_{\mathcal{D}}$ 
2  $\mathbf{W}_{\text{old}} \leftarrow [\boldsymbol{\omega}_0, \dots, \boldsymbol{\omega}_0] \in \mathbb{R}^{m_0 \times P_{\mathcal{M}}}$  // Initial adaptive weights, still represented on the full support  $\mathcal{Z}_{\mathcal{D}}$ 
3  $\chi_{\text{feas}} \leftarrow \text{true}$ 
4 while  $\chi_{\text{feas}} = \text{true}$  do
5    $\tilde{\mathbf{U}}^j \leftarrow (\mathbf{U}_j)_{\mathcal{I}_{\text{act}}} \in \mathbb{R}^{|\mathcal{I}_{\text{act}}| \times \bar{n}_c^j}, j = 1, \dots, P_{\mathcal{M}}$  // Restrict local systems to the current active support
6    $(\chi_{\text{feas}}, \mathcal{I}_{\text{rem}}, \mathbf{W}_{\text{new}}, S_{\text{Dir}}) \leftarrow \text{PRUNESWEEP}(\mathcal{I}_{\text{act}}, \mathbf{W}_{\text{old}}, \{(\tilde{\mathbf{U}}^j, \mathbf{b}_j)\}_{j=1}^{P_{\mathcal{M}}}, \mathcal{K}_{\mathcal{G}}, \alpha_{\mathcal{G}}, n_{\text{try}})$  // Algorithm 2
7   if  $\chi_{\text{feas}} = \text{true}$  then
8      $\mathcal{I}_{\text{act}} \leftarrow \mathcal{I}_{\text{rem}}$  // Update the current active support after successful pruning
9      $\mathbf{W}_{\text{old}} \leftarrow \mathbf{W}_{\text{new}}$  //  $\mathbf{W}_{\text{old}} \in \mathbb{R}^{|\mathcal{I}_{\text{act}}| \times P_{\mathcal{M}}}$ , now supported on  $\mathcal{Z}_{\mathcal{D}}(\mathcal{I}_{\text{act}})$ 
10  else
11    stop // No further feasible pruning step was found
12  end
13 end
14  $\mathcal{Z}_{\omega} \leftarrow \{z_i\}_{i \in \mathcal{I}_{\text{act}}}$ 
15  $\mathbf{W} \leftarrow \mathbf{W}_{\text{old}}$ 

```

---

calls (Line 6) to the candidate-sweep procedure described above. The process terminates when no further feasible pruning step can be found (Line 11). The resulting matrix  $\mathbf{W} \in \mathbb{R}^{|\mathcal{Z}_{\omega}| \times P_{\mathcal{M}}}$  contains the sampled

representation of the desired manifold-adaptive weight fields supported on the retained set of Gauss points  $\mathcal{Z}_\omega = \{g_1, \dots, g_m\} \subseteq \mathcal{Z}_\mathcal{D}$ . More precisely, the  $i$ -th row of  $\mathbf{W}$  stores the samples over the latent space of the weight field associated<sup>4</sup> with the retained Gauss point  $g_i \in \mathcal{Z}_\omega$ .

#### 5.4. Regression of the adaptive weight fields

The matrix  $\mathbf{W}$  is not itself the final outcome of the procedure; it only provides the sampled data required for constructing the continuous manifold-adaptive weight fields entering Eq. (51). More specifically, for each retained Gauss point  $g_i \in \mathcal{Z}_\omega$ ,  $i = 1, \dots, m$ , one seeks to construct a nonlinear map

$$\mathbf{q}_M \mapsto \omega_{g_i}(\mathbf{q}_M), \quad (79)$$

from the pairs

$$\left\{ (\mathbf{q}_M^{(j)}, (\mathbf{W})_{i,j}) \right\}_{j=1}^{P_\mathcal{M}} \quad (80)$$

Generic regression procedures for Eq. 79 do not guarantee positivity or exact preservation of the total volume, particularly outside the convex hull of the training data. Possible strategies for mitigating this issue include, on the one hand, enriching the regression dataset with additional low-fidelity samples outside the FE training region<sup>5</sup>, or, on the other hand, employing regression procedures explicitly designed to preserve positivity, volume, or related geometric constraints [35]. The study of such constraint-preserving regression procedures lies beyond the scope of the present work; in the numerical examples discussed in the sequel, we simply employ the same unconstrained regression techniques used for the nonlinear decoder.

#### 5.5. Summary

By way of conclusion, Box 5.1 summarizes the complete offline construction of the MAW-ECM rule.

## 6. Benchmark 1: hyperelastic metamaterial cell

### 6.1. Problem setting

The first numerical benchmark considers the two-dimensional periodic “metamaterial” unit cell<sup>6</sup> displayed in Fig. 1. The material is modeled as an isotropic compressible Neo-Hookean solid under plane-strain conditions, with Young’s modulus  $E = 1628$  MPa and Poisson’s ratio  $\nu = 0.4$ . The domain is discretized using  $N_{\text{el}} = 1315$  bilinear quadrilateral elements (average size  $0.1$  mm) and 1602 nodes;  $m_{\text{gp}}^e = 4$  Gauss points are employed per element, giving a total of  $M_{\text{gp}} = N_{\text{el}} m_{\text{gp}}^e = 5260$  integration points.

The cell is deformed exclusively through its external boundary  $\partial\Omega_0$  via imposed displacements (thus  $\mathbf{F}^{\text{ext}} = \mathbf{0}$  in Eq. 2). Furthermore, the boundary term  $\hat{\mathbf{d}}$  in Eq. 1 is assumed to be linear with the input parameter  $\boldsymbol{\mu}$ , i.e.,  $\hat{\mathbf{d}} = \bar{\mathbf{U}}\boldsymbol{\mu}$ , the columns of  $\bar{\mathbf{U}}$  defining the prescribed displacement patterns for unit variations of each component of  $\boldsymbol{\mu}$ . To give this parametrization a concrete mechanical interpretation,  $\bar{\mathbf{U}}$  is constructed from the periodic boundary conditions of first-order computational homogenization<sup>7</sup> at finite strains; see, e.g., Ref. [40]. More specifically, the input parameters are identified with two components (thus  $n_\mu = 2$ ) of

<sup>4</sup>It should be noted that, with a mild abuse of notation, the symbol  $\mathbf{W}$  is used both for the sparse solution of the original global sparsification problem (66) (represented on the initial support  $\mathcal{Z}_\mathcal{D}$  by inserting zeros at the eliminated points, hence  $\mathbf{W} \in \mathbb{R}^{m_0 \times P_\mathcal{M}}$ ) and for its compressed dense representation restricted to the retained support  $\mathcal{Z}_\omega$ , for which  $\mathbf{W} \in \mathbb{R}^{|\mathcal{Z}_\omega| \times P_\mathcal{M}}$ .

<sup>5</sup>The additional samples could be generated using the manifold HROM with fixed weights. Incidentally, the same multi-fidelity philosophy could be applied to the decoder by augmenting its training set with additional low-fidelity snapshots generated, for instance, from a standard fixed-weight HROM.

<sup>6</sup>A *metamaterial* is an architected material formed by the repetition of unit cells whose effective macroscopic response differs markedly from that of the constituent solid alone. In the present case, snap-through of the curved ligaments produces negative incremental stiffness under compression and a pronounced asymmetry between the tensile and compressive responses, making the problem a challenging benchmark for model reduction. Further details on mechanical metamaterials may be found in Refs. [36, 37, 38].

<sup>7</sup>Lest this modeling choice be mistaken for an endorsement of first-order homogenization for bistable media, we emphasize that it is adopted here for simplicity rather than fidelity. Its kinematic content is too limited to represent the full complexity of bistable unit cells. Richer nonlinear multiscale formulations endowed with micromorphic variables are more appropriate for that purpose; see, e.g., Ref. [39].

1. Generate the FE training trajectories over  $\mathcal{P}^{\text{tr}}$  and collect the independent displacement snapshots, together with the Gauss-point and history data required to reconstruct the internal-force and output integrands; see Section 2 and Eq. (17).
2. Construct the decoder  $\mathcal{D}$  and encoder  $\mathcal{E}$ , identify the latent coordinates of the training states, and evaluate the tangent basis  $\Phi_{\mathcal{D}}(\mathbf{q}_M)$ . For the factorized decoder adopted here, this amounts to determining  $\Phi$ ,  $\tau(\mathbf{q}_M)$ , and its Jacobian  $\mathbf{J}_{\tau}(\mathbf{q}_M)$ ; see Eqs. (13), (14), (18), and (28).
3. At each training state, assemble the generic manifold-integrand block  $\mathbf{A}_j$  associated with projected equilibrium and/or output evaluation, and concatenate the resulting blocks into the global matrix  $\mathbf{A}$ ; cf. Eqs. (34), (41), and (42).
4. Compress the column space of  $\mathbf{A}$  to obtain  $\mathbf{V}$ , and compute the initial fixed-weight ECM rule from Problem (46). This provides the candidate support  $\mathcal{Z}_{\mathcal{D}} = \text{supp}(\mathbf{w}^{\mathcal{D}})$ , containing  $m_0$  Gauss points, together with the corresponding positive weights; see Eqs. (45), (48), and (50).
5. Select the representative parameter subset  $\mathcal{P}^{\mathcal{M}}$  and its latent states according to Eq. (55). Initialize the sampled adaptive weights by restricting and replicating the fixed-weight rule, as in Eq. (56), and construct the local exactness systems

$$\mathbf{U}_j^T \boldsymbol{\omega}_j = \mathbf{b}_j, \quad j = 1, \dots, P_{\mathcal{M}},$$

by projecting the local integrand blocks, adjoining the prescribed invariant conditions, and removing linearly dependent constraints; cf. Eqs. (57)–(63).

6. Construct the latent graph  $\mathcal{G}_{\mathcal{M}}$ , its graph operator  $\mathcal{K}_{\mathcal{G}}$ , and choose the regularization parameter  $\alpha_{\mathcal{G}}$ .
7. Apply Algorithm 3 (MAWPRUNE) to the initial fixed-weight ECM rule. The algorithm returns the retained support  $\mathcal{Z}_{\omega} \subseteq \mathcal{Z}_{\mathcal{D}}$  and the matrix  $\mathbf{W} \in \mathbb{R}^{|\mathcal{Z}_{\omega}| \times P_{\mathcal{M}}}$  containing the sampled adaptive weights; see Section 5.3.
8. For each retained Gauss point  $g_i \in \mathcal{Z}_{\omega}$ , regress the  $i$ -th row of  $\mathbf{W}$  over the sampled latent states to construct the continuous weight field

$$\mathbf{q}_M \mapsto \omega_{g_i}(\mathbf{q}_M), \quad i = 1, \dots, m,$$

from the data pairs in Eq. (80); see Section 5.4. The resulting online cubature rule has a fixed common support  $\mathcal{Z}_{\omega}$ , while its weights vary over the latent manifold.

Box 5.1: Offline construction of a MAW–ECM rule.

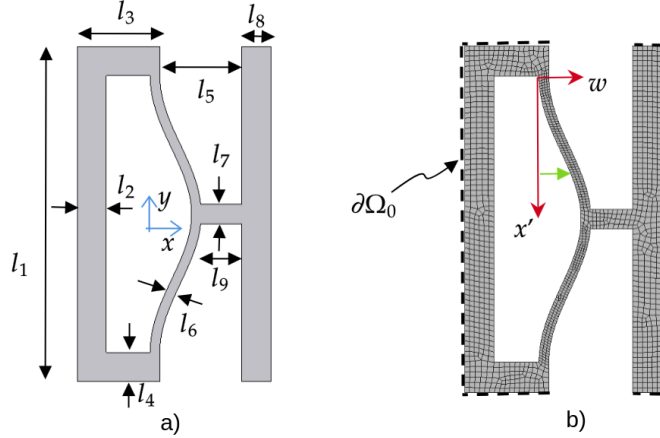


Figure 1: Geometry and finite element discretization of the metamaterial unit cell (the geometry is based on the design proposed in Ref. [38]). (a) Reference domain, with  $l_1 = 60.40$  mm,  $l_2 = l_4 = l_8 = 5.20$  mm,  $l_3 = 14.50$  mm,  $l_5 = 14.90$  mm,  $l_6 = 1.76$  mm,  $l_7 = 3.50$  mm, and  $l_9 = 7.49$  mm. The centerline of the lower curved ligament is described in the local coordinates  $(x', w)$  by  $w(x') = \frac{h}{2}[1 - \cos(\pi x'/L)]$ ,  $0 \leq x' \leq 2L$ , where  $2L = l_1 - 2l_4 = 50.00$  mm is its projected span and  $h = l_5 - l_9 - l_6 = 5.65$  mm is its maximum transverse rise. (b) Finite element mesh; the dashed markers identify the external boundary  $\partial\Omega_0$ , whose opposite sides are paired for the imposition of periodic boundary conditions.

the macroscopic displacement gradient  $\mathbf{G}^{\text{macro}} \in \mathbb{R}^{2 \times 2}$ : the normal component along the  $x$ -direction and the shear component, i.e.:

$$\boldsymbol{\mu} = \begin{bmatrix} G_x^{\text{macro}} \\ G_{xy}^{\text{macro}} \end{bmatrix}. \quad (81)$$

The transverse normal component is set to zero, and  $\mathbf{G}^{\text{macro}}$  is assumed symmetric, so that no macroscopic

rigid-body rotation is prescribed; thus:

$$\mathbf{G}^{\text{macro}}(\boldsymbol{\mu}) = \begin{bmatrix} G_x^{\text{macro}} & G_{xy}^{\text{macro}} \\ G_{xy}^{\text{macro}} & 0 \end{bmatrix} = \begin{bmatrix} \mu_1 & \mu_2 \\ \mu_2 & 0 \end{bmatrix}. \quad (82)$$

Nonzero entries in  $\bar{\mathbf{U}}$  appear only for slave boundary nodes and corner nodes; such entries are determined by considering that, for a node  $\mathbf{X} = (X, Y)^T$ , the macroscopic contribution is defined as  $\bar{\mathbf{u}}(\mathbf{X}) = \mathbf{G}^{\text{macro}} \mathbf{X}$ , which in the present case yields  $\bar{\mathbf{u}}(\mathbf{X}) = (\mu_1 X + \mu_2 Y, \mu_2 X)^T$ . Consistently with this interpretation,  $\mathbf{d} \in \mathbb{R}^N$  in Eq. 1 collects the degrees of freedom of the interior nodes and the independent boundary nodes on the designated master sides for imposing periodic conditions (here  $N = 3018$ ). On the other hand, the lifting operator  $\mathbf{T}$  in Eq. 1 transfers the periodic fluctuation displacements from the master sides to their paired slave sides.

As customary in first-order computational homogenization, the output of interest is the homogenized first Piola–Kirchhoff stress,

$$\mathbf{P}^{\text{macro}} = \frac{1}{|\Omega_0|} \int_{\Omega_0} \mathbf{P} \, d\Omega_0, \quad (83)$$

which is a volumetric output  $\mathbf{y}_y$  of the form introduced in Eq. 12; hence,  $\mathbf{r}^y = \mathbf{P}/|\Omega_0|$  and  $n_c = 4$ .

## 6.2. Training data

The training set  $\mathcal{P}^{\text{tr}}$  (see Eq. 16) is restricted to a two-dimensional rectangle, in order to keep the regression problem simple and to allow a direct graphical inspection of the adaptive weights. The compression/stretching component along the soft direction of the cell is varied in the interval  $G_x^{\text{macro}} \in [-0.42, 0.42]$ , whereas the shear component is varied in  $G_{xy}^{\text{macro}} \in [-0.05, 0.05]$ . As illustrated in Fig. 2, the lower bound  $(G_x^{\text{macro}})_{\min} = -0.42$  corresponds to the most compressed configuration before self-contact occurs<sup>8</sup>. Although these cells are primarily intended to operate under compression, the parameter domain is kept symmetric, with  $(G_x^{\text{macro}})_{\max} = 0.42$ . The purpose is to test the reduced model in a prediction regime in which input parameters of the same magnitude but opposite sign generate outputs of markedly different magnitude (as shown later, in Figure 3b, the corresponding stress levels in the  $x$  direction may differ by up to a factor of 25, since buckling of the curved ligaments leads to a substantially softer response under compression).

We sample the rectangular parameter domain using a Cartesian grid with 230 and 20 uniform intervals in the  $G_x^{\text{macro}}$ - and  $G_{xy}^{\text{macro}}$ -directions, respectively. This gives  $P = 231 \cdot 21 = 4851$  distinct parameter states. The FE simulations corresponding to each of these parameters is obtained by traversing this grid along two continuous zig-zag loading trajectories, covering the nonnegative and nonpositive shear regions, respectively (see Fig. 2).

## 6.3. Standard HROM

We begin by assessing the efficiency of the standard HROM. This model is recovered as a particular case of the manifold HROM introduced in Section 3.1 by identifying, in the decoder factorization of Eq. 18,  $\mathbf{q}_M$  with the standard reduced coordinates  $\mathbf{q}$ , and by taking  $\boldsymbol{\tau}(\mathbf{q}) = \mathbf{q}$ . It then follows that  $\mathbf{J}_\tau = \mathbf{I}$  in Eq. 28 and  $\boldsymbol{\Phi}_{\mathcal{D}} = \boldsymbol{\Phi}$  in Eq. 33.

The accuracy and complexity of this model are governed solely by the truncation tolerances  $\varepsilon^D$ , for determining the displacement basis  $\boldsymbol{\Phi}$ , see Eq. 19, and  $\varepsilon^A$ , for the integrand matrix  $\mathbf{V}$ , see Eq. 45. Both bases are computed using the Sequential Randomized SVD of Ref. [22], briefly introduced in Section 3.5.<sup>9</sup> We then performed a systematic tolerance study and selected the least expensive configuration for which the normalized maximum error in the homogenized stress remained below 0.1% over the complete training set. The study revealed that this criterion requires  $\varepsilon^D = 10^{-6}$ , yielding a basis matrix  $\boldsymbol{\Phi}$  with  $n_\Phi = 52$  modes, and  $\varepsilon^A = 10^{-4}$ , which results in an ECM rule with 1340 Gauss points. For illustration, Fig. 3(a) shows the

<sup>8</sup>Our model does not include any representation of self-contact; hence, extrapolation beyond this compression level is physically meaningless.

<sup>9</sup>For  $n_\Phi = 52$ , the integrand matrix  $\mathbf{A}$  has 5260 rows, one per candidate Gauss point, and  $(n_\Phi + n_y)P = (52 + 4) \times 4851 = 271656$  columns. Its explicit storage in double precision requires approximately 11.4 GB, excluding factorization costs. The SRSVD is therefore particularly useful under memory limitations, as it processes the state-wise blocks  $\mathbf{A}_j$  sequentially without assembling the complete matrix.

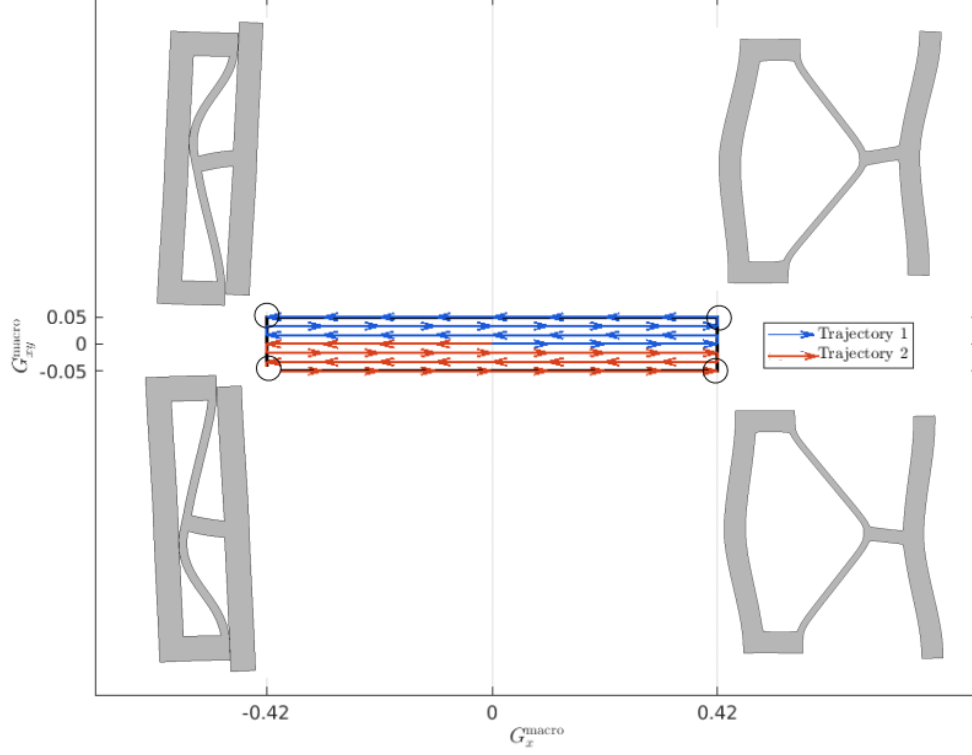


Figure 2: Training parameter domain and zig-zag trajectories used to generate the FE displacement snapshots. The corner configurations correspond to the extreme parameter values.

671 finite elements containing these points. The tight displacement truncation tolerance  $\varepsilon^D = 10^{-6}$  required to meet the accuracy criterion for the stresses reflects the previously mentioned markedly different scales of the tensile and compressive responses shown in Fig. 3(b). Indeed, modes associated with small singular values contribute little to the snapshot energy, yet are necessary to capture local details related to internal buckling mechanisms governing the low-stress compressive branch.

In summary, even though the problem is governed by  $n_\mu = 2$  parameters,  $n_\Phi = 52$  linear displacement modes are required to reproduce the nonlinear response with the desired accuracy. This illustrates the poor linear compressibility of the problem under study. The limitation is even more apparent at the hyperreduction level, where the fixed ECM rule retains 1340 of the original  $M_{\text{gp}} = 5260$  integration points (approximately 25%).

#### 6.4. Manifold HROM

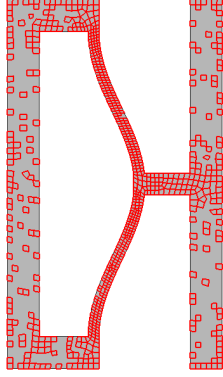
##### 6.4.1. Identification of latent coordinates

The poor linear compressibility observed above motivates replacing the linear reduced representation with the nonlinear-manifold approximation of Eq. 22, introduced in Section 3.2. We first address the identification of the latent coordinates. As established in Eq. 21, these are obtained from the modal coefficients through  $\mathbf{q}_M = \mathbf{H}_M \mathbf{q}$ ; following the input-informed strategy discussed in Section 1.4, the task here is therefore to determine the transformation matrix  $\mathbf{H}_M$  by exploiting the particular parametrization of the present benchmark.

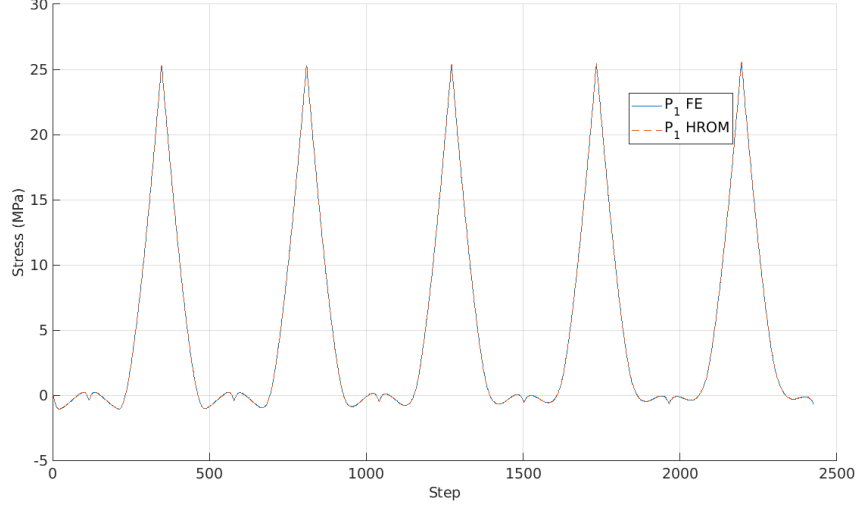
The problem under consideration is static (no inertial effects) and the material response is hyperelastic (no path dependence). It can be argued that, under such circumstances, the displacement state  $\mathbf{d}$  admits a single-valued parametrization in terms of prescribed input  $\boldsymbol{\mu}$ . This implies that the dimension of the solution manifold is equal to the dimension of the input parameter space ( $n = n_\mu = 2$ ), and, furthermore, that there is a one-to-one mapping between input parameters and the desired latent coordinates:

$$\mathbf{q}_M = \varphi(\boldsymbol{\mu}). \quad (84)$$





(a) Elements containing the selected Gauss points when using standard HROM.



(b) Homogenized  $P_x$  response.

Figure 3: Standard-HROM results for  $\varepsilon^D = 10^{-6}$  and  $\varepsilon^A = 10^{-4}$ . (a) The ECM rule comprises 1340 Gauss points distributed over 671 finite elements. (b) FE and standard-HROM homogenized PK1 stress histories along the second trajectory in Fig. 2; the HROM uses  $n_\Phi = 52$  displacement modes.

For the present benchmark,  $\mathbf{H}_M$  is determined by requiring that the above map approximate the identity map over the training set. In view of Eq. 21, this amounts to making  $\mathbf{q}_M(\boldsymbol{\mu}_j) - \boldsymbol{\mu}_j = \mathbf{H}_M \mathbf{q}(\boldsymbol{\mu}_j) - \boldsymbol{\mu}_j$  as small as possible for every training input  $\boldsymbol{\mu}_j$  ( $j = 1, 2 \dots P$ ). Collecting the modal coefficients and the corresponding input parameters into

$$\mathbf{Q}_{ROM} := \Phi^T \mathbf{D}, \quad \mathbf{M}_\mu := [\boldsymbol{\mu}_1 \quad \boldsymbol{\mu}_2 \quad \cdots \quad \boldsymbol{\mu}_P], \quad (85)$$

(here,  $\mathbf{D}$  is the displacement snapshot matrix introduced in Eq. 17.), the matrix  $\mathbf{H}_M$  is accordingly obtained from the least-squares problem

$$\mathbf{H}_M = \arg \min_{\widetilde{\mathbf{H}_M}} \left\| \widetilde{\mathbf{H}_M} \mathbf{Q}_{ROM} - \mathbf{M}_\mu \right\|_F^2. \quad (86)$$

Assuming  $\mathbf{Q}_{ROM}$  has full rank, the solution of Problem 86 is unique and given by

$$\mathbf{H}_M = \mathbf{M}_\mu \mathbf{Q}_{ROM}^\dagger \quad (87)$$

where  $\mathbf{Q}_{ROM}^\dagger$  denotes the Moore–Penrose pseudoinverse.

The relative value of the least-squares residual at the optimum for the problem under consideration is

$$\frac{\|\mathbf{H}_M \mathbf{Q}_{ROM} - \mathbf{M}_\mu\|_F}{\|\mathbf{M}_\mu\|_F} = 3.09 \cdot 10^{-7}. \quad (88)$$

Thus, to numerical accuracy, the latent coordinates associated with the training states can be identified with the input parameters:

$$\mathbf{q}_M \approx \boldsymbol{\mu}. \quad (89)$$

**Remark 2.** This identification has an important consequence for the online stage of the present benchmark. Because  $\mathbf{q}_M$  may be obtained directly from the prescribed input  $\boldsymbol{\mu}$ , the reduced equilibrium equations 54 need not be solved to determine the latent state. The decoder can instead be evaluated directly at  $\mathbf{q}_M \approx \boldsymbol{\mu}$ . Hyperreduction through the proposed adaptive cubature rule is therefore required only for evaluating the homogenized stresses in Eq. 83 (i.e., the equilibrium-residual integrands need not be included in the cubature training problem).

#### 6.4.2. Nonlinear closure

Once  $\mathbf{H}_M$  has been determined, the master basis  $\Phi_M$  and the scaling matrix  $\mathbf{Z}_M$  follow from the SVD of  $\Phi \mathbf{H}_M^T$  introduced in Section 3.2, together with Eq. 25. The slave basis  $\Phi_S$  is subsequently obtained from the orthogonality conditions in Eq. 23. For the present problem, this decomposition yields the  $n = 2$  master modes whose deformed shapes are depicted in Figures 4a and 4b, and  $n_\Phi - n = 50$  slave modes (the deformed shapes of some of these modes are in turn shown in Figures 4c to 4f).

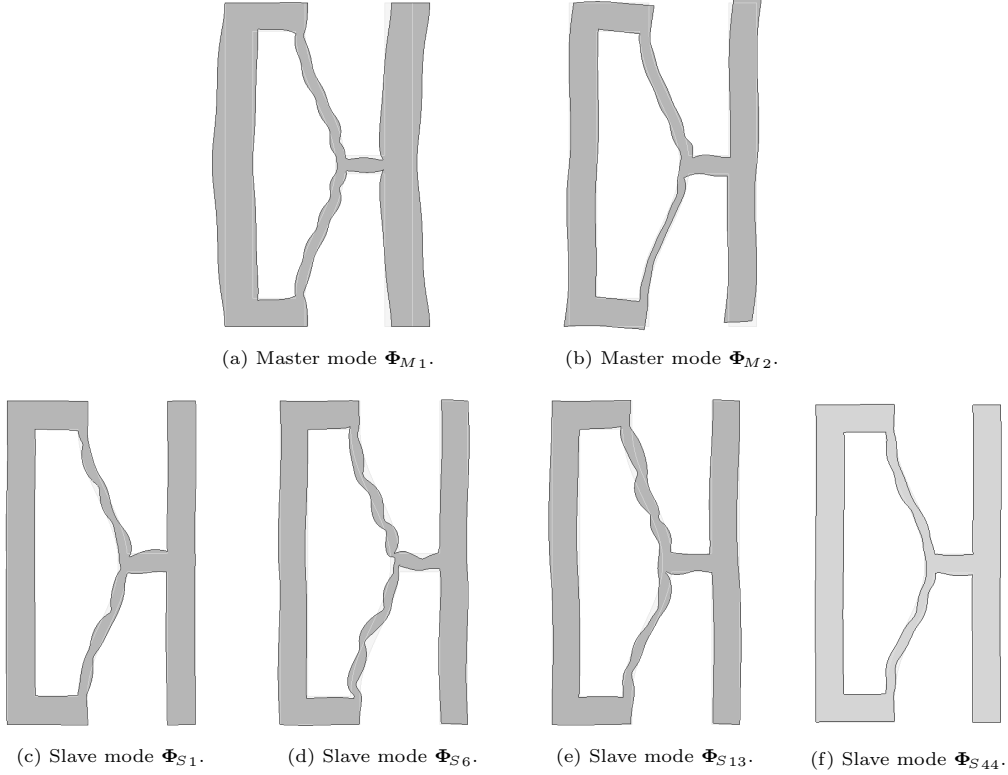


Figure 4: Deformed shapes of the  $n = 2$  master modes  $\Phi_M$  and four of the  $n_\Phi - n = 50$  slave modes  $\Phi_S$ . The modal vectors, originally defined on the independent degrees of freedom, are extended to the complete nodal space through the lifting operator  $\mathbf{T}$  for visualization (see Eq. 1).

It remains to identify the nonlinear closure mapping the  $n = 2$  latent coordinates onto the  $n_\Phi - n = 50$  slave coordinates,  $\mathbf{q}_S \approx \mathcal{N}_S(\mathbf{q}_M)$ , from the training pairs  $(\mathbf{q}_M, \mathbf{q}_S)$  obtained from the displacement snapshots. To this end, we employ anisotropic Gaussian radial basis functions (RBFs) augmented with polynomial terms (quadratic). The regression coefficients are determined through standard-form Tikhonov-regularized least squares, using the identity regularization matrix (see Refs. [41, 42]). Only a subset of the available snapshot pairs is retained as RBF centers. A structured subsampling frequency of  $[3, 1]$  is used: every third sample is retained along the first latent direction (compression/extension), whereas all samples are retained along the second direction (shear). This results in 1540 centers out of  $P = 4851$  data points. The remaining states are employed to assess the accuracy of the closure away from the retained centers. The characteristic lengths of the anisotropic Gaussian kernels are set to  $\ell_1 = 3.3013 \cdot 10^{-2}$  and  $\ell_2 = 1.5789 \cdot 10^{-2}$ , and the Tikhonov regularization parameter to  $\lambda_{\text{RBF}} = 10^{-10}$ . With these settings, the relative reconstruction error is  $4.30 \cdot 10^{-6}$  at the retained centers and  $1.03 \cdot 10^{-4}$  over the complete dataset.

Figure 5 shows the resulting mappings for four representative slave coordinates. The reconstructed surfaces are smooth and single-valued over the sampled latent domain (the black markers indicate the retained RBF centers).

#### 6.4.3. Fixed-weight ECM

With the manifold coordinates and the slave closure in place, we next construct the fixed-weight ECM rule, discussed in Section 3.4. As noted in Remark 2, the latent state is prescribed directly by the macroscopic

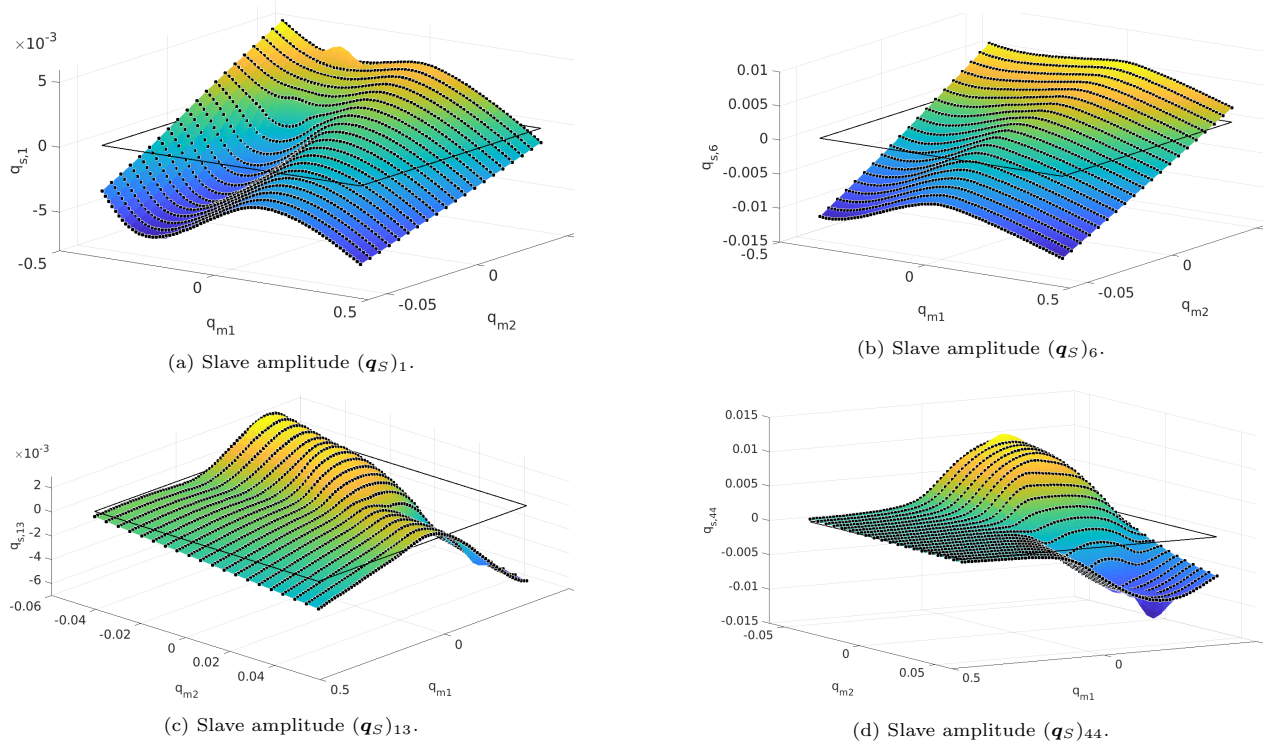


Figure 5: RBF reconstructions of four representative slave amplitudes over the two-dimensional latent space. The black markers denote the retained RBF centers.

input, and no reduced equilibrium problem needs to be solved for the evaluation of the homogenized response. We therefore adopt the output-only case described after Eq. 40 in Section 3.4 and construct the cubature rule exclusively for the homogenized stress in Eq. 83, that is:  $\mathbf{r}_g = \mathbf{P}_g/|\Omega_0|$  ( $g = 1, 2 \dots M_{gp}$ ), which implies that there are  $n_c = s_\sigma = 4$  integration conditions per state. The same  $P_M = 1540$  retained latent states used for the RBF closure described previously in Section 6.4.2 are employed to assemble the output-integrand matrix  $\mathbf{A}$  defined in Eq. 42; hence<sup>10</sup>  $\mathbf{A} \in \mathbb{R}^{M_{gp} \times (s_\sigma P_M)} = \mathbb{R}^{5260 \times 6160}$ . Using the SRSVD with  $\varepsilon^A = 10^{-4}$  to compress  $\mathbf{A}$ , and augmenting the resulting basis with the volume constraint in Eq. 49, the ECM yields a fixed rule with  $m_0 = 54$  Gauss points distributed over 48 finite elements.

Table 1 compares the accuracy of the fixed-weight manifold approximation with those of the standard ROM and HROM. Here,  $e_d$  denotes the relative Frobenius error in the independent displacement histories, whereas  $e_P$  is the corresponding relative Frobenius error in the homogenized PK1 stress histories, in both cases using the FE solution as reference. The errors are computed separately over each complete loading trajectory and then averaged over the two trajectories.

Table 1: Online complexity (in terms of number of master/slave coordinates, and number of integration points) and accuracy (relative Frobenius norms in displacements,  $e_d$  and stresses  $e_P$ ) of the fixed-weight manifold HROM, together with the standard ROM and HROM.

| Model                | Master coords. | Slave coords. | Integ. points | $e_d$                | $e_P$                |
|----------------------|----------------|---------------|---------------|----------------------|----------------------|
| ROM, full Gauss rule | 52             | 0             | 5260          | $2.52 \cdot 10^{-6}$ | $5.05 \cdot 10^{-5}$ |
| HROM, ECM            | 52             | 0             | 1340          | $1.85 \cdot 10^{-5}$ | $7.13 \cdot 10^{-5}$ |
| M-HROM, fixed ECM    | 2              | 50            | 54            | $7.87 \cdot 10^{-5}$ | $2.98 \cdot 10^{-4}$ |

<sup>10</sup>This matrix is markedly smaller than its standard-HROM counterpart, which contains  $(n_\Phi + n_y)P = 271656$  columns; see Footnote 9 in Section 6.3. The number of scalar fields processed by the SRSVD is therefore reduced by a factor of approximately 44, substantially alleviating the memory demands associated with the construction of the ECM basis.

Table 1 shows that the manifold HROM increases the homogenized-stress error by a factor of approximately four relative to the standard HROM. This loss of accuracy is due to the subsampling and subsequent regression (recall that the error is evaluated over all  $P = 4851$  states, whereas the closure is fitted using only  $P_{\mathcal{M}} = 1540$  retained centers). The resulting accuracy–compression trade-off is nevertheless strongly favorable: the number of independent coordinates decreases from  $n_{\Phi} = 52$  to  $n = 2$ , and the number of integration points from 1340 to  $m_0 = 54$ . Interestingly, the corresponding compression ratios are almost identical, namely 26 and approximately 25, respectively.

### 6.5. Manifold-adaptive weight ECM

The preceding fixed-weight manifold HROM completes the kinematic compression, but its cubature rule remains global: the same  $m_0 = 54$  weights must represent the homogenized-stress integrands over the entire latent domain. The final stage of the compression process is therefore to exploit the manifold structure also at the cubature level, by allowing the weights to vary with the latent state. Following the formulation of Section 4, the fixed-weight ECM rule provides the initial candidate support  $\mathcal{Z}_{\mathcal{D}}$  and the feasible initial weight vector  $\boldsymbol{\omega}_0$ . The greedy strategy of Section 5 then progressively removes integration points and redistributes the remaining weights over the sampled manifold while preserving the local exactness and positivity conditions.

In the present output-only setting, the state-dependent integrands are the  $s_{\sigma} = 4$  components of the PK1 stress entering the homogenized response. At each retained latent state, these four integration conditions are supplemented with the invariant volume constraint. This gives a lower bound of  $m_{\min} = 4 + 1 = 5$  points (see Eq. 73) for the desired adaptive rule .

The same  $P_{\mathcal{M}} = 1540$  states retained as RBF centers for the decoder are used as the nodes of the latent graph  $\mathcal{G}_{\mathcal{M}}$ . Here, the numerical identification  $\mathbf{q}_M \approx \boldsymbol{\mu}$ , together with the Cartesian sampling of the input parameter domain (see Figure 3), makes the graph construction immediate: the retained latent states inherit the connectivity of the structured parameter grid. Adjacent states are connected to define an auxiliary quadrilateral mesh in latent space, and the graph operator  $\mathcal{K}_{\mathcal{G}} \in \mathbb{R}^{1540 \times 1540}$  is assembled from the standard finite element discretization of an isotropic Laplacian using bilinear shape functions, consistently with the graph-regularized redistribution described in Section 5.

#### 6.5.1. Influence of the fixed-weight initialization and pruning efficiency

We first assess the reduction achieved by the MAW–ECM pruning strategy of Algorithm 3, and its sensitivity to the fixed-weight initialization. The regularity of the resulting adaptive weight fields, their regression over the latent domain, and the associated online accuracy are examined subsequently.

Four initial cubature rules are considered, obtained with  $\varepsilon^A \in \{10^{-4}, 10^{-5}, 10^{-6}, 10^{-7}\}$  in the truncation of the training integrand matrix  $\mathbf{A}$ . The case  $\varepsilon^A = 10^{-4}$  is precisely the  $m_0 = 54$ -point rule assessed in Table 1. The other three rules are introduced here solely to determine whether the final adaptive support and the pruning efficiency depend on the size and accuracy of the fixed-weight initialization. The graph-regularization parameter and the number of candidate trials are held fixed throughout the study at  $\alpha_{\mathcal{G}} = 10^4$ , appearing in Algorithm 1, and  $n_{\text{try}} = 5$ , appearing in Algorithm 2. These values were selected after preliminary experimentation as a compromise between regularity of the adaptive weight fields and computational efficiency during pruning.

Table 2: Influence of the fixed-weight initialization on the MAW–ECM pruning process for different values of the SVD truncation tolerance  $\varepsilon^A$  for the integrand matrix  $\mathbf{A}$ . Here,  $m_0$  denotes the number of points of the resulting initial fixed-weight ECM rule, and  $m$  the final number of points after pruning. *First-stage limit* denotes the minimum number of points reached before positivity enforcement and graph regularization are activated in Algorithm 1. *Total time* corresponds to the wall-clock time of Algorithm 3, whereas *Full-regularized time* denotes the wall-clock time obtained when regularization is applied during all pruning steps.

| $\varepsilon^A$ | $m_0$ | First-stage limit | $m$ | Total time (s) | Full-regularized time (s) |
|-----------------|-------|-------------------|-----|----------------|---------------------------|
| $10^{-4}$       | 54    | 20                | 10  | 50.0           | 1172.0                    |
| $10^{-5}$       | 87    | 18                | 9   | 44.0           | 7682.0                    |
| $10^{-6}$       | 135   | 18                | 10  | 58.0           | —                         |
| $10^{-7}$       | 187   | 19                | 10  | 89.0           | —                         |

Table 2 summarizes the results obtained with the four initial cubature rules. The most striking observation is the weak dependence of the final adaptive rule on the initialization. As the fixed-weight tolerance becomes

more stringent, the size of the initial rule increases markedly, from  $m_0 = 54$  to  $m_0 = 187$ . Nevertheless, the pruning algorithm consistently reduces these rules to either  $m = 9$  or  $m = 10$  integration points. Thus, the final support appears to be nearly independent of the initial one. Moreover, the resulting adaptive rules contain only about twice the theoretical lower bound,  $m_{\min} = 5$ .

An equally important observation concerns the behavior of the first pruning stage, namely the stage in which positivity enforcement is not explicitly activated. As can be gleaned from the third column of Table 2, independently of the initial number of points, this stage systematically reduces the number of points down to approximately 18–20 before positivity violations begin to appear (Line 18 of Algorithm 1). Thus, not only the final number of cubature points, but also the number of points at which positivity violations appear, seems to be essentially independent of the initial number of points.

This behavior has a direct impact on computational efficiency (see last two columns of Table 2). For instance, increasing the initial support from 54 to 87 points produces no increase in wall-clock time (which remains below one minute in both cases<sup>11</sup>). This confirms the rationale behind Algorithm 1 and the proposed two-stage strategy: the computational cost is governed primarily not by the initial number of candidate points, but by the number of pruning iterations requiring activation of the graph-regularized positivity-enforcement stage.

In order to quantify the computational gains introduced by the proposed two-stage strategy, we compare the performance of the algorithm against a modified version in which the first pruning stage is artificially suppressed. This is achieved by temporarily disabling, for testing purposes, the conditional skip in Line 22 of Algorithm 1 (namely, the line stating “Successful pruning step, no need for regularization”). Under this modification, graph regularization is enforced throughout the entire pruning process, leading to a dramatic increase in computational cost. In particular, the case  $m_0 = 54$  increases from 50 seconds to approximately 20 minutes, whereas the case  $m_0 = 87$  increases from 44 seconds to more than two hours.

### 6.5.2. Adaptive cubature rule and weight-field regression

Having quantified the level of reduction afforded by MAW-ECM, we next examine the structure of the resulting adaptive rule in greater detail. In particular, we focus on the case  $m_0 = 54$ , and analyze both the spatial distribution of the retained integration points and the associated distribution of weights. Figure 6 compares the initial fixed-weight rule with the final adaptive rule containing  $m = 10$  points (for the particular state  $q_{M1} = 0$  and  $q_{M2} = 0$ ). The upper row displays the spatial location of the selected integration points and their associated cubature weights, whereas the lower row identifies the finite elements containing such points. Notice that the initial rule in Fig. 6a is already highly non-uniform: one of the 54 integration points carries approximately 85% of the total volume. As discussed in Ref. [22], this lack of uniformity is a characteristic feature of ECM-type cubature rules, reflecting the greedy, sparsity-promoting nature of the algorithm (which prioritizes error reduction over weight uniformity). It can be seen in Fig. 6b that the adaptive pruning process preserves this dominant point (labelled as point 1), with its relative weight increasing from 85% to 87% in the final rule.

Once the support of the adaptive cubature rule has been identified, the next step consists of constructing continuous approximations of the associated adaptive weight fields,  $\omega_g = \omega_g(\mathbf{q}_M)$ ,  $g = 1, \dots, m$ . Here, we use the same anisotropic RBFs employed for the decoder, but with constant polynomial correction, Tikhonov regularization parameter  $\lambda = 10^{-10}$  and characteristic length scales  $\ell_1 = 2.2 \times 10^{-2}$  and  $\ell_2 = 1.053 \times 10^{-2}$ . Figure 7 shows the resulting weight fields (normalized by the volume  $|\Omega_0|$ ), together with the sampled manifold states (black dots).

The first weight field,  $\omega_1$  remains nearly constant throughout the latent manifold and accounts for more than 87% of the total volume. This observation further reinforces the interpretation of this point as primarily responsible for enforcing the volume-preservation constraint. The nontrivial variations are therefore concentrated in the remaining nine weight fields. Their average magnitude ranges from approximately 3.0% of the total volume for  $\omega_2$  down to only 0.05% for  $\omega_{10}$ . Furthermore, the amplitude of the oscillations tends to increase as the average value of the corresponding weight decreases. Their average magnitude ranges from approximately 3.0% of the total volume for  $\omega_2$  down to only 0.05% for  $\omega_{10}$ . At the same time, the spatial structure of the corresponding fields becomes progressively richer as their average magnitude decreases.

<sup>11</sup>All computations were performed in MATLAB on a Linux laptop, equipped with a 13th-generation Intel Core i9-13900H processor and 16 GiB of RAM.

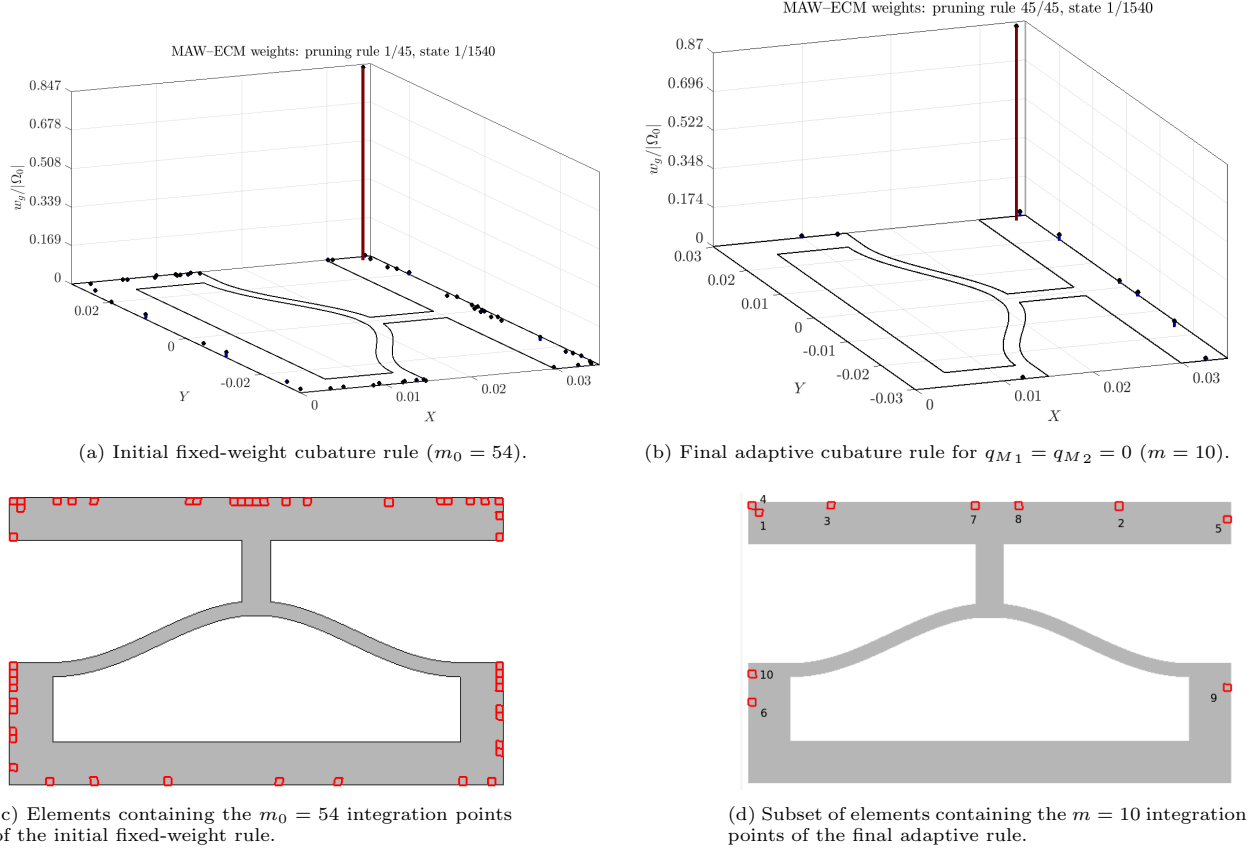


Figure 6: Comparison between the initial fixed-weight manifold ECM rule and the final adaptive cubature rule obtained after pruning. The upper row shows the associated cubature weights, whereas the lower row identifies the elements containing the selected Gauss points. The adaptive rule shown in the right column corresponds to latent state  $q_{M1} = q_{M2} = 0$ . Heights in the upper row are proportional to the corresponding cubature weights.

Nevertheless, all weight fields remain smooth functions of the latent coordinates, with no evidence of abrupt localized variations. The RBF reconstruction yields a relative interpolation error of  $3.89 \cdot 10^{-9}$  over the complete dataset.

An alternative way of visualizing the weight variability is to exploit the fact that the latent coordinates are approximately equal to the input macro-gradients ( $q_{M1} \approx G_x^{\text{macro}}$  and  $q_{M2} \approx G_{xy}^{\text{macro}}$ ). Thus, the adaptive weights can be interpreted directly as deformation-dependent quadrature weights. Figure 8 shows this interpretation along a purely compressive path, with  $G_{xy}^{\text{macro}} = 0$  and  $G_x^{\text{macro}} < 0$ . The deformed configuration of the unit cell is displayed together with the corresponding adaptive cubature weights.

### 6.5.3. Effect of graph regularization

Let us now qualitatively examine the effect of the graph regularization parameter  $\alpha_G$  on the smoothness of the resulting weight fields. To this end, Fig. 9 compares the sampled values of the third weight field for  $\alpha_G = 10^4$  (the same field previously shown in Fig. 7) and for  $\alpha_G = 0$ , i.e., in the absence of regularization. The benefits of the regularization are evident: for  $\alpha_G = 0$ , the sampled values clearly exhibit stronger local fluctuations and a less coherent evolution over the latent manifold.

A more quantitative assessment of the role of graph regularization is obtained by comparing the online errors shown in Table 3. The first row reproduces the results previously reported for the fixed-weight M-HROM in Table 1, and is included here solely to facilitate a direct comparison with the adaptive cubature rules. The displacement error  $e_d$  remains unchanged across all three cases because the displacement is reconstructed directly from the decoder using (since  $\mathbf{q}_M \approx \boldsymbol{\mu}$ ), and, therefore does not depend on the particular cubature rule employed. As for the error  $e_P$  in predicting homogenized PK1 stresses, we see that reducing the support from the original  $m_0 = 54$  integration points to only  $m = 10$  points leads to an error

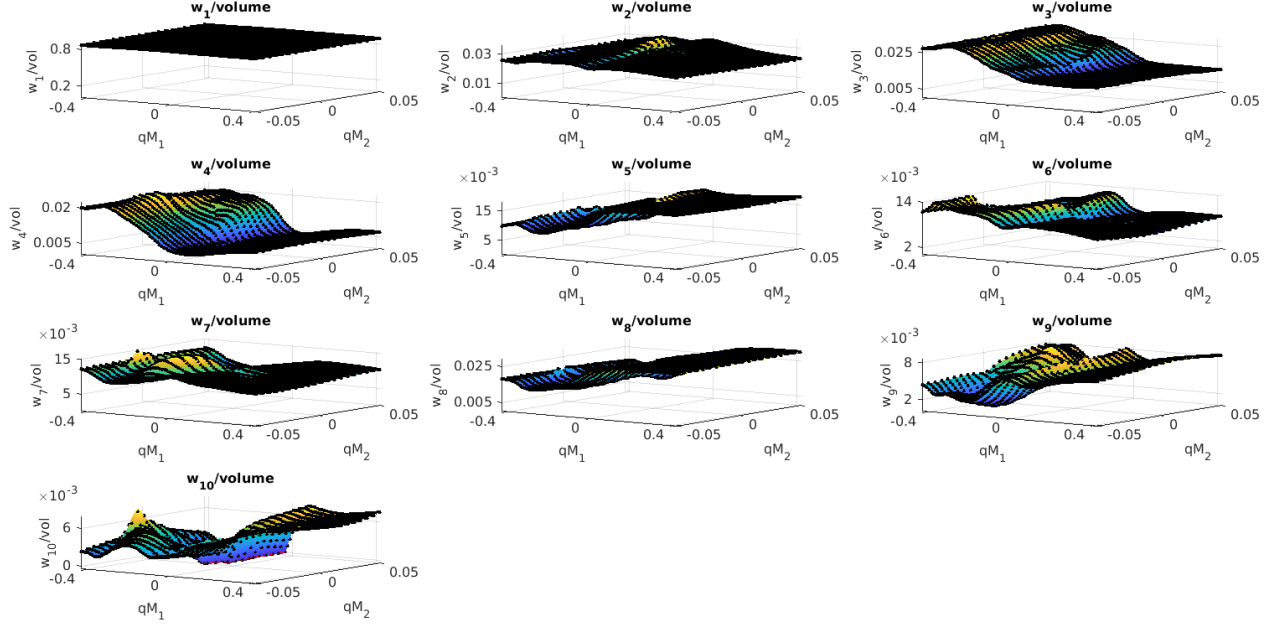


Figure 7: RBF reconstruction of the  $m = 10$  adaptive weight fields  $\omega_g = \omega_g(\mathbf{q}_M)$ ,  $g = 1, \dots, m$  (normalized by the volume  $|\Omega_0|$ ). Black dots denote the sampled manifold states used by the adaptive cubature procedure. The spatial location of the integration point associated with each weight field is shown in Fig. 6d.

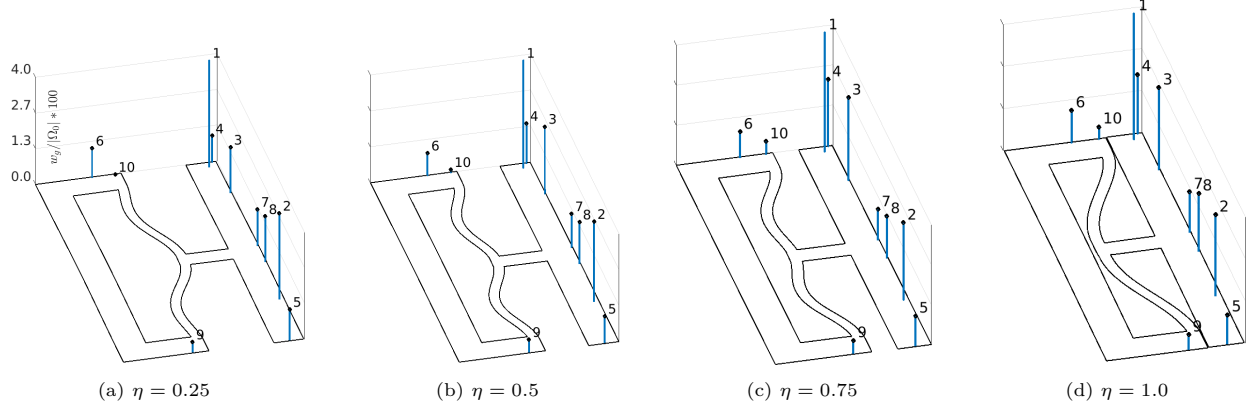


Figure 8: Evolution of the adaptive cubature weights along a purely compressive loading path ( $q_{M2} = G_{xy}^{\text{macro}} = 0$ ,  $q_{M1} = G_x^{\text{macro}} < 0$ ). The quantity  $\eta = \eta(q_{M1})$  denotes the compression level ( $\eta = 0$  for the undeformed configuration, and  $\eta = 1$  for maximum compression). Each subfigure shows the deformed configuration of the unit cell together with the location of the  $m = 10$  integration points and their associated weights, which evolve according to the regressed fields shown in Fig. 7. The height of each bar represents the percentage of the total volume carried by the corresponding integration point. To make the redistribution among the smaller weights visible, the plotted height is truncated at 4% of the total volume. Consequently, the dominant weight  $\omega_1$  is displayed only partially; its actual value is approximately 88.37%, 87.83%, 87.21%, and 86.99% for  $\eta = 0.25, 0.5, 0.75$ , and  $1.0$ , respectively.

Table 3: Influence of graph regularization on the accuracy of the adaptive cubature rule. Errors correspond to averages over the two training trajectories.

| Model                         | $m_{\text{ECM}}$ | $e_d$                | $e_P$                |
|-------------------------------|------------------|----------------------|----------------------|
| M-HROM, fixed-weight ECM      | 54               | $7.87 \cdot 10^{-5}$ | $2.98 \cdot 10^{-4}$ |
| MAW-ECM ( $\alpha_G = 10^4$ ) | 10               | $7.87 \cdot 10^{-5}$ | $7.21 \cdot 10^{-4}$ |
| MAW-ECM ( $\alpha_G = 0$ )    | 10               | $7.87 \cdot 10^{-5}$ | $5.21 \cdot 10^{-3}$ |

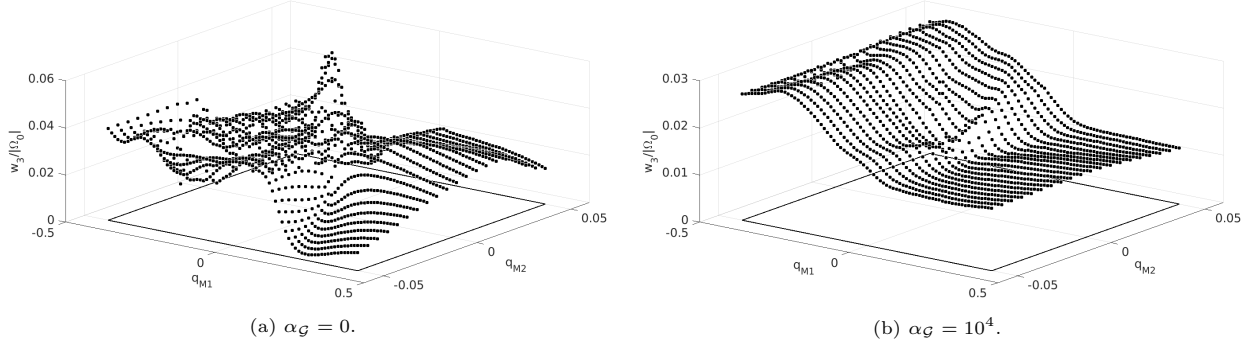


Figure 9: Sampled values of the third adaptive weight field obtained without graph regularization ( $\alpha_G = 0$ , left) and with graph regularization ( $\alpha_G = 10^4$ , right).

increase with respect to the original fixed-weight rule of  $7.21/2.98 \approx 2.4$  when the regularization is active. In the absence of regularization, this error level rises up to  $52.1/2.98 \approx 18$ .

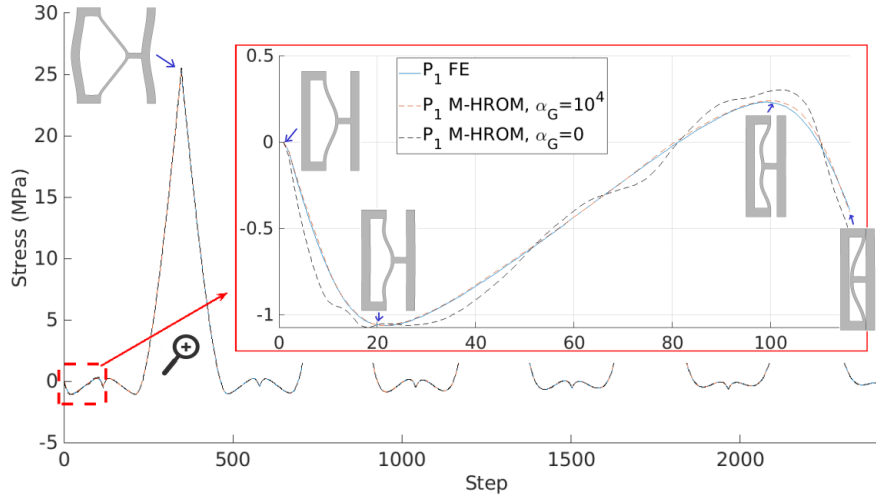


Figure 10: Comparison of the first component of the homogenized PK1 stress predicted by the FE model, the regularized MAW-ECM rule ( $\alpha_G = 10^4$ ), and the unregularized MAW-ECM rule ( $\alpha_G = 0$ ) along the second zig-zag trajectory used for training. The inset shows a magnified view of the first portion of the trajectory (steps 1–116), corresponding to the compression path previously shown in Fig. 8. The deformed configurations illustrate the progressive buckling of the two internal beams and the associated negative-stiffness response of the unit cell.

To better understand the origin of the accuracy degradation caused by the absence of regularization, we compare in Fig. 10 the evolution of the first component of the homogenized PK1 stress along the second zig-zag trajectory used for training. The inset provides a magnified view of the first 116 loading steps, corresponding to the purely compressive path previously shown in Fig. 8. The accompanying deformed configurations illustrate the progressive buckling of the two internal beams responsible for the apparent negative-stiffness response of the cell.

It can be readily seen that the deterioration associated with the unregularized rule is not uniformly distributed. Rather, it is concentrated in the compressive regime<sup>12</sup>, as can be appreciated in the enlarged view. Although the unregularized M-HROM prediction ( $\alpha_G = 0$ ) captures the overall trend of the response, it exhibits noticeable local deviations from the FE solution. By contrast, the regularized prediction remains

<sup>12</sup>It is worth noting that the global Frobenius-type error measure employed in Table 3 tends to attenuate the impact of these localized discrepancies because the unit cell is approximately twenty-five times stiffer in tension than in compression. Interestingly, this asymmetry is also reflected in Fig. 9, where the strongest departures from smooth behavior occur precisely in the compressive regime ( $q_{M1} \equiv G_x^{\text{macro}} < 0$ ).



in close agreement with the reference response throughout the entire compression path.

In summary, the preceding results evidence that graph regularization does improve the smoothness of the adaptive weight fields and, in doing so, also the quality of the resulting stress predictions.

## 7. Benchmark 2: continuum damage problem

### 7.1. Problem setting

Having examined the performance of the proposed methodology in a problem involving geometric nonlinearities and elastic material behavior, we next consider a problem in which the nonlinear response originates from inelastic constitutive effects. The benchmark selected for this purpose is the plane-strain deformation of a plate containing a circular hole under small-strain conditions, governed by a simple isotropic damage model with linear hardening. Owing to symmetry, only one quarter of the structure is modeled, as illustrated in Figure 11(a). A uniformly distributed traction is applied along the right boundary; the amplitude of such a traction, denoted by  $\mu$ , is selected as the input parameter in the problem. Displacement boundary conditions are homogeneous, and thus  $\bar{\mathbf{d}} = \mathbf{0}$  in Eq. (1); likewise  $\mathbf{T}$  in the same equation reduces to a Boolean mapping matrix from the unconstrained degrees of freedom to the full nodal space.

The output of interest is not a derived quantity, but the state variable itself, namely the unconstrained entries nodal displacement vector  $\mathbf{d}$ . Unlike the Neo-Hookean-based benchmark considered in the previous section, the constitutive memory embodied in Eq. (90) prevents the solution from being parameterized by the current value of the load factor  $\mu$  alone. Consequently, the structural response must be recovered through the solution of the reduced-order equilibrium equations.

Although the employed damage model is classical, see e.g. Ref. [43], it is convenient to briefly recall their main features, since they will later motivate several aspects of the proposed input-informed decoder. In an isotropic damage model, the constitutive response at each Gauss point is driven by a single scalar internal variable,  $r_\epsilon$ , whose evolution equation can be integrated explicitly as follows:

$$r_\epsilon(t) = \max \left\{ r_\epsilon(0), \max_{0 \leq s \leq t} \sqrt{\boldsymbol{\varepsilon}(s)^T \mathbb{C} \boldsymbol{\varepsilon}(s)} \right\}, \quad (90)$$

where  $\boldsymbol{\varepsilon}$  and  $\mathbb{C}$  denote the infinitesimal strain vector and the elasticity matrix, respectively. The damage variable  $d_\epsilon \in [0, 1]$  is then recovered as  $d_\epsilon = 1 - q_\epsilon / r_\epsilon$ , where  $q_\epsilon = H \dot{r}_\epsilon$ ,  $H > 0$  being the hardening modulus. The corresponding stress vector is finally obtained as

$$\boldsymbol{\sigma} = (1 - d_\epsilon) \mathbb{C} \boldsymbol{\varepsilon}. \quad (91)$$

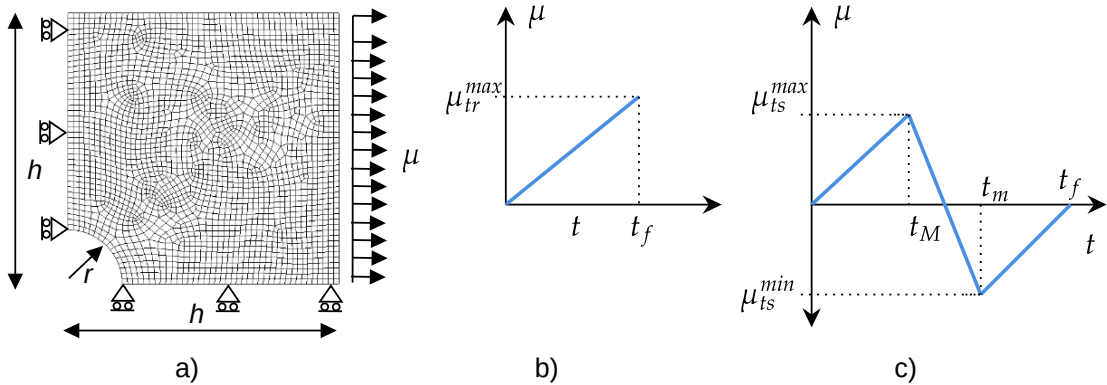


Figure 11: Damage benchmark and loading trajectories. (a) Quarter model of a plate with a circular hole ( $h = 80$  mm and  $r = 20$  mm). Symmetry conditions are imposed along the left and lower boundaries, while a uniform traction of amplitude  $\mu$  is applied on the right boundary. (b) Training trajectory used for the construction of the HROMs. (c) Testing trajectory employed for model validation, involving loading, unloading, and reloading stages.

The finite element mesh consists of  $N_{el} = 2450$  quadratic quadrilateral elements (average size 2 mm) and 9997 nodes. The material parameters are: Young's modulus  $E = 70$  GPa, Poisson ratio  $\nu = 0.3$ , damage threshold  $\sigma_c = 70$  MPa, and hardening parameter  $H = 0.01$ .

Concerning training and generalization, we adopt a deliberately parsimonious strategy: the reduced-order models considered in this work—namely the standard HROM, the manifold HROM, and the proposed manifold HROM equipped with adaptive weights—are trained exclusively on a monotonic tensile loading path (Figure 11(b)). No unloading, reloading, or load-reversal information is supplied as data<sup>13</sup>. The reduced-order models will be then challenged with a deformation history that does contain unloading and reloading stages (Figure 11(c)).

For the training stage, the loading parameter increases monotonically from zero to  $\mu^{\max} = 70$  MPa over a pseudo-time interval  $t_f = 1$  s, discretized using  $P = 1000$  increments. The generalization test is performed using the non-monotonic loading history shown in Figure 11(c), with  $\mu_{ts}^{\max} = 0.9\mu^{\max}$  s,  $\mu_{ts}^{\min} = -0.95\mu^{\max}$  s,  $t_f = 1$  s  $t_M = 0.45t_f$ ,  $t_m = 0.9t_f$  and 1500 increments.

## 7.2. Standard HROM

We begin the assessment by constructing the standard HROM (as pointed out in Section 6.3, this model is recovered by setting  $\mathbf{q}_M = \mathbf{q}$  and  $\boldsymbol{\tau} = \mathbf{q}$  in the manifold HROM of Section 3.1). As customary, the unconstrained displacement snapshots are stored in the matrix  $\mathbf{D} \in \mathbb{R}^{N \times P}$ , where, in this case,  $N = 19832$ . The reduced basis  $\boldsymbol{\Phi}$  is then sought as a linear combination of the columns of  $\mathbf{D}$ . In contrast to the metamaterial example, however,  $\boldsymbol{\Phi}$  is not obtained directly from the singular value decomposition of  $\mathbf{D}$ . Instead, we follow the procedure introduced by the first author in Ref. [44] for elastoplastic problems, and herein adapted to the present damage setting, whereby the displacement vector is decomposed into a linear reference contribution and an orthogonal nonlinear correction:

$$\mathbf{d} = \mathbf{d}_L^{lin} + \mathbf{d}_L^{non}, \quad \text{where} \quad \mathbf{d}_L^{linT} \mathbf{d}_L^{non} = 0. \quad (92)$$

The component  $\mathbf{d}_L^{lin}$  represents the projection of the solution onto the undamaged subspace, whereas  $\mathbf{d}_L^{non}$  contains the remaining information associated with damage. Separate reduced bases are then constructed for each contribution. This requires distinguishing between the undamaged and damaged portions of the training response. Accordingly, the displacement snapshots are partitioned into two submatrices:

$$\mathbf{D} = [\mathbf{D}^{lin} \quad \mathbf{D}^{non}] \quad (93)$$

corresponding to the undamaged regime,  $\mathbf{D}^{lin}$ , and the subsequent nonlinear regime,  $\mathbf{D}^{non} \in \mathbb{R}^{N \times \hat{P}}$  (for this case,  $\hat{P} = 678$ ). The snapshots in  $\mathbf{D}^{lin}$  are compressed without truncation, yielding the orthonormal basis  $\boldsymbol{\Phi}^{lin}$ , which in the present case consists of a single mode (because there is only parameter defining the input loads). The remaining snapshots  $\mathbf{D}^{non}$  are then projected onto the orthogonal complement of  $\boldsymbol{\Phi}^{lin}$ , producing the residual matrix  $\tilde{\mathbf{D}}_L^{non}$ . A truncated singular value decomposition is subsequently applied to  $\tilde{\mathbf{D}}_L^{non}$ , and the corresponding left singular vectors define the basis  $\boldsymbol{\Phi}^{non}$ . Hence,

$$\mathbf{d}_L^{lin} = \boldsymbol{\Phi}^{lin} q_M^{lin}, \quad \mathbf{d}_L^{non} \approx \boldsymbol{\Phi}^{non} \mathbf{q}^{non}. \quad (94)$$

where  $q_M^{lin} \in \mathbb{R}$  and  $\mathbf{q}^{non} \in \mathbb{R}^{n_\Phi - 1}$  are the corresponding linear and nonlinear amplitudes.

Using a relative truncation tolerance  $\varepsilon^D = 10^{-4}$ , this procedure yields 15 damage-correction modes. The resulting reduced basis and associated generalized coordinates  $\mathbf{q}$  are therefore

$$\boldsymbol{\Phi} = [\boldsymbol{\Phi}^{lin} \quad \boldsymbol{\Phi}^{non}], \quad \mathbf{q} = \begin{bmatrix} q_M^{lin} \\ \mathbf{q}^{non} \end{bmatrix}, \quad \mathbf{q} = \boldsymbol{\Phi}^T \mathbf{d}, \quad (95)$$

with a total dimension of  $n_\Phi = 16$ . Note that, by construction,

$$\boldsymbol{\Phi}^{linT} \boldsymbol{\Phi}^{non} = \mathbf{0}, \quad \boldsymbol{\Phi}^{linT} \boldsymbol{\Phi}^{lin} = \mathbf{I}, \quad \boldsymbol{\Phi}^{nonT} \boldsymbol{\Phi}^{non} = \mathbf{I}. \quad (96)$$

<sup>13</sup>By stating that no unloading, reloading, or load-reversal information is supplied as data, we mean that such trajectories are not included among the training snapshots. Naturally, this deliberate choice relies on prior knowledge of the problem; for instance, the assumptions of small strains and isotropic damage tells us that the model is symmetric with changes of the sign of  $\mu$ , so there is no need to include compression data.

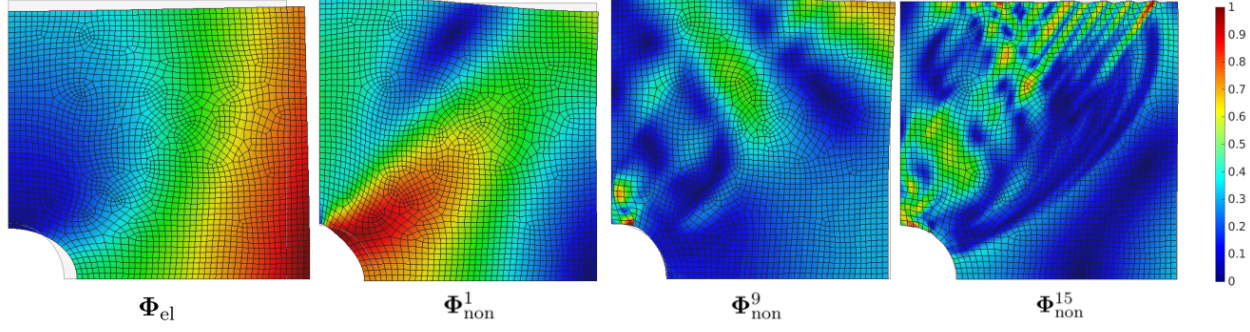


Figure 12: Elastic mode  $\Phi^{\text{lin}} \in \mathbb{R}^N$  and three representative nonlinear modes from  $\Phi^{\text{non}} \in \mathbb{R}^{N \times n_\Phi - 1}$  ( $N = 19832$ ,  $n_\Phi = 16$ ), obtained from the displacement snapshots generated along the monotonic training trajectory shown in Figure 11(b). The deformed configurations are displayed together with contour plots of the Euclidean norm, normalized by its maximum value.

By way of illustration, we show in Figure 12 the linear mode  $\Phi^{\text{lin}}$  together with three representative modes from  $\Phi^{\text{non}}$ .

We next construct the standard fixed-weight ECM rule following Sections 3.4 and 3.5. The only quantity entering the cubature training matrix are the projected internal-force densities; the size of this matrix in this case is thus  $M_{\text{gp}} \cdot n_\Phi P$ , where  $M_{\text{gp}} = 2450 \cdot 9 = 22050$  and  $n_\Phi P = 16 \cdot 1000 = 16000$ . Using the SRSVD with  $\varepsilon^A = 10^{-5}$ , we get 2369 left singular vectors. Augmenting this basis with the volume constraint in Eq. 49, the ECM selects 2370 Gauss points distributed over 1140 finite elements; see Fig. 13(b).

Despite the apparent simplicity of the problem —driven by the history of a single input parameter—, the resulting reduction in terms of number of integration points is rather low (from  $M_{\text{gp}} = 22050$  to 2370, around 10.7%). The explanation behind this poor reduction may lie in the fact that the growth of damage produces an evolving nonlinear region whose spatial support changes throughout the loading process. From the standpoint of linear approximation, this is analogous to the difficulties encountered in transport- and propagation-dominated problems, where the movement or evolution of localized features is known to result in a slow decay of singular values (see e.g. Refs. [3, 4]). The contour plot of the damage variable  $d_\epsilon$  at the end of the loading process and the spatial distribution of the resulting ECM points, shown in Figures 13(a)–(b), corroborate this interpretation, as most selected Gauss points cluster in the region where damage is most severe.

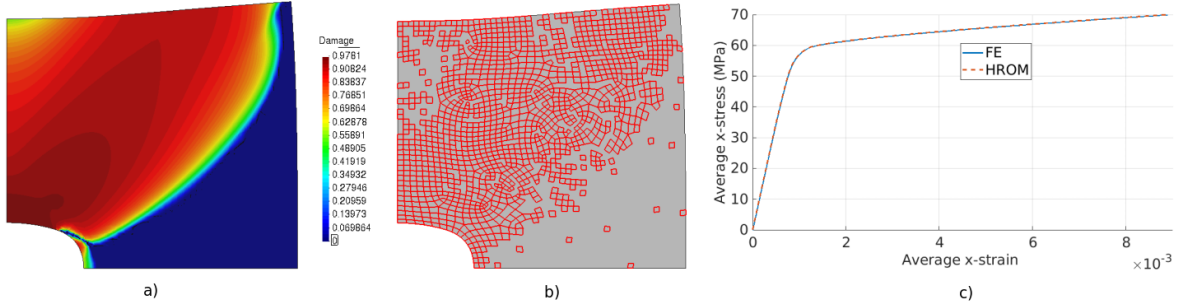


Figure 13: Standard HROM for the damage benchmark. (a) Damage field at the maximum training load. (b) Elements containing the 2370 Gauss points selected by the fixed ECM rule. (c) FE and HROM average stress–strain responses along the monotonic training trajectory.

The accuracy of the resulting HROM is assessed using the relative Frobenius displacement error  $e_d$ , defined as in Section 6.4.3. On the training trajectory, the standard HROM gives  $e_d = 1.38 \cdot 10^{-4}$ . The agreement between the FE and HROM responses can be further appreciated in Figure 13(c), where we plot the FE and HROM average stress–strain curves in the loading direction. The average stress is computed by dividing the applied force  $\mu$  by the length of the loaded boundary  $h$ , while the average strain is obtained from the predicted mean horizontal displacement of the right boundary. This graph also helps to identify the main stages of the response: an initial elastic branch is followed by a relatively sharp transition associated with

damage initiation and, subsequently, by a nonlinear hardening regime.

As for the generalization test of Figure 11(c), the relative displacement error remains of the same order, namely  $e_d = 1.28 \cdot 10^{-4}$ . This result corroborates our initial assumption that, for the present benchmark, training on the monotonic tensile branch is sufficient to capture the subsequent unloading and reloading behavior, including the (unseen) compression regime. The distinct stages of the cycle are illustrated in the average stress-strain graph of Figure 14.(a), and in the evolution of the amplitudes of the three first modes, in Figure 14(b).

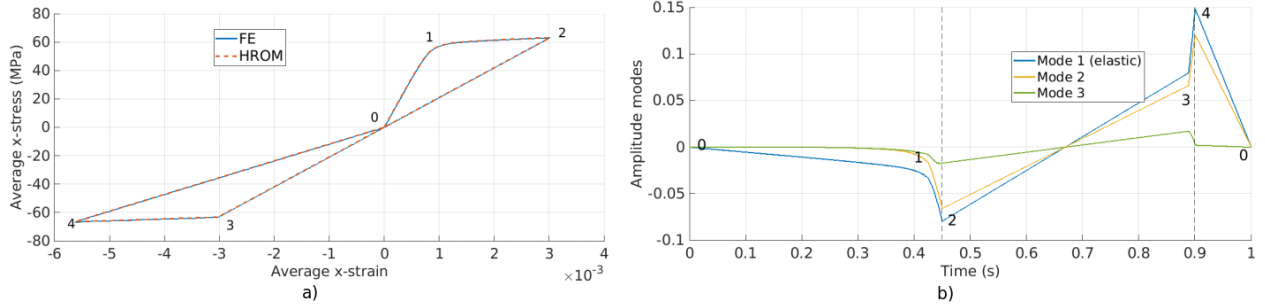


Figure 14: Generalization test under cyclic loading. (a) Average stress-strain response obtained during the loading cycle. Comparison between FE and standard HROM. The labels indicate the successive stages of the loading history: primary tensile loading (0–2), unloading from the tensile state (2–3), reverse loading into compression (3–4), and unloading from the compressive state (4–0). (b) Evolution of the amplitudes corresponding to the first three modes  $\Phi_1 = \Phi^{\text{lin}}$ ,  $\Phi_2$  and  $\Phi_3$ .

### 7.3. Manifold HROM

#### 7.3.1. Identification of latent variables

We now turn to the manifold HROM. We begin with the construction of the input-informed decoder (as discussed in Section 1.4), for which we adopt the same master-slave structure introduced in Eq. 22, but adapted to the linear/nonlinear decomposition of Eq. 92. The latent coordinate  $q_M^{\text{lin}}$  and the corresponding basis matrix  $\Phi^{\text{lin}}$  for the linear component  $\mathbf{d}_L^{\text{lin}}$  have already been identified. It remains to determine how many additional latent coordinates are required to represent the nonlinear component  $\mathbf{d}_L^{\text{non}}$ , how to identify them by exploiting the available loading information, and, finally, how to construct the map from the resulting master coordinates to the slave coordinates,  $\mathbf{q}_S = \mathcal{N}_S(\mathbf{q}_M)$ .

The constitutive response at each Gauss point is governed by a single scalar internal variable,  $r_\epsilon$ , defined in Eq. (90). This feature, together with the fact that the evolution of these local variables along the training trajectory is ultimately induced by the single scalar input  $\mu$ , suggests that one additional latent coordinate may suffice to parameterize the nonlinear component  $\mathbf{d}_L^{\text{non}}$ . We denote this additional coordinate by  $q_M^{\text{non}}$ . In accordance with the hypothesis embodied in Eq. 21, we seek  $q_M^{\text{non}}$  as a linear combination of the nonlinear amplitudes  $\mathbf{q}^{\text{non}} = \Phi^{\text{non}T} \mathbf{d}$ , that is:

$$q_M^{\text{non}} = \mathbf{H}_M^{\text{non}} \mathbf{q}^{\text{non}}. \quad (97)$$

Thus, the problem of identifying  $q_M^{\text{non}}$  amounts to determining the row matrix  $\mathbf{H}_M^{\text{non}} \in \mathbb{R}^{1 \times (n_\Phi - 1)}$ . It is worth noting that, with the above choice, the transformation matrix introduced in Eq. 21 adopts the form

$$\mathbf{H}_M = \begin{bmatrix} 1 & \mathbf{0} \\ 0 & \mathbf{H}_M^{\text{non}} \end{bmatrix}. \quad (98)$$

The conditions used to determine  $\mathbf{H}_M^{\text{non}}$  will not be imposed directly on Eq. 97, but on its normalized counterpart, obtained by dividing both sides of that equation by  $q_M^{\text{lin}}$ :

$$\hat{q}_M^{\text{non}} = \mathbf{H}_M^{\text{non}} \hat{\mathbf{q}}^{\text{non}}, \quad (99)$$

where

$$\hat{q}_M^{\text{non}} := \frac{q_M^{\text{non}}}{q_M^{\text{lin}}}, \quad \hat{\mathbf{q}}^{\text{non}} := \frac{\mathbf{q}^{\text{non}}}{q_M^{\text{lin}}}, \quad q_M^{\text{lin}} \neq 0. \quad (100)$$

The rationale behind this normalization is that these normalized variables remain unchanged during purely elastic loading or unloading and evolve only when damage develops at one or more Gauss points. The normalization therefore filters out the reversible dependence on the loading amplitude and isolates changes associated with the evolution of damage. This property follows directly from the nature of the employed damage model. Whenever the internal variable  $r_\epsilon$  in Eq. (90), and consequently the damage variable  $d_\epsilon$ , remain frozen at all Gauss points, the structure responds elastically with the fixed degraded stiffness implied by Eq. 91. The reduced amplitudes are then proportional to the input force  $\mu$  and vanish at the same zero-load state (as can be appreciated in Fig. 14(b), between states 2 and 3) ; hence, the ratios defined in Eq. 100 remain constant.

Having clarified the role of the normalization, we now turn to the determination of  $\mathbf{H}_M^{\text{non}}$ . We restrict attention to the portion of the training trajectory over which damage evolves, represented by the nonlinear snapshot block  $\mathbf{D}^{\text{non}}$ , see Eq. 93, whose columns corresponds to the ordered load levels

$$\mu_1^{\text{non}} < \mu_2^{\text{non}} < \dots < \mu_P^{\text{non}}, \quad (101)$$

where  $\mu_1^{\text{non}}$  marks the onset of damage, and  $\mu_P^{\text{non}} = \mu^{\text{max}}$  is the maximum load. The desired latent coordinate  $\hat{\mathbf{q}}_M^{\text{non}} = \mathbf{H}_M^{\text{non}} \hat{\mathbf{q}}^{\text{non}}$  must parameterize this one-dimensional trajectory so that every component of  $\hat{\mathbf{q}}^{\text{non}}$  can be expressed as a single-valued function of  $\hat{q}_M^{\text{non}}$ . This requires  $\hat{q}_M^{\text{non}}$  to be injective with respect to the applied load  $\mu$ , which in this case amounts to require strict monotonicity of  $\hat{q}_M^{\text{non}}$  with respect to  $\mu$ .

Monotonicity is the desired outcome, but not the quantity that we enforce directly. Instead, we seek  $\mathbf{H}_M^{\text{non}}$  as the coefficients associated with the smoothest linear combination, on the grounds that oscillatory coordinates are incompatible with the monotonicity requirement. The resulting optimization problem leads to the generalized eigenvalue problem

$$(\hat{\mathbf{Q}}_{\text{non}} \hat{\mathbf{K}}_{\mathcal{G}} \hat{\mathbf{Q}}_{\text{non}}^T) \mathbf{v}_i = \lambda_i (\hat{\mathbf{Q}}_{\text{non}} \hat{\mathbf{Q}}_{\text{non}}^T) \mathbf{v}_i, \quad i = 1, \dots, n_\Phi - 1. \quad (102)$$

Here,  $\hat{\mathbf{K}}_{\mathcal{G}} \in \mathbb{R}^{\hat{P} \times \hat{P}}$  denotes the graph-Laplacian operator associated with the sequence of loads  $\{\mu_i^{\text{non}}\}_{i=1}^{\hat{P}}$ , whereas  $\hat{\mathbf{Q}}_{\text{non}} \in \mathbb{R}^{(n_\Phi-1) \times \hat{P}}$  is obtained by dividing each column of  $\mathbf{Q}_{\text{non}} = \Phi^{\text{non}T} \mathbf{D}^{\text{non}}$  by the corresponding entries of  $\mathbf{Q}_{\text{lin}} = \Phi_M^{\text{lin}T} \mathbf{D}^{\text{non}}$ . The desired coefficients are finally chosen as

$$\mathbf{H}_M^{\text{non}} = \frac{\mathbf{v}_1^T}{\|\Phi^{\text{non}} \mathbf{v}_1\|}, \quad (103)$$

where  $\mathbf{v}_1$  is the eigenvector associated with the smallest eigenvalue. The normalization by  $\|\Phi^{\text{non}} \mathbf{v}_1\|$  ensures that  $\mathbf{Z}_M = \mathbf{I}$  in Eq. 22.

**Remark 3.** *The eigenvalue problem 102 is a standard construction in spectral graph theory; see, e.g., [45]. In the present problem, it arises from interpreting the ordered load samples  $\{\mu_i^{\text{non}}\}_{i=1}^{\hat{P}}$  as the nodes of a one-dimensional graph, and the rows of  $\hat{\mathbf{Q}}_{\text{non}}$  as scalar fields defined on those nodes. Within this interpretation, the quadratic form appearing in the left-hand side of Eq. 102 represents the graph-Dirichlet energy (the same energy introduced when dealing with the regularization of the MAW-ECM in section 5.1.1), whereas the quadratic form in the right-hand side provides a measure of its magnitude. The objective is therefore to find the linear combination with minimum graph-Dirichlet energy among all linear combinations of unit magnitude. This constrained minimization problem takes the form of a Rayleigh quotient [45], whose optimality conditions are precisely Eq. 102. In the present one-dimensional setting, the graph-Laplacian operator  $\hat{\mathbf{K}}_{\mathcal{G}}$  can be interpreted as the stiffness (Laplacian) matrix of a one-dimensional finite-element mesh whose nodes coincide with the ordered load samples.*

Figure 15 provides an a posteriori illustration of the identification. The components of  $\hat{\mathbf{q}}^{\text{non}}$  shown in Fig. 15a fail to satisfy the monotonicity requirement and are therefore unsuitable as scalar latent coordinates (although not shown, the remaining components also fail to satisfy the monotonicity requirement). By contrast, the linear combination associated with the smallest generalized eigenvalue, shown in Fig. 15b, exhibits the desired monotone evolution. Thus, for the present benchmark, the adopted smoothness criterion successfully yields an injective parametrization of the normalized nonlinear trajectory.

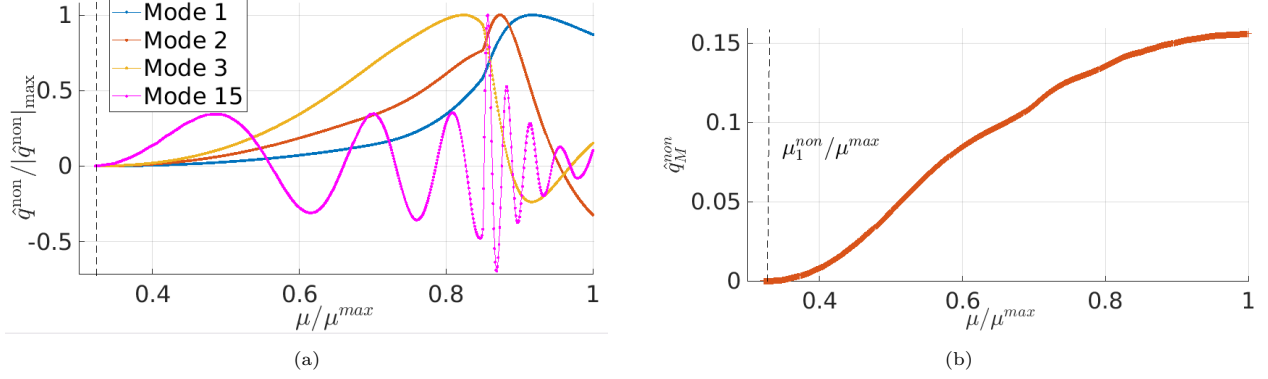


Figure 15: Identification of the nonlinear latent coordinate for the damage benchmark. The horizontal axis represents the normalized load parameter  $\mu/\mu^{\text{max}}$  for the nonlinear portion of the training tensile test (see expression 101). (a) Components 1, 2, 3, and 15 of the normalized nonlinear coordinate vector  $\hat{\mathbf{q}}_M^{\text{non}} \in \mathbb{R}^{n_\Phi-1}$ , where  $n_\Phi = 16$ ; see Eq. 100. (b) Normalized latent variable  $\hat{q}_M^{\text{non}}$  obtained as a linear combination of  $\hat{\mathbf{q}}_M^{\text{non}}$  ( $\hat{q}_M^{\text{non}} = \mathbf{H}_M^{\text{non}} \hat{\mathbf{q}}_M^{\text{non}}$ ). The matrix of coefficients in this linear combination,  $\mathbf{H}_M^{\text{non}}$ , is determined by solving the generalized eigenvalue problem (102).

### 7.3.2. Nonlinear closure

Consistently with the normalization introduced above, we express the closure map  $\mathcal{N}_S = \mathcal{N}_S(q_M^{\text{lin}}, q_M^{\text{non}})$  appearing in the decoder of Eq. 22 in the multiplicative form

$$\mathcal{N}_S(q_M^{\text{lin}}, q_M^{\text{non}}) = q_M^{\text{lin}} \tilde{\mathcal{N}}_S(\hat{q}_M^{\text{non}}). \quad (104)$$

The map  $\tilde{\mathcal{N}}_S = \tilde{\mathcal{N}}_S(\hat{q}_M^{\text{non}})$  is constructed from the input–output pairs  $\{\hat{q}_M^{\text{non}}(\mu_j^{\text{non}}), \mathbf{g}_S^{\text{non}}(\mu_j^{\text{non}})\}_{j=1}^{\hat{P}}$ , where  $\mathbf{g}_S^{\text{non}}(\mu_j^{\text{non}})$  contains the normalized slave amplitudes:

$$\mathbf{g}_S^{\text{non}}(\mu_j^{\text{non}}) = \frac{\Phi_S^T \mathbf{d}(\mu_j^{\text{non}})}{q_M^{\text{lin}}(\mu_j^{\text{non}})}. \quad (105)$$

The condition  $q_M^{\text{lin}} = 0$  occurs at zero applied load. Apart from the initial state, it is encountered when the load passes through zero during unloading. Since  $\hat{q}_M^{\text{non}}$  remains constant throughout unloading, we resolve the resulting indeterminacy by retaining its value from the previous converged increment:

$$\hat{q}_M^{\text{non}}(t_{n+1}) = \hat{q}_M^{\text{non}}(t_n), \quad \text{if} \quad q_M^{\text{lin}}(t_{n+1}) = 0, \quad (106)$$

with  $\hat{q}_M^{\text{non}}(0) = 0$ , where  $t_n$  denotes the pseudo-time of the  $n$ -th load increment.

With the multiplicative closure in Eq. 104, the generic coefficient map in Eq. 27 specializes to

$$\boldsymbol{\tau}(\mathbf{q}_M) = \begin{bmatrix} q_M^{\text{lin}} & q_M^{\text{non}} & q_M^{\text{lin}} \tilde{\mathcal{N}}_S(\hat{q}_M^{\text{non}})^T \end{bmatrix}^T, \quad (107)$$

where  $\hat{q}_M^{\text{non}} = q_M^{\text{non}}/q_M^{\text{lin}}$  for  $q_M^{\text{lin}} \neq 0$ , with its value at  $q_M^{\text{lin}} = 0$  defined by Eq. 106. Its Jacobian, see Eq. 28, follows directly from the chain rule:

$$\mathbf{J}_\tau = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ \tilde{\mathcal{N}}_S(\hat{q}_M^{\text{non}}) - \hat{q}_M^{\text{non}} \tilde{\mathcal{N}}_S'(\hat{q}_M^{\text{non}}) & \tilde{\mathcal{N}}_S'(\hat{q}_M^{\text{non}}) \end{bmatrix}. \quad (108)$$

Interestingly, the dependence on the master coordinates enters only through the normalized coordinate  $\hat{q}_M^{\text{non}}$ . Hence, the tangent basis  $\Phi_{\mathcal{D}} = \Phi \mathbf{J}_\tau$ , defined in Eq. 33 and used to project the equilibrium residual, see Eq. 32, remains unchanged during elastic unloading and reloading at frozen damage, since we have argued that  $\hat{q}_M^{\text{non}}$  remains constant along these branches.

Figure 16 displays four representative normalized slave amplitudes,  $g_{S,i}^{\text{non}}$  ( $i = 1, 2, 3, 14$ ), as functions of  $\hat{q}_M^{\text{non}}$ . They vary smoothly over most of the training range, with localized oscillations and kinks appearing only

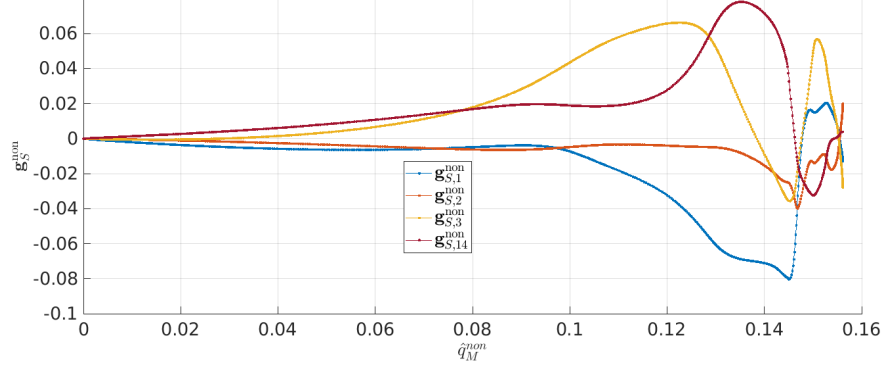


Figure 16: Normalized slave amplitudes  $\mathbf{g}_S^{\text{non}}$ , defined in Eq. 105, as functions of the normalized latent coordinate  $\hat{q}_M^{\text{non}}$ . This dataset is used for regressing the mapping component  $\tilde{\mathcal{N}}_S = \tilde{\mathcal{N}}_S(\hat{q}_M^{\text{non}})$  in Eq. 104 .

near the final, strongly damaged states. To prevent these irregularities from producing oscillatory decoder derivatives in Eqs. 108, we fit  $\tilde{\mathcal{N}}_S$  using 100 uniformly distributed samples from the  $\hat{P} = 678$  snapshots and cubic least-squares B-splines with 90 knots. The resulting smoothing error will be assessed in the following subsection (when discussing Table 4). Outside the training interval, the spline is extended by a quadratic Taylor expansion at the corresponding endpoint.

### 7.3.3. Fixed-weight ECM

Having defined the latent coordinates and the associated slave closure, we now construct the fixed-weight ECM rule of Section 3.4. Since the present benchmark involves no volumetric output of interest, we adopt the equilibrium-only case of Eq. 40, for which  $\mathbf{r}_g = \mathbf{r}_g^D$  and  $n_c = n = 2$ . Accordingly, the training matrix  $\mathbf{A}$  of Eq. 42 contains only the projected internal-force densities and has dimensions  $\mathbf{A} \in \mathbb{R}^{22050 \times 2000}$ . Using the SRSVD with  $\varepsilon^A = 10^{-5}$ , and augmenting the resulting basis with the volume constraint in Eq. 49, the ECM yields a fixed rule with  $m_0 = 277$  Gauss points distributed over 213 finite elements, see Fig. 17(a).

Table 4 compares the online complexity and accuracy of the resulting manifold HROM with those of the standard HROM. The errors are the relative Frobenius displacement errors over the monotonic training trajectory and the cyclic generalization trajectory, respectively.

Table 4: Online complexity and displacement accuracy of the fixed-weight manifold HROM for the damage benchmark, together with the standard HROM.

| Model             | Master coords. | Slave coords. | Integ. points | $(e_d)^{\text{train}}$ | $(e_d)^{\text{test}}$ |
|-------------------|----------------|---------------|---------------|------------------------|-----------------------|
| HROM, ECM         | 16             | 0             | 2370          | $1.38 \cdot 10^{-4}$   | $1.28 \cdot 10^{-4}$  |
| M-HROM, fixed ECM | 2              | 14            | 277           | $1.78 \cdot 10^{-3}$   | $1.81 \cdot 10^{-3}$  |

The manifold HROM reduces the number of independent coordinates from  $n_\Phi = 16$  to  $n = 2$ , and the number of integration points from 2370 to 277. As in the first benchmark, the corresponding compression factors are nearly identical, namely 8 and approximately 8.6, respectively.

On the other hand, according to Table 4, the displacement errors increase by approximately one order of magnitude relative to the standard HROM, but remain nearly identical over the training and generalization trajectories. These results indicate that the dominant additional error arises from the smoothing introduced in the closure regression discussed in Section 7.3.2, rather than from a loss of generalization over the unseen cyclic trajectory. Moreover, quadratic Newton convergence was observed at every load increment.

Figure 17(b) further compares the latent coordinates predicted by the manifold HROM along the generalization trajectory with those obtained by applying the encoder in Eq. 29 to the FE solutions. The linear coordinates are virtually indistinguishable, whereas  $\hat{q}_M^{\text{non}}$  exhibits only minor discrepancies near the onset of damage.

**Remark 4.** Although introduced solely to parameterize the nonlinear closure in Eq. 104,  $\hat{q}_M^{\text{non}}$  exhibits in Figure 17(b) a behavior that admits a compelling physical interpretation: as anticipated, it remains constant



during elastic unloading and reloading and, furthermore, increases steadily as damage progresses. This is the same behavior exhibited by the variable  $r_\epsilon$  at the Gauss-point level; see Eq. 90. Thus, in seeking a maximally compressed parametrization of the displacement manifold, we have identified the structural-scale counterpart of the local internal variable  $r_\epsilon$ .

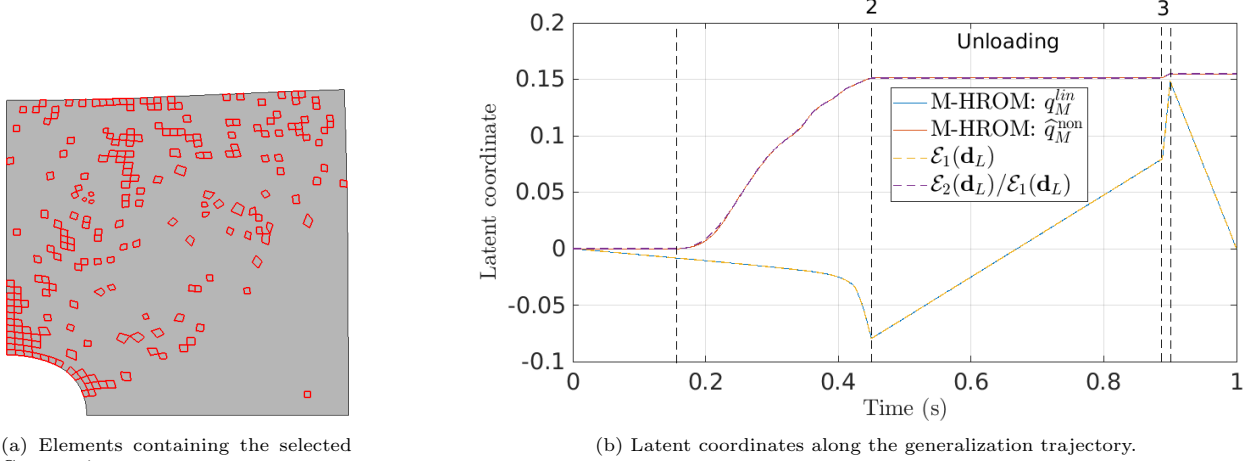


Figure 17: Fixed-weight manifold HROM for the damage benchmark. (a) Elements containing the 277 Gauss points selected by the ECM. (b) Evolution of  $q_M^{lin}$  and  $\hat{q}_M^{non}$  along the generalization trajectory of Fig. 11(c), compared with the coordinates obtained by applying the encoder in Eq. 29 to the FE solutions.

#### 7.4. Manifold-adaptive weight ECM

We now apply the MAW–ECM methodology summarized in Box 5.1 to the manifold HROM of the present damage benchmark. Consistently with the normalized manifold parametrization, we choose the adaptive weights to depend only on  $\hat{q}_M^{non}$ :

$$\omega_g = \omega_g(\hat{q}_M^{non}), \quad g \in \mathcal{Z}_\omega. \quad (109)$$

This choice assigns the same weight vector to all states of the virgin elastic branch, for which  $\hat{q}_M^{non} = 0$  while  $q_M^{lin}$  varies. The adaptive rule must therefore reproduce exactly the projected internal forces over this branch. To enforce this requirement, we partition the training matrix into its elastic and nonlinear blocks,  $\mathbf{A} = [\mathbf{A}^{lin} \quad \mathbf{A}^{non}]$ , analogously to the displacement-snapshot decomposition in Eq. 93. Let  $\mathbf{U}_{el}$  denote a basis for the column space of  $\mathbf{A}^{lin}$ , which, owing to the linearity of the response in this regime, has dimension  $n = 2$ . We include this basis among the invariant constraints of Eq. 58:

$$\mathbf{U}_{inv} = [\mathbf{U}_{el}, \mathbf{1}]. \quad (110)$$

The nonlinear block  $\mathbf{A}^{non} \in \mathbb{R}^{M_{gp} \times \hat{P}n}$  supplies the state-dependent conditions along the damaged portion of the training trajectory. In summary, at each sampled value  $\hat{q}_M^{non}(\mu_j^{non})$ , the local system comprises the  $n_c = n = 2$  conditions associated with the projected internal forces, the  $n = 2$  elastic invariants in  $\mathbf{U}_{el}$ , and the volume constraint. Hence, the lower bound defined in Eq. 73 becomes

$$m_{min} = n_c + n_{inv} = n + n + 1 = 5. \quad (111)$$

In other words, the desired adaptive rule satisfying the local constraints must contain at least  $m_{min} = 5$  integration points.

**Remark 5.** A key consequence of the proposed parametrization for the adaptive weights is that no additional constraints are required to describe unloading. Indeed, since  $\hat{q}_M^{non}$  remains constant during unloading, the Jacobian matrix  $\mathbf{J}_\tau$  of Eq. (108) remains constant as well. Furthermore, according to Eq. (107), the decoder is linear with respect to  $q_M^{lin}$  when  $\hat{q}_M^{non}$  is fixed. Hence, all states belonging to the same unloading branch differ only through a scalar scaling factor and, as a result, enforcing the local integration constraints at a given damage state of the training set automatically guarantees the same level of integration accuracy along any unloading trajectory emanating from that state.



#### 7.4.1. Influence of the fixed-weight initialization and pruning efficiency

We now proceed with the assessment of the efficiency of the pruning process of Algorithm 3 for different initializations, namely three initial cubature rules containing  $m_0 = 97$ ,  $m_0 = 277$ , and  $m_0 = 534$  integration points, respectively. These rules are obtained by varying the truncation tolerance  $\varepsilon^A$  in Eq. 45, applied to the internal-force training matrix  $\mathbf{A}$ . The initial rule displayed previously in Figure 17a corresponds to the intermediate case,  $\varepsilon^A = 10^{-5}$ , yielding  $m_0 = 277$  integration points. The ordered samples  $\{\hat{q}_M^{\text{non}}(\mu_j^{\text{non}})\}_{j=1}^{P_M}$  ( $P_M = \hat{P}$ ) define the nodes of a one-dimensional latent mesh. Accordingly, the graph operator  $\mathcal{K}_G$  entering the pruning algorithm is assembled from the standard finite element discretization of the one-dimensional Laplacian using linear shape functions, analogously to the construction of  $\hat{\mathcal{K}}_G$  in Eq. 102. The graph-regularization parameter is fixed at  $\alpha_G = 0.1$ , and the pruning algorithm performs  $n_{\text{try}} = 5$  trials per branch.

Table 5: Influence of the fixed-weight initialization on the MAW-ECM pruning process for different values of the SVD truncation tolerance  $\varepsilon^A$  employed in the construction of the internal-force matrix  $\mathbf{A}_D$ . Here,  $m_0$  denotes the number of points of the initial fixed-weight ECM rule, whereas *Unregularized pruning* reports the percentage of integration points eliminated without activating graph regularization or explicit positivity enforcement;  $m$  denotes the final number of integration points, and *Total time* corresponds to the wall-clock time of the complete MAW-ECM pruning algorithm.

| $\varepsilon^A$ | $m_0$ | Unregularized pruning (%) | $m$ | Total time (s) |
|-----------------|-------|---------------------------|-----|----------------|
| $10^{-4}$       | 97    | 80.4                      | 8   | 46.1           |
| $10^{-5}$       | 277   | 87.0                      | 7   | 131.0          |
| $10^{-6}$       | 534   | 96.4                      | 7   | 63.9           |

Table 5 summarizes the results of the three pruning tests. Despite the large variability in the size of the initial fixed-weight rules, all three pruning processes converge to adaptive cubature rules containing only  $m = 7$ –8 integration points, i.e., just two or three points above the theoretical lower bound  $m_{\min} = 5$  established in Eq. (111). This is essentially the same behavior observed in the metamaterial benchmark (Table 2) and corroborates that the number of integration points is dictated primarily by the intrinsic dimension of the solution manifold, rather than by the size of the initial fixed-weight cubature rule.

Table 5 also confirms another trend observed in the metamaterial benchmark discussed in Section 6.5.1: graph regularization and explicit positivity enforcement are required only during a relatively small fraction of the pruning iterations. Indeed, in the three studied cases, between 80% and 96% of the integration points are eliminated without activating the positivity-enforcement stage of Algorithm 1. It is worth noting that the tests also revealed an intermittent activation of the regularization stage, with the pruning algorithm switching between regularized and unregularized iterations before reaching the final rule. Such behavior was not observed in the metamaterial benchmark, where the first activation of the positivity-enforcement stage invariably led to a permanent transition to the regularized regime.

By way of illustration, Figure 18 displays, in the top row, the finite elements containing the integration points of the initial fixed-weight rules considered in Table 5; the lower row, on the other hand, shows the location of the integration points of the final sparsest rule, superposed on the contour plot of the damage field predicted by the FE model at the final step of the training trajectory. It is interesting to note that the spatial distribution of the retained points exhibits, in all three cases, a remarkably similar pattern: one point is invariably located in the undamaged region (point 1 in Fig. 18(b)); three or four points are placed in the vicinity of the hole boundary, where damage nucleates and the stress concentration is highest (points 3, 4, 6, and 7 in Fig. 18(b)); and the other two or three points are located within the damaged region, but away from the hole.

On the other hand, Figure 19(a) displays the sampled adaptive weights  $\{(\mathbf{W})_{g,j}\}_{j=1}^P$ ,  $g = 1, 2, \dots, m$  (see Eq. (80)) computed by the MAW-ECM for the case  $m_0 = 277$ . The weights are represented as functions of the corresponding latent states  $\{\hat{q}_M^{\text{non}}(\mu_j^{\text{non}})\}_{j=1}^{\hat{P}}$ . For illustration purposes, the finite elements containing the integration points of the corresponding adaptive rule, already shown in Fig. 18(b), are reproduced as an inset. Inspection of these graphs reveals a smooth dependence on  $\hat{q}_M^{\text{non}}$  throughout most of the sampled interval, with some irregularities appearing near  $\hat{q}_M^{\text{non}} = 0$  (which may be attributed to boundary effects inherent to the graph-regularization procedure). Nevertheless, such irregularities are relatively minor when compared with the behavior of the graphs obtained when the regularization stage of Algorithm 1 is disabled, in the same spirit as the comparison reported in Fig. 9 for the metamaterial benchmark. The corresponding results for the present problem are shown in Fig. 19(b). We can see that the sampled weights exhibit abrupt

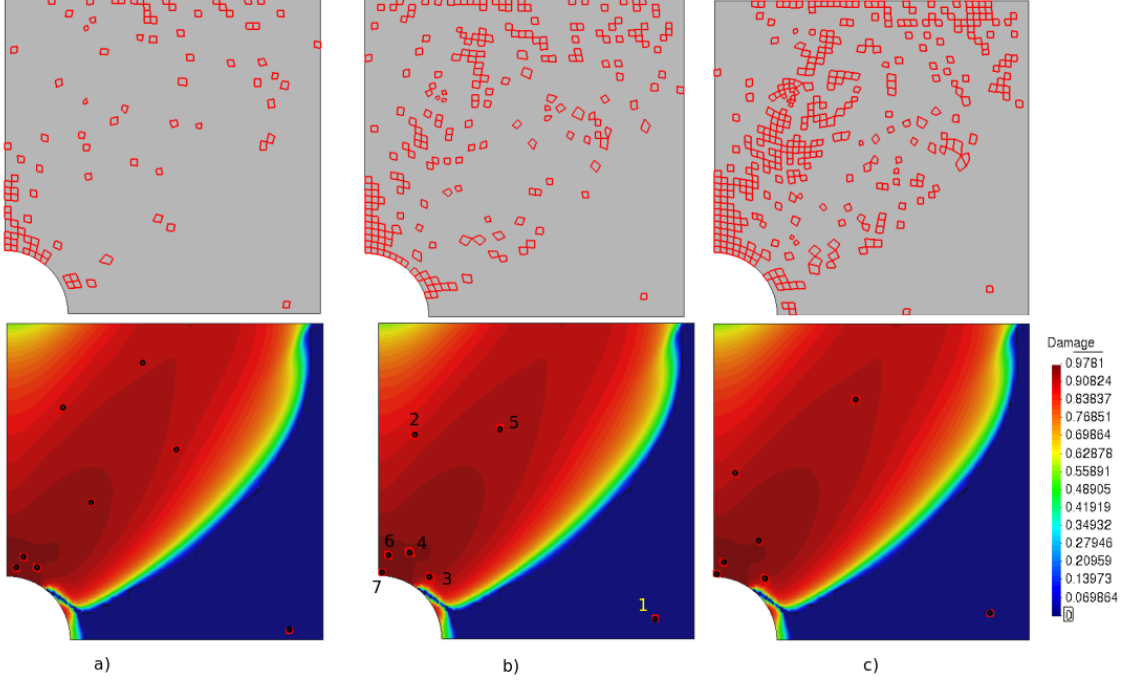


Figure 18: Spatial distribution of the integration points associated with the initial and final cubature rules corresponding to the three fixed-weight initializations reported in Table 5. The upper row shows the elements containing the integration points of the initial fixed-weight rules. The lower row shows the retained integration points of the final adaptive rules, superposed on the damage field  $d_\epsilon$  at the end of the training trajectory. The three columns correspond to the initializations  $m_0 = 97$ ,  $m_0 = 277$ , and  $m_0 = 534$ , respectively. The labels in panel (b) identify the seven retained points used in the subsequent analysis.

variations and a markedly less coherent dependence on  $\hat{q}_M^{\text{non}}$  that in the regularized case, confirming that graph regularization does play a crucial role in generating weight distributions which are relatively smooth, and thus amenable to subsequent regression.

The adaptive nature of the proposed cubature rules can be appreciated more clearly in Fig. 20, which displays the sampled weights associated with four intermediate rules (with  $m = 14, 12, 10$ , and  $8$  integration points) extracted along the pruning process that led to the  $m = 7$  adaptive weight fields shown previously in Fig. 19(a). To visualize the progressive evolution of the spatial distribution of the integration points, each panel includes an inset showing the corresponding integration rule, with circles marking the integration points that will be removed in the subsequent pruning stage. We can see that, as the rule is progressively pruned, the variability of the weight curves increases. Indeed, for  $m = 14$  and  $m = 12$ , the weights exhibit only mild variations along the latent manifold, whereas for  $m = 10$  and, even more markedly, for  $m = 8$ , they vary more significantly with  $\hat{q}_M^{\text{non}}$  to compensate for the discarded integration points.

**Remark 6.** *It should be recalled that all the rules shown in Fig. 20 are feasible solutions of the original optimization problem, i.e., they satisfy the local exactness and positivity constraints. Consequently, if the optimal (i.e., the sparsest) rule is deemed excessively irregular for subsequent regression, one may deliberately step back along the pruning path and select a denser, yet smoother, rule instead.*

#### 7.4.2. Online accuracy assessment

We finally assess whether the substantial reduction, in terms of integration points, achieved by MAW-ECM compromises the online accuracy of the manifold HROM. To this end, we consider the adaptive rule shown in Figure 19(a), obtained by pruning the  $m_0 = 277$  fixed-weight rule down to only  $m = 7$  integration points. The corresponding fixed-weight manifold HROM of Section 7.3.3 (see Table 6) is retained as reference, so that the comparison isolates the effect of replacing the fixed cubature rule by the adaptive one.

For online evaluation, the sampled adaptive weights  $\omega_g = \omega_g(\hat{q}_M^{\text{non}})$  are represented using the same spline framework employed for the decoder in Section 7.3.2. The regressions use 500 uniformly distributed samples

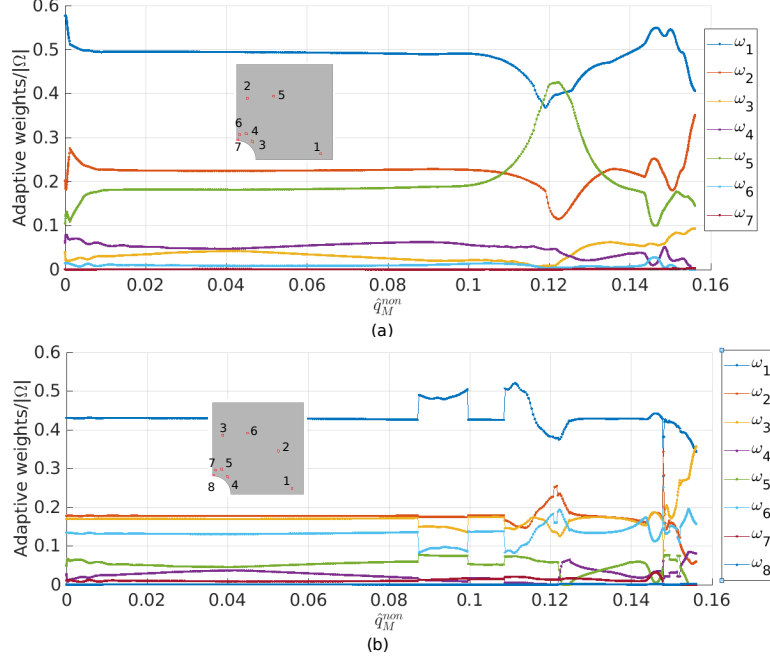


Figure 19: Sampled adaptive weights obtained from the pruning algorithm for the damage benchmark with  $m_0 = 277$ . The insets show the retained supports and the labels used to identify the individual weight samples. (a) Graph-regularized pruning procedure. (b) Pruning without regularization.

Table 6: Relative displacement errors for the fixed- and adaptive-weight manifold HROMs.

| Model                      | Number of points | $(e_d)^{train}$        | $(e_d)^{test}$         |
|----------------------------|------------------|------------------------|------------------------|
| Fixed-weight manifold HROM | 277              | $1.78 \cdot 10^{-3}$   | $1.81 \cdot 10^{-3}$   |
| Manifold HROM with MAW-ECM | 7                | $1.8125 \cdot 10^{-3}$ | $1.6538 \cdot 10^{-3}$ |

selected from the  $\hat{P} = 678$  available snapshots. Outside the sampled interval, each weight is held constant at its nearest endpoint value, thereby preserving positivity and the exact volume constraint inherited from the adaptive cubature construction.

The resulting displacement errors over the training and generalization trajectories are reported in Table 6. Inspection of this table shows that the adaptive rule essentially preserves the accuracy of the fixed-weight manifold HROM: the small variations observed in the training and test errors are negligible relative to the approximation error already introduced by the decoder.

A more detailed assessment is provided in Figure 21, which compares the latent coordinates—the same comparison carried out in Figure 17(b) for the fixed-weight case—and the macroscopic stress–strain response along the generalization trajectory. As in the fixed-weight case,  $q_M^{lin}$  in Figure 21a is virtually indistinguishable from its encoded FE counterpart. The normalized nonlinear coordinate  $\hat{q}_M^{non}$ , by contrast, exhibits a small but visible departure from its encoded FE counterpart around  $t = 0.2$  s, at the onset of damage. However, once damage evolution is fully established, the manifold-HROM curve rejoins the reference FE curve, and the two solutions again become indistinguishable.

The origin of the above mentioned deviation appears to lie in the weight regression: it occurs at the onset of damage, that is, when  $\hat{q}_M^{non} \approx 0$ , precisely where the sampled weights in Figure 19(a) exhibit their largest local irregularities. Nevertheless, this local deviation has practically no influence on the global displacement error, as confirmed by Table 6, nor on the macroscopic stress–strain response shown in Figure 21b, where the complete macroscopic cycle—including primary loading, unloading, reverse loading in compression, and final unloading—is reproduced with no visible discrepancy.

Finally, Figure 22 compares the von Mises stress histories predicted by the FE model and the adaptive manifold HROM at the seven selected Gauss points along the generalization trajectory. With the exception

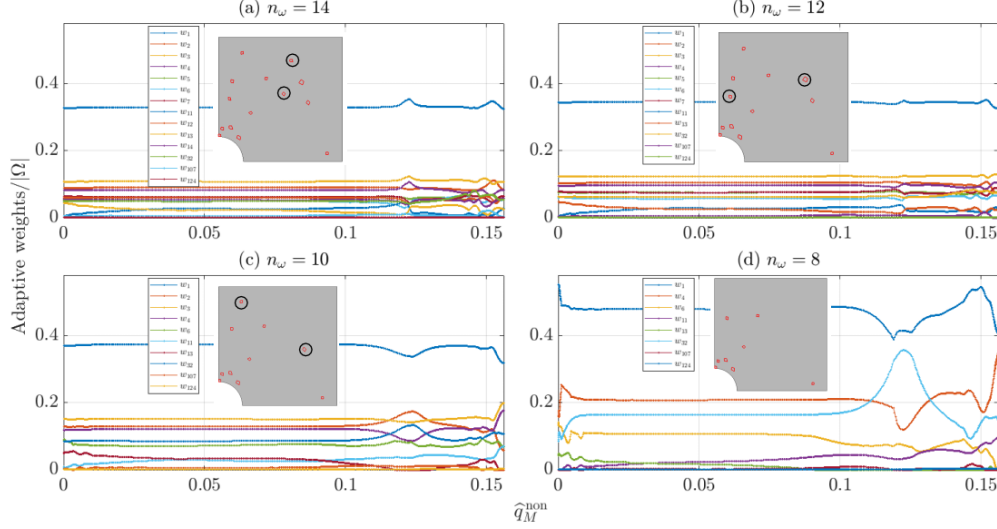


Figure 20: Adaptive weights as a function of the normalized, nonlinear latent coordinate  $\hat{q}_M^{\text{non}}$ , corresponding to four feasible cubature rules extracted from different stages of the pruning process (containing  $m = 14, 12, 10$ , and  $8$  integration points). The insets show the elements containing the integration points of each rule. Integration points enclosed by circles are those selected for removal in the subsequent pruning stage.

of a small discrepancy at point 6 during the initial damage transition, approximately over  $0.2 \lesssim t \lesssim 0.3$ , the FE and manifold-HROM stress histories practically coincide. The agreement is maintained not only during the tensile branch represented in the training data, but also throughout unloading, reverse loading in compression, and the final return to the unloaded state. Particularly noteworthy is point 7, located in the vicinity of the hole and subjected to the largest stress concentration. Despite exhibiting the highest stresses and the largest variation over the cycle, its complete history is reproduced at plotting resolution.

## 8. Concluding remarks

The central premise examined in this work is that the conventional use of fixed weights in sampling-and-weighting hyperreduction becomes unnecessarily restrictive once the state is represented on a nonlinear manifold. The two benchmarks of Sections 6 and 7 support this premise. In both cases, replacing the linear ROM by a nonlinear-manifold representation reduces the number of integration points by a factor remarkably close to the reduction in the number of independent state coordinates. In the metamaterial problem, the latter decreases from 52 to 2, a factor of 26, while the corresponding fixed cubature rule decreases from 1340 to 54 Gauss points, a factor of approximately 25. In the damage problem, the analogous factors are 8 and approximately 8.6, respectively. Allowing these weights to adapt to the latent state then provides a further substantial reduction: MAW-ECM eliminates approximately 80% of the remaining points in the metamaterial benchmark, while in the damage problem the reduction exceeds 97%, from 277 points to only 7–8, with essentially no loss of accuracy relative to the corresponding fixed-weight manifold HROM.

Of particular relevance is the weak dependence of these final cardinalities on the fixed ECM rule used for initialization. Initial supports varying from 54 to 187 points in the first benchmark, and from 97 to 534 points in the second, ultimately collapse to approximately the same number of retained points. Moreover, the final rules lie only a few points above the lower bounds imposed by the local integration conditions. These observations suggest that, once the weights are allowed to adapt, the required support is controlled primarily by the dimension of the local constraint space—which in the present examples is itself tied to the intrinsic dimension of the solution manifold—rather than by the dimension of the global span of all sampled integrands. This is perhaps the most significant practical consequence of replacing global fixed weights by manifold-adaptive ones.

Adaptivity, however, is useful only if the resulting weight fields can be represented reliably online. The numerical experiments confirm the role played by the proposed graph regularization: without it, the sampled weights develop local irregularities that substantially deteriorate the accuracy of their regression. On the

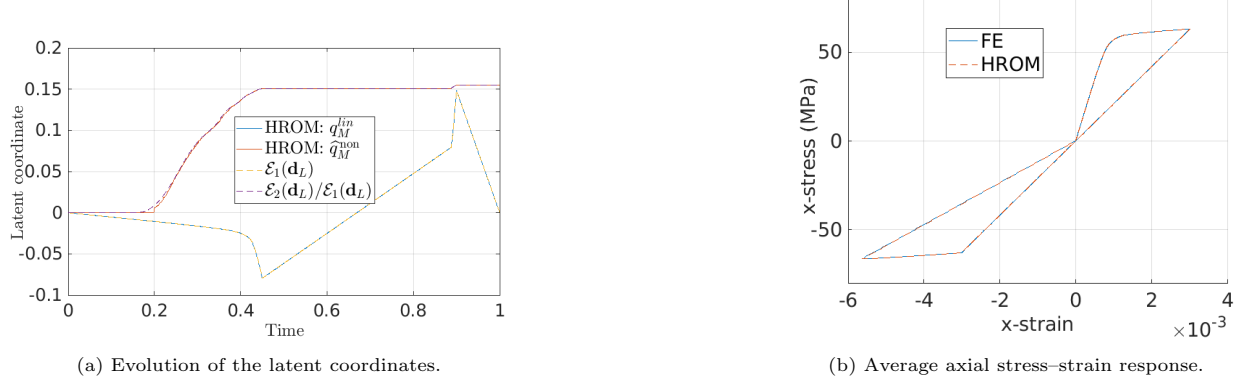


Figure 21: Online response of the manifold HROM with the adaptive rule of  $m = 7$  integration points along the generalization trajectory of Fig. 11(c). (a) Evolution of  $q_M^{lin}$  and  $\hat{q}_M^{non}$ , compared with the coordinates obtained by applying the encoder in Eq. 29 to the FE solutions (see Fig. 17(b) for the analogous fixed-weight comparison). (b) Average axial stress-strain response predicted by the FE and manifold-HROM models (the same comparison carried out in Fig. 14(a) for the standard HROM).

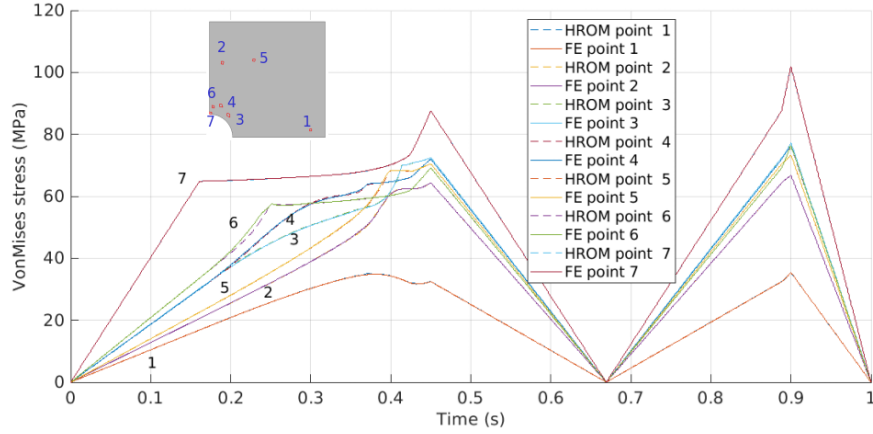


Figure 22: Evolution of the von Mises equivalent stress during the generalization test at the seven Gauss points selected by MAW-ECM. The inset identifies the location and numbering of the selected points.

other hand, the idea of performing the redistribution locally whenever positivity is preserved and activates the graph-coupled stage only when positivity enforcement has proved successful: the experiments show that most points can be removed in this inexpensive regime. In the metamaterial benchmark, for example, this strategy reduces pruning times from more than two hours to less than one minute in one of the studied cases.

The present study is nevertheless exploratory in several respects. Both examples involve only two independent latent coordinates. Extension to substantially higher-dimensional parameter and latent spaces will require revisiting the globally coupled graph operations entering the regularized redistribution. Local graph constructions, partitions, or clusters of the latent domain appear natural alternatives. Similarly, the graph-based identification used for the history coordinate in the damage problem has here been demonstrated only for an ordered one-dimensional damage trajectory; its extension to genuinely multidimensional loading histories remains to be investigated.

A second issue concerns the continuous representation of the adaptive weights after pruning. Positivity and the invariant integration conditions are enforced at the sampled latent states, but generic regression and extrapolation do not preserve these properties automatically. Constraint-preserving regression of the weight fields, including their extrapolation outside the training domain, is therefore an immediate subject for further work. The consistent linearization derived in Appendix A also contains an additional tangent contribution arising from the derivatives of the adaptive weights. Although no convergence difficulties were encountered in the numerical experiments reported here, the implications of this term for robustness deserve a more systematic investigation. Finally, extensions to dynamics and to the other settings will require corresponding

invariant constraints to be identified with care. In particular, if adaptive cubature is applied to inertial terms, preservation of the appropriate mass, first-moment, and inertia properties should be examined so that state-dependent reweighting does not introduce spurious inertial effects.

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## Appendix A. Consistent linearization of the MAW–ECM equilibrium equations

This appendix derives the consistent Newton–Raphson linearization of the MAW–ECM reduced equilibrium equations introduced in Eq. 54. The residual depends on the latent coordinates through the constitutive response, the manifold tangent basis, and additionally through the adaptive cubature weights. The resulting tangent operator therefore comprises the standard and manifold-curvature contributions, together with a new term arising from the variation of the weight fields.

For convenience, Eq. 54 is first recast in residual form. Defining

$$\mathbf{R}_{\mathcal{D}}(\mathbf{q}_M; \boldsymbol{\mu}) := \sum_{g \in \mathcal{Z}_{\omega}} \mathbf{r}_g^{\mathcal{D}}(\mathbf{q}_M; \boldsymbol{\mu}) \omega_g(\mathbf{q}_M) - \mathbf{F}_{\mathcal{D}}^{\text{ext}}(\mathbf{q}_M; \boldsymbol{\mu}), \quad (\text{A.1})$$

where  $\mathbf{r}_g^{\mathcal{D}}$  and  $\mathbf{F}_{\mathcal{D}}^{\text{ext}}$  are given by Eqs. 32 and 36, respectively, the reduced equilibrium equations read

$$\mathbf{R}_{\mathcal{D}}(\mathbf{q}_M; \boldsymbol{\mu}) = \mathbf{0}. \quad (\text{A.2})$$

The consistent tangent matrix,  $\mathbf{K}_{\mathcal{D}} \in \mathbb{R}^{n \times n}$ , is obtained by differentiating Eq. A.1 with respect to the latent coordinates. Using the factorized decoder introduced in Eq. 18, together with Eqs. 33 and 36, the result can be written as

$$\mathbf{K}_{\mathcal{D}} = \mathbf{K}_{\mathcal{D}}^{\text{std}} + \mathbf{K}_{\mathcal{D}}^{\text{curv}} + \mathbf{K}_{\mathcal{D}}^{\omega}. \quad (\text{A.3})$$

The three terms account, respectively, for the constitutive linearization, the curvature of the approximation manifold, and the variation of the adaptive cubature weights.

Let

$$\mathbf{k}_g^{\text{FE}} := \frac{\partial \mathbf{r}_g^{\text{FE}}}{\partial \mathbf{d}}$$

denote the consistent Gauss-point tangent operator. The standard contribution is

$$\mathbf{K}_{\mathcal{D}}^{\text{std}} = \mathbf{J}_{\tau}^T \left( \sum_{g \in \mathcal{Z}_{\omega}} \boldsymbol{\Phi}^T \mathbf{k}_g^{\text{FE}} \boldsymbol{\Phi} \omega_g \right) \mathbf{J}_{\tau}. \quad (\text{A.4})$$

The curvature contribution is expressed componentwise as

$$(\mathbf{K}_{\mathcal{D}}^{\text{curv}})_{ab} = \left[ \frac{\partial^2 \boldsymbol{\tau}}{\partial q_{M,a} \partial q_{M,b}} \right]^T \mathbf{R}_{\tau}^D, \quad a, b = 1, \dots, n, \quad (\text{A.5})$$

where

$$\mathbf{R}_{\tau}^D := \sum_{g \in \mathcal{Z}_{\omega}} \boldsymbol{\Phi}^T \mathbf{r}_g^{\text{FE}} \omega_g - \hat{\mathbf{f}}_{\text{ext}}^{\tau} \boldsymbol{\mu}. \quad (\text{A.6})$$

Notice that the residual in Eq. (A.1) satisfies

$$\mathbf{R}_{\mathcal{D}} = \mathbf{J}_{\tau}^T \mathbf{R}_{\tau}^D. \quad (\text{A.7})$$

Finally, the dependence of the cubature weights on the latent coordinates gives

$$\begin{aligned}\mathbf{K}_{\mathcal{D}}^{\omega} &= \mathbf{J}_{\tau}^T \sum_{g \in \mathcal{Z}_{\omega}} (\Phi^T \mathbf{r}_g^{\text{FE}}) \frac{\partial \omega_g}{\partial \mathbf{q}_M} \\ &= \sum_{g \in \mathcal{Z}_{\omega}} \mathbf{r}_g^{\mathcal{D}} \frac{\partial \omega_g}{\partial \mathbf{q}_M}.\end{aligned}\tag{A.8}$$

Here,  $\partial \omega_g / \partial \mathbf{q}_M \in \mathbb{R}^{1 \times n}$  is obtained by differentiating the regression model used to represent the corresponding weight field.

For a fixed-weight rule,  $\partial \omega_g / \partial \mathbf{q}_M = \mathbf{0}$ , and therefore  $\mathbf{K}_{\mathcal{D}}^{\omega} = \mathbf{0}$ . Likewise,  $\mathbf{K}_{\mathcal{D}}^{\text{curv}}$  vanishes when the decoder is affine, since  $\mathbf{J}_{\tau}$  is then constant and  $\partial^2 \tau / \partial \mathbf{q}_M^2 = \mathbf{0}$ . In the standard linear-ROM case,  $\mathbf{J}_{\tau} = \mathbf{I}$ .

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